UPSC ISS · STATISTICS PAPER-I · OBJECTIVE

Numerical Analysis

Every previous-year question from 2018 to 2026, with answer, exam shortcut, tips & tricks and a step-by-step solution.

Questions in this volume	**180** of 720 across the four units
Papers covered	9 — 2018 to 2026
Per paper	2018: 20 · 2019: 20 · 2020: 20 · 2021: 20 · 2022: 20 · 2023: 20 · 2024: 20 · 2025: 20 · 2026: 20
Syllabus topics	8
Syllabus concepts	22
Layout of every entry	QUESTION → OPTIONS → ANSWER → EXAM SHORTCUT → TIPS & TRICKS → STEP-BY-STEP SOLUTION
Dataset version	1.0

What is authentic and what is not. The questions and their four options are reproduced word for word from the original UPSC ISS Statistics Paper-I booklets. Wording, option text, option order and question numbers are unaltered.

None of the source booklets carried an official answer key. Every ANSWER, EXAM SHORTCUT, TIPS & TRICKS note and STEP-BY-STEP SOLUTION in this volume was worked out for this project and is marked *AI-derived explanation* where it appears. They are a study aid, not an official UPSC solution.

Source defects. 2 question(s) in this volume have a defect in the printed paper. None has been silently corrected: the original text is reproduced exactly and the problem is described under the question.

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N1 · NUMERICAL ANALYSIS

Finite Differences: Operators, Factorial Polynomials, Separation of Symbols

62 questions · 2018: 9 2019: 7 2020: 6 2021: 3 2022: 11 2023: 10 2024: 5 2025: 5 2026: 6

Weight. 62 questions (34.4% of Numerical Analysis) · appeared in 9 of 9 papers · peak year 2022 (11)

Examiner's pattern. The single largest topic in the entire paper (62 items, roughly 7 per paper). Separation of symbols is the engine: everything reduces to E = 1 + Delta. Recurring drills are (i) evaluate (Delta^2/E) f(x), (ii) act with 1/(E - a) on a^x-type functions, (iii) Delta^n applied to a product of linear/polynomial factors — the answer is n! (leading coefficient) h^n, or 0 if the degree is below n, (iv) differences of zero Delta^n 0^m, (v) factorial-polynomial conversion, (vi) delta and mu operator identities, (vii) missing-term problems in an equal-interval table.

Must-know. E = e^(hD); Delta = E - 1; nabla = 1 - E^(-1); delta = E^(1/2) - E^(-1/2); mu = (E^(1/2) + E^(-1/2))/2; Delta - nabla = Deltanabla; Delta^n(a^x) = a^x (a^h - 1)^n; for degree-n poly, Delta^n f = n! a_n h^n and Delta^(n+1) f = 0; Delta x^(n) = n h x^(n-1).

1. 2018 · Q41

Factorial Representation of a Polynomial · Numerical

QUESTION

Given $f(1)=12, f(2)=40, f(3)=90, f(4)=168, f(5)=280, f(6)=432$, the degree of the polynomial $f(x)$ is at least

OPTIONS

(a) 2

(b) 3

(c) 4

(d) 5

ANSWER (AI-derived — no official key exists for this paper)

(b) 3

EXAM SHORTCUT (AI-derived explanation)

Build the difference table: first differences 28, 50, 78, 112, 152; second 22, 28, 34, 40; third 6, 6, 6 - constant. A constant third difference means degree 3.

TIPS & TRICKS (AI-derived explanation)

- Rule: the order of the FIRST constant difference is the degree of the polynomial. Stop differencing the moment a row is constant.
- You do not need the whole table - as soon as three equal entries appear in a row you can stop, since a fourth difference of zero confirms degree 3.
- The word 'at least' is a hint that the data are consistent with degree 3 but any higher-degree polynomial would also fit; the minimum degree is what is asked.
- Arithmetic tip: differences of 12, 40, 90, 168, 280, 432 are all even, so halving them first (14, 25, 39, 56, 76) speeds up the mental subtraction.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. List the values: 12, 40, 90, 168, 280, 432.
2. First differences: 40-12 = 28, 90-40 = 50, 168-90 = 78, 280-168 = 112, 432-280 = 152.
3. Second differences: 50-28 = 22, 78-50 = 28, 112-78 = 34, 152-112 = 40.
4. Third differences: 28-22 = 6, 34-28 = 6, 40-34 = 6. These are constant.
5. Fourth differences are therefore 0, so the data are generated by a polynomial of degree exactly 3.
6. Hence the degree is at least 3, option (b).

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2. 2018 · Q42 Factorial Representation of a Polynomial · Numerical

QUESTION

$f(x)=x^{3}$ is represented as the factorial polynomial $f(x)=Ax^{(3)}+Bx^{(2)}+Cx+D$ with interval of differencing $h=2$. The values of A,B,C,D are

OPTIONS

(a) $1,2,6,0$

(b) $1,6,4,0$

(c) $2,4,6,0$

(d) $2,2,6,1$

ANSWER (AI-derived — no official key exists for this paper)

(b) $1,6,4,0$

EXAM SHORTCUT (AI-derived explanation)

With h = 2, x^(3) = x(x-2)(x-4) = x^3 - 6x^2 + 8x and x^(2) = x(x-2) = x^2 - 2x. Matching coefficients of x^3 gives A = 1, then x^2 gives B = 6 and x gives C = 4, with D = 0.

TIPS & TRICKS (AI-derived explanation)

- Factorial notation with spacing h: x^(n) = x(x-h)(x-2h)...(x-(n-1)h). Here h = 2, so the factors step down by 2 - not by 1. Using h = 1 gives A,B,C,D = 1,3,1,0 and no matching option.
- D must be 0 because x^3 has no constant term and every factorial term vanishes at x = 0 - read that off before any algebra.
- The leading coefficient always transfers unchanged: A = 1 because both x^(3) and x^3 are monic.
- Cross-check with a value: at x = 2, x^(3) = 0 and x^(2) = 0, so f(2) = 2C = 8, giving C = 4 in a single substitution.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With interval of differencing h = 2, the factorial polynomials are x^(3) = x(x-2)(x-4) and x^(2) = x(x-2).
2. Expand: x(x-2)(x-4) = x(x^2 - 6x + 8) = x^3 - 6x^2 + 8x, and x(x-2) = x^2 - 2x.
3. So A x^(3) + B x^(2) + Cx + D = A(x^3 - 6x^2 + 8x) + B(x^2 - 2x) + Cx + D, which must equal x^3.

4. Coefficient of x^3: A = 1. Coefficient of x^2: -6A + B = 0, so B = 6.
5. Coefficient of x: 8A - 2B + C = 0, i.e. 8 - 12 + C = 0, so C = 4. Constant: D = 0.
6. Hence (A, B, C, D) = (1, 6, 4, 0), option (b).

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3. 2018 · Q45 Finite Differences of Different Orders · Conceptual

QUESTION

The images $f(n)$ of $n \in \{1,\dots,99\} \setminus \{50\}$ are given. To estimate $f(50)$ from a forward difference table, the working hypothesis is

OPTIONS

(a) f is a polynomial of degree 50

(b) degree 99

(c) degree 98

(d) degree not more than 97

ANSWER (AI-derived — no official key exists for this paper)

(d) degree not more than 97

EXAM SHORTCUT (AI-derived explanation)

The set {1,...,99} minus {50} has 98 points. A unique interpolating polynomial through 98 points has degree at most 97.

TIPS & TRICKS (AI-derived explanation)

- Counting rule: k data points determine a polynomial of degree at most k - 1. Count the points first, subtract one, and you are done.
- The classic slip is to count 99 points and forget that f(50) is the value being ESTIMATED, not a datum - that gives 98 and lands on option (c).
- 'Degree not more than' is the mathematically correct phrasing; the data could accidentally lie on a lower-degree polynomial, so an exact degree cannot be asserted.
- In practice a difference table for missing-value problems is truncated far earlier; the theoretical maximum degree is what this item tests.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The images f(n) are given for n = 1, 2, ..., 99 with n = 50 omitted.
2. That is 99 - 1 = 98 known data points.
3. Through 98 distinct points there passes exactly one polynomial of degree at most 98 - 1 = 97.
4. The forward-difference (missing-value) technique assumes precisely this: that f is a polynomial whose degree does not exceed 97, so that the 98th difference vanishes.
5. Hence the working hypothesis is that the degree is not more than 97, option (d).

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4. 2018 · Q46 Factorial Representation of a Polynomial · Numerical

QUESTION

Table:

x	1	2	3	4	5
$f(x)$	2	5	7	α	32

The missing term α is

OPTIONS

(a) 14

(b) 15

(c) 16

(d) 17

ANSWER (AI-derived — no official key exists for this paper)

(a) 14

EXAM SHORTCUT (AI-derived explanation)

Four values are known, so assume a cubic and set the fourth difference to zero: f4 - 4f3 + 6f2 - 4f1 + f0 = 0, i.e. 32 - 4a + 42 - 20 + 2 = 0, giving a = 14.

TIPS & TRICKS (AI-derived explanation)

- Missing-value template: with one gap among k+1 tabulated points, set Delta^{k} f_0 = 0 and solve. The binomial coefficients with alternating signs are the whole method.
- The fourth-difference operator expands with coefficients 1, -4, 6, -4, 1. Memorise the row of Pascal's triangle with alternating signs - it appears in every missing-value item.
- Arithmetic care: 6f2 = 6 x 7 = 42 and -4f1 = -20. Combining the knowns gives 56, so 4a = 56 and a = 14 in one step.
- Sanity check the pattern: with a = 14 the first differences are 3, 2, 7, 18 and the third differences are constant at 9, confirming a genuine cubic fit.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The tabulated values are f0 = 2, f1 = 5, f2 = 7, f3 = alpha, f4 = 32 at x = 1, 2, 3, 4, 5 with h = 1.
2. Only four values are actually known, so the standard assumption is that f is a polynomial of degree 3; then the fourth difference vanishes.
3. Delta^{4} f0 = f4 - 4 f3 + 6 f2 - 4 f1 + f0 = 0.
4. Substituting: 32 - 4 alpha + 6(7) - 4(5) + 2 = 0, i.e. 32 - 4 alpha + 42 - 20 + 2 = 0.
5. So 56 - 4 alpha = 0, giving alpha = 14.
6. Hence option (a) is correct.

5. 2018 · Q47 Delta, E and D Operators · Numerical

QUESTION

Given u_{10}=9,u_{20}=39,u_{30}=74,u_{40}=116,u_{50}=167, the value of $E^{3/2}(u_{10})$ is

OPTIONS

(a) 35.75

(b) 45.75

(c) 55.75

(d) 65.75

ANSWER (AI-derived — no official key exists for this paper)

(c) 55.75

EXAM SHORTCUT (AI-derived explanation)

E^(3/2) advances 1.5 intervals from u10, i.e. it estimates u25. Expand (1+Delta)^(3/2) = 1 + 1.5 Delta + 0.375 Delta^2 - 0.0625 Delta^3 and apply to u10: 9 + 45 + 1.875 - 0.125 = 55.75.

TIPS & TRICKS (AI-derived explanation)

- E^p = $(1 + \text{Delta})^p$, so a fractional shift becomes a binomial series in Delta. The coefficients for p = 3/2 are 1, 3/2, 3/8, -1/16, ... - the fourth one is NEGATIVE, which is where careless work goes wrong.
- Build the difference table first: Delta u10 = 30, Delta^2 = 5, Delta^3 = 2, Delta^4 = 0. The series terminates automatically at the third difference.
- The four options differ by exactly 10, and 1.5 x 30 = 45 dominates the sum. Getting the leading term right (9 + 45 = 54) already points to (c).
- Interpretation check: with a tabular interval of 10, E^(3/2) applied to u10 gives the value at 10 + 1.5(10) = 25 - a genuine interpolation, which is why the answer lies between u20 = 39 and u30 = 74... note it does not, because the expansion is a formal operator identity, so trust the algebra.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The tabulated values are u10 = 9, u20 = 39, u30 = 74, u40 = 116, u50 = 167.
2. First differences: 30, 35, 42, 51. Second differences: 5, 7, 9. Third differences: 2, 2. Fourth difference: 0.
3. Since E = 1 + Delta, we expand E^(3/2) = (1 + Delta)^(3/2) = 1 + (3/2)Delta + [(3/2)(1/2)/2!]Delta^2 + [(3/2)(1/2)(-1/2)/3!]Delta^3 + ...
4. The coefficients are 1, 1.5, 0.375 and -0.0625; all later terms vanish because Delta^4 u10 = 0.
5. E^(3/2) u10 = 9 + 1.5(30) + 0.375(5) - 0.0625(2) = 9 + 45 + 1.875 - 0.125.
6. Summing gives 55.75, option (c).

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6. 2018 · Q48 Delta, E and D Operators · Numerical

QUESTION

The value of k satisfying $\Delta^2(1+k\Delta^3)0^5=630$ is

OPTIONS

(a) 4

(b) 5

(c) 6

(d) 7

ANSWER **(AI-derived — no official key exists for this paper)**

(b) 5

EXAM SHORTCUT **(AI-derived explanation)**

Differences of zero: Delta^2 0^5 = 30 and Delta^5 0^5 = 120. So the expression is 30 + 120k = 630, giving k = 5.

TIPS & TRICKS **(AI-derived explanation)**

- Delta^n 0^m = n! S(m, n), where S is a Stirling number of the second kind. The values you need most often are Delta^2 0^5 = 30, Delta^3 0^5 = 150, Delta^4 0^5 = 240 and Delta^5 0^5 = 120.
- Note that Delta^2 (k Delta^3) 0^5 = k Delta^5 0^5 - the operators simply add their orders, so the second term is a FIFTH difference, not a third.
- Delta^n 0^n = n! always (here 5! = 120), which is the fastest of the two values to recall.
- Build Delta^2 0^5 quickly from the table of x^5 at x = 0,1,2: 0, 1, 32 gives first differences 1, 31 and second difference 30.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. The operator expression is Delta^2 (1 + k Delta^3) 0^5 = Delta^2 0^5 + k Delta^5 0^5.
2. Differences of zero satisfy Delta^n 0^m = n! S(m, n). For m = 5: Delta^2 0^5 = 2! S(5,2) = 2 x 15 = 30.
3. Equivalently, from the values of x^5 at x = 0, 1, 2 (namely 0, 1, 32) the first differences are 1 and 31, so the second difference at 0 is 30.
4. Also Delta^5 0^5 = 5! S(5,5) = 120 x 1 = 120.
5. The equation becomes 30 + 120k = 630, so 120k = 600 and k = 5.
6. Hence option (b) is correct.

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7. **2018 · Q49** Sub-division of Intervals · Theoretical

QUESTION

Consider the following statements (interval of differencing =1): 1. $(E\text{-}2)^2(x\cdot 2^x)=0$. 2. $(E\text{-}3)^3(x\cdot 3^x)\neq 0$.

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(a) 1 only

EXAM SHORTCUT (AI-derived explanation)

For any base a, (E - a)(x a^x) = a^(x+1), and (E - a) applied to a^(x+1) gives 0. So $(E-a)^2$ annihilates x a^x. Statement 1 is true; statement 2 claims non-vanishing and is therefore false.

TIPS & TRICKS (AI-derived explanation)

- Annihilator rule: (E - a)^(k+1) kills x^k a^x. Here k = 1, so the SECOND power already gives zero - and so does every higher power.
- Statement 2 is false not because the cube is wrong but because the square already vanishes; anything annihilated by $(E-3)^2$ is certainly annihilated by $(E-3)^3$.
- Do the algebra once with a general base a: E(x a^x) = (x+1)a^(x+1), so (E-a)(x a^x) = (x+1)a^(x+1) - a . x a^x = a^(x+1). One derivation covers both statements.
- This mirrors the differential-operator result (D - a)^(k+1) [x^k e^(ax)] = 0 - the same annihilator theory in shift form.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Work with a general base a and interval of differencing 1. E(x a^x) = (x + 1) a^(x+1).
2. (E - a)(x a^x) = (x+1) a^(x+1) - a (x a^x) = (x+1) a^(x+1) - x a^(x+1) = a^(x+1).
3. Now (E - a) a^(x+1) = a^(x+2) - a . a^(x+1) = a^(x+2) - a^(x+2) = 0.
4. So $(E - a)^2$ (x a^x) = 0 for every a.
5. Statement 1: with a = 2, $(E - 2)^2$ (x 2^x) = 0. TRUE.
6. Statement 2: with a = 3, $(E - 3)^2$ (x 3^x) = 0, hence $(E - 3)^3$ (x 3^x) = 0 as well. The claim that it is non-zero is FALSE. Only statement 1 holds, option (a).

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8. 2018 · Q52 Delta, E and D Operators · Theoretical

QUESTION

If δ, μ are the central difference and average operators, $\delta[f(x)g(x)]$ equals

OPTIONS

(a) $\mu\ f(x)\mu\ g(x)+\delta\ f(x)\delta\ g(x)$

(b) $\delta\ f(x)\delta\ g(x)-\mu\ f(x)\mu\ g(x)$

(c) $\mu\ f(x)\delta\ g(x)-\mu\ g(x)\delta\ f(x)$

(d) $\mu\ f(x)\delta\ g(x)+\mu\ g(x)\delta\ f(x)$

ANSWER (AI-derived — no official key exists for this paper)

(d) $\mu f(x) \delta g(x) + \mu g(x) \delta f(x)$

EXAM SHORTCUT (AI-derived explanation)

The central-difference product rule mirrors the derivative product rule with mu playing the role of 'value': delta(fg) = mu f . delta g + mu g . delta f.

TIPS & TRICKS (AI-derived explanation)

- Expand once and remember it: with f+ = f(x+h/2) and f- = f(x-h/2), mu f . delta g + mu g . delta f = f+ g+ - f- g- = delta(fg). The cross terms cancel exactly.
- Structural check: delta(fg) must be ANTI-symmetric in the sense that swapping f and g leaves it unchanged. Option (c) has a minus sign and changes sign on swapping, so it is impossible.
- Option (a) contains no delta at all on one term and is dimensionally wrong: every term of a delta-identity must carry exactly one delta.
- Companion identities to keep together: mu(fg) = mu f mu g + (1/4) delta f delta g, and mu^2 = 1 + delta^2/4.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write f+ = f(x + h/2), f- = f(x - h/2), and similarly for g. Then delta f = f+ - f- and mu f = (f+ + f-)/2.
2. By definition, delta[f(x)g(x)] = f+ g+ - f- g-.
3. Now compute mu f . delta g + mu g . delta f = (1/2)(f+ + f-)(g+ - g-) + (1/2)(g+ + g-)(f+ - f-).
4. Expanding the first product: (1/2)[f+g+ - f+g- + f-g+ - f-g-]. Expanding the second: (1/2)[f+g+ - f-g+ + f+g- - f-g-].
5. Adding, the terms -f+g- and +f+g- cancel, as do +f-g+ and -f-g+, leaving f+g+ - f-g-.
6. This equals delta(fg), so delta[f(x)g(x)] = mu f . delta g + mu g . delta f, option (d).

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9. 2018 · Q55 Delta, E and D Operators · Theoretical

QUESTION

The central difference δ is equivalent to which of the following? 1. $2\sinh(\frac{hD}{2})$ 2. $E^{-1/2}\Delta$ 3. $\Delta(1+\Delta)^{-1/2}$

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

All three are standard: delta = E^(1/2) - E^(-1/2) = e^(hD/2) - e^(-hD/2) = 2 sinh(hD/2); also delta = E^(-1/2)(E - 1) = E^(-1/2) Delta; and since E = 1 + Delta, E^(-1/2) Delta = Delta(1 + Delta)^(-1/2).

TIPS & TRICKS (AI-derived explanation)

- Anchor everything on the two identities E = e^(hD) and E = 1 + Delta. Every operator relation in this chapter is one substitution away from those two.
- delta = E^(1/2) - E^(-1/2) is the definition; factor out E^(-1/2) to get E^(-1/2)(E - 1) = E^(-1/2) Delta, which settles statements 2 and 3 together.
- The hyperbolic form follows because e^(t) - e^(-t) = 2 sinh t with t = hD/2 - note the HALF in the argument, a frequent slip.
- Related identities worth carrying: mu = cosh(hD/2), Delta = E - 1, nabla = 1 - E^(-1), and delta^2 = Delta nabla.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. By definition the central difference operator is delta = E^(1/2) - E^(-1/2), where E is the shift operator.
2. Statement 1: since E = e^(hD), we have E^(1/2) = e^(hD/2) and E^(-1/2) = e^(-hD/2), so delta = e^(hD/2) - e^(-hD/2) = 2 sinh(hD/2). TRUE.
3. Statement 2: delta = E^(1/2) - E^(-1/2) = E^(-1/2)(E - 1) = E^(-1/2) Delta, since Delta = E - 1. TRUE.
4. Statement 3: because E = 1 + Delta, we have E^(-1/2) = (1 + Delta)^(-1/2), so from statement 2, delta = Delta (1 + Delta)^(-1/2). TRUE.
5. All three representations of delta are valid.
6. Hence option (d) is correct.

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10. 2019 · Q51 Factorial Representation of a Polynomial · Numerical

QUESTION

Consider the following table:

x	2	3	4	5	6
y	45	49.2	54.1	k	67.4

What is the value of k?

OPTIONS

(a) 60.05

(b) 61.05

(c) 62.05

(d) 63.05

ANSWER **(AI-derived — no official key exists for this paper)**

(a) 60.05

EXAM SHORTCUT **(AI-derived explanation)**

Four values are known, so set the fourth difference to zero: y4 - 4y3 + 6y2 - 4y1 + y0 = 0 gives 67.4 - 4k + 324.6 - 196.8 + 45 = 0, so 4k = 240.2 and k = 60.05.

TIPS & TRICKS **(AI-derived explanation)**

- Missing-value technique: with one gap among five equally spaced points, assume a cubic and impose Delta^4 y0 = 0. The coefficients are 1, -4, 6, -4, 1.
- Collect the three known contributions first: 67.4 + 324.6 - 196.8 + 45 = 240.2, then divide by 4. Doing the arithmetic in one sweep avoids sign errors.
- The four options differ by exactly 1.00, so a single arithmetic slip moves you a whole option - carry the decimals carefully.
- Plausibility: the y-values rise by roughly 4 to 6 per step, so a value near 60 between 54.1 and 67.4 is entirely reasonable.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. The tabulated values at x = 2, 3, 4, 5, 6 are y0 = 45, y1 = 49.2, y2 = 54.1, y3 = k, y4 = 67.4, equally spaced with h = 1.
2. Only four values are known, so the standard assumption is that y is a cubic in x, whence the fourth difference vanishes.
3. Delta^4 y0 = y4 - 4y3 + 6y2 - 4y1 + y0 = 0.
4. Substituting: 67.4 - 4k + 6(54.1) - 4(49.2) + 45 = 0, i.e. 67.4 - 4k + 324.6 - 196.8 + 45 = 0.
5. Collecting the constants: 240.2 - 4k = 0, so k = 60.05.
6. Hence option (a) is correct.

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11. 2019 · Q52 Delta, E and D Operators · Numerical

QUESTION

What is the value of $\Delta^{10}[(1\text{-}ax)(1\text{-}bx^2)(1\text{-}cx^3)(1\text{-}dx^4)]$?

OPTIONS

(a) 0

(b) $10!$

(c) $abcd\times(10)!$

(d) $abcd$

ANSWER **(AI-derived — no official key exists for this paper)**

(c) $abcd\times(10)!$

EXAM SHORTCUT (AI-derived explanation)

The product has degree 1 + 2 + 3 + 4 = 10 with leading coefficient (-a)(-b)(-c)(-d) = abcd. For h = 1, Delta^n of a degree-n polynomial is (leading coefficient) x n!, so the answer is abcd x 10!.

TIPS & TRICKS (AI-derived explanation)

- Key identity: for h = 1, Delta^n [a_n x^n + lower terms] = a_n . n!. All lower-degree terms are annihilated.
- Get the degree by ADDING the degrees of the factors (1+2+3+4 = 10) - do not multiply them out.
- The leading coefficient is the product of the leading coefficients: (-a)(-b)(-c)(-d) = +abcd. Four negatives give a positive, which is why no minus sign appears.
- If the operator order had exceeded the degree the answer would be 0 (option a); here they match exactly, which is what makes the factorial appear.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Expand the degrees: (1 - ax) has degree 1, (1 - bx^2) degree 2, (1 - cx^3) degree 3, (1 - dx^4) degree 4.
2. The product is therefore a polynomial of degree 1 + 2 + 3 + 4 = 10.
3. Its leading term comes from multiplying the highest-degree term of each factor: (-ax)(-bx^2)(-cx^3)(-dx^4) = abcd x^{10}, so the leading coefficient is abcd.
4. For an interval of differencing h = 1, the standard result is $\text{Delta}^n(x^n)$ = n!, and all terms of degree below n vanish under Delta^n.
5. Hence Delta^{10} of the product equals abcd x 10!.
6. Option (c) is correct.

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12. 2019 · Q53 Delta, E and D Operators · Numerical

QUESTION

What are the values of $\Delta^5 0^3$ and $\Delta^3 0^5$ respectively?

OPTIONS

(a) 0 and 0

(b) 144 and 0

(c) 0 and 150

(d) 0 and 96

ANSWER (AI-derived — no official key exists for this paper)

(c) 0 and 150

EXAM SHORTCUT (AI-derived explanation)

$\text{Delta}^n 0^m$ = 0 whenever n > m, so $\text{Delta}^5 0^3$ = 0. And $\text{Delta}^3 0^5$ = 3! S(5,3) = 6 x 25 = 150.

TIPS & TRICKS (AI-derived explanation)

- First rule: Delta^{n} 0^{m} vanishes when n > m, because 0^{m} behaves like a polynomial of degree m. This kills half the item instantly.
- Second rule: Delta^{n} 0^{m} = n! S(m, n) with S the Stirling number of the second kind. The values you need most are S(5,2) = 15, S(5,3) = 25, S(5,4) = 10.
- Direct check for Delta^{3} 0^{5}: from x^{5} at x = 0,1,2,3 the values are 0, 1, 32, 243; first differences 1, 31, 211; second 30, 180; third 150.
- Since Delta^{5} 0^{3} = 0, options (b) and (d) fall immediately, and only the second value has to be computed.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The symbol Delta^{n} 0^{m} denotes the nth forward difference of x^{m} evaluated at x = 0 with h = 1.
2. Since x^{3} is a polynomial of degree 3, its fifth difference is identically zero: Delta^{5} 0^{3} = 0.
3. For Delta^{3} 0^{5}, tabulate x^{5} at x = 0, 1, 2, 3: values 0, 1, 32, 243.
4. First differences: 1, 31, 211. Second differences: 30, 180. Third difference: 180 - 30 = 150.
5. This agrees with the formula Delta^{n} 0^{m} = n! S(m,n): 3! S(5,3) = 6 x 25 = 150.
6. Hence the values are 0 and 150, option (c).

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13. 2019 · Q54 Summation of Series · Numerical

QUESTION

For interval of differencing as unity: $f(0)=41$, $f(1)=61$, $f(5)=20$, $f(6)=50$. What is the value of $S=\sum_{k=0}^{4} \Delta^{2} f(k)$?

OPTIONS

(a) -10

(b) 10

(c) 50

(d) Cannot be determined

ANSWER (AI-derived — no official key exists for this paper)

(b) 10

EXAM SHORTCUT (AI-derived explanation)

The sum of second differences telescopes: sum over k = 0 to 4 of Delta^{2} f(k) = Delta f(5) - Delta f(0) = (50 - 20) - (61 - 41) = 30 - 20 = 10.

TIPS & TRICKS (AI-derived explanation)

- Telescoping rule: the sum of Delta g(k) from k = a to b equals g(b+1) - g(a). Apply it with g = Delta f to collapse a sum of second differences into two first differences.

- This is exactly why only f(0), f(1), f(5) and f(6) are supplied - the intermediate values cancel, so option (d) 'cannot be determined' is the planted trap.
- Delta f(5) needs f(6) and f(5); Delta f(0) needs f(1) and f(0). Identify which four values you need BEFORE worrying about the missing ones.
- The same telescoping idea underlies the 'summation of series whose general term is a first difference' topic in the syllabus - one technique, two exam applications.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. By definition Delta^2 f(k) = Delta[Delta f(k)], so the sum in question is the sum of the first differences of the function g(k) = Delta f(k).
2. The sum of Delta g(k) for k = 0 to 4 telescopes to g(5) - g(0).
3. Hence S = Delta f(5) - Delta f(0).
4. Delta f(5) = f(6) - f(5) = 50 - 20 = 30.
5. Delta f(0) = f(1) - f(0) = 61 - 41 = 20.
6. Therefore S = 30 - 20 = 10, option (b).

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14. 2019 · Q55 Factorial Representation of a Polynomial · Numerical

QUESTION

Which one of the following is equal to $1+x+x^2+x^3$?

OPTIONS

(a) $x^{(3)}+3x^{(2)}+4x^{(1)}+1$

(b) $x^{(3)}+x^{(2)}+3x^{(1)}+1$

(c) $3x^{(3)}+4x^{(2)}+x^{(1)}+1$

(d) $x^{(3)}+4x^{(2)}+3x^{(1)}+1$

ANSWER (AI-derived — no official key exists for this paper)

(d) $x^{(3)}+4x^{(2)}+3x^{(1)}+1$

EXAM SHORTCUT (AI-derived explanation)

With h = 1, x^(2) = x^2 - x and x^(3) = x^3 - 3x^2 + 2x. Matching coefficients from the top: x^3 gives A = 1, x^2 gives B = 4, x gives C = 3, constant gives D = 1.

TIPS & TRICKS (AI-derived explanation)

- Work downwards from the highest power: the leading coefficient transfers unchanged, then each lower coefficient is corrected for the terms the factorial polynomials contribute.
- The constant is read off instantly: every factorial polynomial vanishes at x = 0, so D = f(0) = 1.
- Fast check at x = 1: x^(2) = 0 and x^(3) = 0, so the expression reduces to C + D. Since f(1) = 1+1+1+1 = 4 and D = 1, we get C = 3 in one line.
- Alternatively build the difference table of f at x = 0,1,2,3 (values 1, 4, 15, 40): the leading differences 1, 3, 8, 6 give the coefficients 1, 3, 4, 1 in the order D, C, B/1!, ... - a useful cross-check.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With interval of differencing h = 1, x^(1) = x, x^(2) = x(x-1) = x^2 - x, and x^(3) = x(x-1)(x-2) = x^3 - $3x^2$ + 2x.
2. Write 1 + x + x^2 + x^3 = A x^(3) + B x^(2) + C x^(1) + D.
3. Coefficient of x^3: A = 1.
4. Coefficient of x^2: -3A + B = 1, so B = 1 + 3 = 4.
5. Coefficient of x: 2A - B + C = 1, i.e. 2 - 4 + C = 1, so C = 3. Constant: D = 1.
6. Hence 1 + x + x^2 + x^3 = x^(3) + 4x^(2) + 3x^(1) + 1, option (d).

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15. 2019 · Q56 Factorial Representation of a Polynomial · Numerical

QUESTION

$f(x)$ is a polynomial of third degree with $f(-2)=-33$, $f(-1)=m_1$, $f(0)=5$, $f(1)=9$, $f(2)=m_2$, $f(3)=47$. What are the values of m_1 and m_2 respectively?

OPTIONS

(a) -10 and 19

(b) -10 and 21

(c) -5 and 19

(d) -5 and 21

ANSWER (AI-derived — no official key exists for this paper)

(c) -5 and 19

EXAM SHORTCUT (AI-derived explanation)

A cubic has zero fourth differences. Applying Delta^4 = 0 at x = -2 and at x = -1 gives m2 - 4m1 = 39 and m1 - 4m2 = -81; solving, m1 = -5 and m2 = 19.

TIPS & TRICKS (AI-derived explanation)

- Two unknowns need two equations, so write the fourth-difference condition at TWO starting points. The coefficient pattern 1, -4, 6, -4, 1 is the same each time.
- Verify at the end by building the table: 45 values -33, -5, 5, 9, 19, 47 give first differences 28, 10, 4, 10, 28 and third differences 12, 12, 12 - constant, as a cubic demands.
- The value f(0) = 5 is the constant term and f(1) - f(0) = 4 is a strong anchor; a candidate value of m1 = -10 would make the first differences 23, 15, 4, ... which cannot come from a cubic with constant third differences.
- Solve the pair by substitution: m2 = 4m1 + 39 into m1 - 4m2 + 81 = 0 gives -15 m1 = 75, so m1 = -5.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a third-degree polynomial the fourth forward difference vanishes identically, with coefficient pattern 1, -4, 6, -4, 1.

2. Applying it at x = -2: f(2) - 4f(1) + 6f(0) - 4f(-1) + f(-2) = 0, i.e. m2 - 36 + 30 - 4m1 - 33 = 0, so m2 - 4m1 = 39.
3. Applying it at x = -1: f(3) - 4f(2) + 6f(1) - 4f(0) + f(-1) = 0, i.e. 47 - 4m2 + 54 - 20 + m1 = 0, so m1 - 4m2 = -81.
4. From the first equation m2 = 4m1 + 39; substituting into the second gives m1 - 16m1 - 156 = -81, i.e. -15m1 = 75, so m1 = -5.
5. Then m2 = 4(-5) + 39 = 19.
6. Check: the values -33, -5, 5, 9, 19, 47 give first differences 28, 10, 4, 10, 28; second differences -18, -6, 6, 18; third differences 12, 12, 12 - constant, confirming a cubic. Option (c) is correct.

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16. 2019 · Q79 Delta, E and D Operators · Theoretical

QUESTION

If δ is the central difference operator, what is $\frac{1}{2}\delta^2 + \delta\sqrt{1+\frac{\delta^2}{4}}$ equal to?

OPTIONS

(a) $E+1$

(b) ∇

(c) Δ

(d) $\Delta+1$

ANSWER (AI-derived — no official key exists for this paper)

(c) Δ

EXAM SHORTCUT (AI-derived explanation)

Recognise sqrt(1 + delta^2/4) = mu. Then (1/2)delta^2 + delta.mu = (E - 2 + E^{-1})/2 + (E - E^{-1})/2 = E - 1 = Delta.

TIPS & TRICKS (AI-derived explanation)

- Memorise mu = sqrt(1 + delta^2/4) - it is the single identity that unlocks this whole family of questions.
- Work in terms of E^(1/2) and E^(-1/2): delta = E^(1/2) - E^(-1/2) and mu = (E^(1/2) + E^(-1/2))/2, so delta.mu = (E - E^(-1))/2 as a difference of squares.
- delta^2 = E - 2 + E^(-1). Adding half of it to delta.mu makes both E^(-1) terms cancel, leaving E - 1.
- Sanity test on f(x) = x with h = 1: delta^2 x = 0 and delta.mu x = 1, so the expression gives 1, which equals Delta x. Quick numerical checks like this settle operator identities fast.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write everything in terms of the shift operator: delta = E^(1/2) - E^(-1/2) and mu = (E^(1/2) + E^(-1/2))/2.
2. A standard identity is mu = sqrt(1 + delta^2/4), so the given expression is (1/2)delta^2 + delta.mu.
3. delta^2 = (E^(1/2) - E^(-1/2))^2 = E - 2 + E^(-1), so (1/2)delta^2 = (E - 2 + E^(-1))/2.

4. delta.mu = (E^(1/2) - E^(-1/2))(E^(1/2) + E^(-1/2))/2 = (E - E^(-1))/2.
5. Adding: (E - 2 + E^(-1))/2 + (E - E^(-1))/2 = (2E - 2)/2 = E - 1.
6. Since Delta = E - 1, the expression equals Delta, option (c).

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17. 2020 · Q63 Delta, E and D Operators · Conceptual

QUESTION

What is $\delta^2 y_5$ equal to? (δ = central difference operator)

OPTIONS

(a) $y_6 - y_5 + y_4$

(b) $y_6 - 2y_5 + y_4$

(c) $y_6 + y_5$

(d) $y_6 - 4y_2 + y_3$

ANSWER (AI-derived — no official key exists for this paper)

(b) $y_6 - 2y_5 + y_4$

EXAM SHORTCUT (AI-derived explanation)

delta^2 y_n = y_(n+1) - 2y_n + y_(n-1). With n = 5 that is y6 - 2y5 + y4.

TIPS & TRICKS (AI-derived explanation)

- delta y_n = y_(n+1/2) - y_(n-1/2), so applying it twice at an integer index gives the symmetric three-point formula y_(n+1) - 2y_n + y_(n-1).
- The coefficients 1, -2, 1 are the same as for Delta^2 and nabla^2; only the INDEXING differs - the central form is symmetric about y_n.
- This is exactly the second-difference stencil used to approximate f''(x) by [f(x+h) - 2f(x) + f(x-h)]/h^2.
- The coefficients of any pure difference operator must sum to zero (a constant has zero second difference): 1 - 2 + 1 = 0. Options (a) and (c) fail this test instantly.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The central difference operator is defined by delta y_n = y_(n + 1/2) - y_(n - 1/2).
2. Applying it twice: delta^2 y_n = delta(y_(n+1/2)) - delta(y_(n-1/2)) = (y_(n+1) - y_n) - (y_n - y_(n-1)).
3. Simplifying: delta^2 y_n = y_(n+1) - 2y_n + y_(n-1).
4. Setting n = 5 gives delta^2 y_5 = y_6 - 2 y_5 + y_4.
5. Check: the coefficients 1, -2, 1 sum to zero, as they must for a second difference to annihilate constants.
6. Hence option (b) is correct.

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18. 2020 · Q66 Delta, E and D Operators · Theoretical

QUESTION

If δ, μ are the central difference and average operators, $\mu\delta$ equals

OPTIONS

(a) $\frac{1}{2}(\Delta+\nabla)$

(b) $\frac{1}{2}(\Delta-\nabla)$

(c) $(\Delta+\nabla)$

(d) $(\Delta-\nabla)$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{1}{2}(\Delta+\nabla)$

EXAM SHORTCUT (AI-derived explanation)

mu delta = (E^(1/2)+E^(-1/2))/2 x (E^(1/2)-E^(-1/2)) = (E - E^(-1))/2. Since Delta = E - 1 and nabla = 1 - E^(-1), their sum is E - E^(-1), so mu delta = (Delta + nabla)/2.

TIPS & TRICKS (AI-derived explanation)

- Convert everything to E: Delta = E - 1, nabla = 1 - E^(-1), delta = E^(1/2) - E^(-1/2), mu = (E^(1/2) + E^(-1/2))/2. Every identity in this chapter falls out of these four.
- mu delta is a difference of squares, which is why it collapses so cleanly to (E - E^(-1))/2.
- Note Delta + nabla = (E - 1) + (1 - E^(-1)) = E - E^(-1) - the constants cancel, which is the crux of the identity.
- Companion result: Delta - nabla = E - 2 + E^(-1) = delta^2, worth memorising alongside this one.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write the operators in terms of the shift operator E: mu = (E^(1/2) + E^(-1/2))/2 and delta = E^(1/2) - E^(-1/2).
2. Their product is mu delta = (1/2)(E^(1/2) + E^(-1/2))(E^(1/2) - E^(-1/2)) = (1/2)(E - E^(-1)).
3. Also Delta = E - 1 and nabla = 1 - E^(-1).
4. Adding these: Delta + nabla = (E - 1) + (1 - E^(-1)) = E - E^(-1).
5. Therefore mu delta = (1/2)(Delta + nabla).
6. Hence option (a) is correct.

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19. 2020 · Q67 Sub-division of Intervals · Numerical

QUESTION

What is $\frac{1}{E^2-9E+18}(12\times5^x)$ equal to? (interval of differencing =1)

OPTIONS

(a) -6×5^x

(b) 6×5^{x}

(c) $-6^{-1} \times 5^{x}$

(d) $6^{-1} \times 5^{x}$

ANSWER (AI-derived — no official key exists for this paper)

(a) -6×5^{x}

EXAM SHORTCUT (AI-derived explanation)

On 5^x the operator E acts as multiplication by 5, so E^2 - 9E + 18 becomes 25 - 45 + 18 = -2. Hence the inverse operator gives $12(5^x)/(-2) = -6\ (5^x)$.

TIPS & TRICKS (AI-derived explanation)

- Key substitution rule: $E(a^x)$ = a^(x+1) = a . a^x, so any polynomial in E acting on a^x is just that polynomial EVALUATED at a.
- Evaluate the quadratic at a = 5: 25 - 45 + 18 = -2. The sign is negative, which is exactly what distinguishes option (a) from option (b).
- The inverse operator 1/phi(E) simply divides by the same number, so the answer is 12/(-2) = -6 times 5^x.
- This is the discrete analogue of the particular-integral rule 1/phi(D) e^(ax) = e^(ax)/phi(a) from differential equations - same technique, shift operator instead of D.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For the exponential function 5^x with interval of differencing 1, $E(5^x)$ = 5^(x+1) = 5 . 5^x, so E acts as multiplication by 5.
2. Consequently, for any polynomial phi, phi(E) 5^x = phi(5) 5^x.
3. Here phi(E) = E^2 - 9E + 18, so phi(5) = 25 - 45 + 18 = -2.
4. Therefore (E^2 - 9E + 18)(5^x) = -2 (5^x).
5. Applying the inverse operator to 12 x 5^x: [1/(E^2 - 9E + 18)](12 . 5^x) = 12 . 5^x/(-2) = -6 . 5^x.
6. Hence option (a) is correct.

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20. 2020 · Q68 Factorial Representation of a Polynomial · Numerical

QUESTION

The value of $(3x+8)^{(4)}$ at $x=2$ is

OPTIONS

(a) 6160

(b) 12320

(c) 13440

(d) 24024

ANSWER **(AI-derived — no official key exists for this paper)**

(a) 6160

EXAM SHORTCUT **(AI-derived explanation)**

For a linear argument the factorial function steps down by the coefficient: (3x+8)^(4) = (3x+8)(3x+5)(3x+2)(3x-1). At x = 2 this is 14 x 11 x 8 x 5 = 6160.

TIPS & TRICKS **(AI-derived explanation)**

- Generalised factorial: (ax+b)^(n) = (ax+b)(ax+b-a)(ax+b-2a)...(ax+b-(n-1)a) when h = 1. The step is a, NOT 1, because that is what makes Delta(ax+b)^(n) = na(ax+b)^(n-1) hold.
- Verify the convention by the difference rule: only the step-a version has a clean forward difference, which is the whole purpose of factorial notation.
- The distractors are built from wrong steps: step 1 gives 14 x 13 x 12 x 11 = 24024 and step 2 gives 14 x 12 x 10 x 8 = 13440. Identify the step before multiplying.
- Compute in pairs: (14 x 11) = 154 and (8 x 5) = 40, then 154 x 40 = 6160 - much safer than a four-way chain.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. With interval of differencing h = 1, the generalised factorial function of a linear argument is (ax+b)^(n) = (ax+b)(ax+b-a)(ax+b-2a)...(ax+b-(n-1)a).
2. This is the definition that yields the standard difference rule Delta[(ax+b)^(n)] = n a (ax+b)^(n-1).
3. Here a = 3, b = 8 and n = 4, so (3x+8)^(4) = (3x+8)(3x+5)(3x+2)(3x-1).
4. At x = 2: 3(2) + 8 = 14, and the successive factors are 14, 11, 8 and 5.
5. Multiplying: (14)(11) = 154 and (8)(5) = 40, so the product is 154 x 40 = 6160.
6. Hence option (a) is correct.

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21. **2020 · Q75** Delta, E and D Operators · Theoretical

QUESTION

Which one of the following is correct?

OPTIONS

(a) $\Delta\nabla \neq \nabla\Delta$

(b) $\Delta\ E \neq\ E\Delta$

(c) $\Delta\text{-}E \neq\ I$

(d) $E\text{-}\Delta \neq\ I$

ANSWER **(AI-derived — no official key exists for this paper)**

(c) $\Delta\text{-}E \neq\ I$

EXAM SHORTCUT (AI-derived explanation)

Delta - E = (E - 1) - E = -I, which is NOT equal to I. So the inequality in option (c) is the true statement; the other three assert inequalities that are actually equalities.

TIPS & TRICKS (AI-derived explanation)

- Convert each option to E-form: Delta nabla = nabla Delta = delta^2, Delta E = E Delta = E^2 - E, E - Delta = I, and Delta - E = -I.
- All four options are phrased as 'not equal' claims, so you are hunting for the one pair that genuinely differs - read the direction of the subtraction carefully.
- The trap is the near-identical pair (c) and (d): E - Delta = I is true, so (d)'s 'not equal' is false, while Delta - E = -I makes (c)'s 'not equal' true.
- All these operators are polynomials in E, and polynomials in a single operator always commute - which is why options (a) and (b) can be dismissed on principle.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. All the operators here are functions of the shift operator E: Delta = E - 1 and nabla = 1 - E^(-1).
2. Option (a): Delta nabla = (E-1)(1 - E^(-1)) = E - 2 + E^(-1) = nabla Delta, since polynomials in E commute. So Delta nabla = nabla Delta and the claimed inequality is FALSE.
3. Option (b): Delta E = (E-1)E = E^2 - E and E Delta = E(E-1) = E^2 - E, so they are equal and the claimed inequality is FALSE.
4. Option (d): E - Delta = E - (E - 1) = 1 = I, so E - Delta does equal I and the claimed inequality is FALSE.
5. Option (c): Delta - E = (E - 1) - E = -1 = -I, which is not equal to I. The claimed inequality is TRUE.
6. Hence option (c) is correct.

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22. 2020 · Q80 Delta, E and D Operators · Numerical

QUESTION

If $y = a(3)^x + b(-2)^x$ with increment $h = 1$, then $(\Delta^2 + \Delta - 6)y$ equals

OPTIONS

(a) 1

(b) 0

(c) $2x$

(d) $4x$

ANSWER (AI-derived — no official key exists for this paper)

(b) 0

EXAM SHORTCUT (AI-derived explanation)

Delta^2 + Delta - 6 = (E-1)^2 + (E-1) - 6 = E^2 - E - 6 = (E-3)(E+2). The factor (E-3) annihilates 3^x and (E+2) annihilates (-2)^x, so the result is 0.

TIPS & TRICKS (AI-derived explanation)

- Always convert the operator polynomial to E first: substituting Delta = E - 1 turns Delta^2 + Delta - 6 into the far friendlier E^2 - E - 6.
- Factorise and read off the roots: (E-3)(E+2) has roots 3 and -2, which are exactly the bases appearing in y. That match is the whole design of the question.
- Annihilator rule: (E - a) kills a^x, since E(a^x) = a^(x+1) = a . a^x. Each factor removes one term of y.
- This is the discrete analogue of solving a linear differential equation whose auxiliary roots are 3 and -2 - the given y is precisely the general solution of the difference equation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With interval h = 1, Delta = E - 1, so Delta^2 + Delta - 6 = (E-1)^2 + (E-1) - 6.
2. Expanding: (E^2 - 2E + 1) + (E - 1) - 6 = E^2 - E - 6.
3. Factorising: E^2 - E - 6 = (E - 3)(E + 2).
4. Apply to the first term: (E - 3)(a 3^x) = a 3^(x+1) - 3a 3^x = 0, so the factor (E-3) annihilates a 3^x.
5. Apply to the second term: (E + 2)(b(-2)^x) = b(-2)^(x+1) + 2b(-2)^x = -2b(-2)^x + 2b(-2)^x = 0, so (E+2) annihilates b(-2)^x.
6. Since the two factors commute, the product annihilates the whole of y, giving 0. Option (b) is correct.

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23. 2021 · Q43 Delta, E and D Operators · Theoretical

QUESTION

$\Delta^3 y_2$ is equal to

OPTIONS

(a) y_2

(b) $\Delta^2 y_2$

(c) $\nabla^3 y_5$

(d) y_5

ANSWER (AI-derived — no official key exists for this paper)

(c) $\nabla^3 y_5$

EXAM SHORTCUT (AI-derived explanation)

Forward and backward differences of the same order are related by Delta^n y_k = nabla^n y_(k+n). With n = 3 and k = 2, Delta^3 y_2 = nabla^3 y_5.

TIPS & TRICKS (AI-derived explanation)

- Memorise Delta^n y_k = nabla^n y_(k+n): the same number sits in the difference table, read forward from y_k or backward from y_(k+n).
- Both expand to y_5 - 3y_4 + 3y_3 - y_2; writing out the four terms once makes the identity obvious.

- Index check: a third difference starting at index 2 must reach index 5, so any answer must involve y_5 - which immediately narrows the options.
- Operator proof: Delta = E - 1 and nabla = 1 - E^(-1) = (E-1)E^(-1) = Delta E^(-1), so nabla^n = Delta^n E^(-n) and nabla^n y_(k+n) = Delta^n y_k.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Expand the forward difference: Delta^3 y_2 = y_5 - 3y_4 + 3y_3 - y_2.
2. Expand the backward difference: nabla^3 y_5 = y_5 - 3y_4 + 3y_3 - y_2.
3. The two expansions are identical, so Delta^3 y_2 = nabla^3 y_5.
4. In operator form, nabla = 1 - E^(-1) = (E - 1)E^(-1) = Delta E^(-1), hence nabla^3 = Delta^3 E^(-3) and nabla^3 y_5 = Delta^3 y_2.
5. Options (a), (b) and (d) equate a third difference with a function value or a lower-order difference, which cannot hold in general.
6. Hence option (c) is correct.

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24. **2021 · Q67** Delta, E and D Operators · Numerical

QUESTION

What is $\frac{\Delta^2}{E}(x^3)$ for $h=1$?

OPTIONS

(a) $6x$

(b) $3x$

(c) $2x$

(d) x

ANSWER (AI-derived — no official key exists for this paper)

(a) $6x$

EXAM SHORTCUT (AI-derived explanation)

Delta^2 x^3 = 6x + 6 with h = 1, and dividing by E means shifting back one step: 6(x-1) + 6 = 6x.

TIPS & TRICKS (AI-derived explanation)

- 1/E = E^(-1) is the BACKWARD shift: it replaces x by x - 1. Treating it as a division rather than a shift is the main trap.
- Compute in two clean stages: Delta x^3 = 3x^2 + 3x + 1, then Delta^2 x^3 = 6x + 6.
- Alternative route: Delta^2/E = Delta^2 E^(-1) = Delta . (Delta E^(-1)) = Delta . nabla, and Delta nabla x^3 = 6x follows the same way.
- Sanity check at x = 1: Delta^2 x^3 at x = 0 is 6(0) + 6 = 6, and shifting to x = 1 gives 6 - matching 6x at x = 1.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With h = 1, Delta $x^3 = (x+1)^3 - x^3 = 3x^2 + 3x + 1$.
2. Applying Delta again: Delta^2 $x^3 = 3[(x+1)^2 - x^2] + 3[(x+1) - x] = 3(2x + 1) + 3 = 6x + 6$.
3. The operator 1/E is E^(-1), the backward shift that replaces x by x - 1.
4. So (Delta^2/E)(x^3) = E^(-1)(6x + 6) = 6(x - 1) + 6 = 6x - 6 + 6 = 6x.
5. Option (a) is correct.

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25. 2021 · Q70 Delta, E and D Operators · Theoretical

QUESTION

Which holds for every $n \geq 0$?

OPTIONS

(a) $\Delta^n x^{(n)} = nh^n$

(b) $\Delta^n x^{(n)} = n(n\text{-}1)h^n$

(c) $\Delta^n x^{(n)} = n!\ h$

(d) $\Delta^n x^{(n)} = n!\ h^n$

ANSWER (AI-derived — no official key exists for this paper)

(d) $\Delta^n x^{(n)} = n!\ h^n$

EXAM SHORTCUT (AI-derived explanation)

Each application of Delta to x^(n) brings down a factor n h and lowers the order by one, so after n applications the result is n! h^n.

TIPS & TRICKS (AI-derived explanation)

- Base rule: Delta x^(n) = n h x^(n-1). Iterating n times gives n(n-1)...(1) h^n = n! h^n - the exact analogue of the nth derivative of x^n.
- Dimensional check: x^(n) is a product of n factors each carrying the dimension of x, and each difference removes one, so the answer must carry h^n. Options (a), (b) and (c) fail this.
- Test with n = 2, h = 1: x^(2) = x(x-1) = x^2 - x, whose second difference is 2 = 2! h^2. Only option (d) matches.
- This identity is the reason factorial notation exists - it makes differencing as mechanical as differentiating.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The factorial function with interval h is x^(n) = x(x-h)(x-2h)...(x-(n-1)h).
2. Its first difference is Delta x^(n) = (x+h)x(x-h)...(x-(n-2)h) - x(x-h)...(x-(n-1)h).
3. Factoring out the common product x(x-h)...(x-(n-2)h) gives [(x + h) - (x - (n-1)h)] = n h, so Delta x^(n) = n h x^(n-1).
4. Applying Delta again gives n h (n-1) h x^(n-2), and so on.

5. After n applications the factorial part is exhausted and the constant accumulated is n(n-1)(n-2)...(1) h^n = n! h^n.
6. Hence Delta^n x^(n) = n! h^n, option (d).

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26. 2022 · Q23 Sub-division of Intervals · Numerical

QUESTION

What is $(\frac{\Delta^2}{E})e^x \times \frac{Ee^x}{\Delta^2 e^x}$ equal to? (interval of differencing $= h$)

OPTIONS

(a) 0

(b) e^{-x}

(c) e^{2x}

(d) e^x

ANSWER (AI-derived — no official key exists for this paper)

(d) e^x

EXAM SHORTCUT (AI-derived explanation)

Delta^2 e^x = $e^x(e^h - 1)^2$, so the first factor is e^(x-h)$(e^h-1)^2$ and the second is e^(x+h)/[$e^x(e^h-1)^2$]. The bracketed powers cancel, leaving e^(x-h) e^(x+h)/e^x = e^x.

TIPS & TRICKS (AI-derived explanation)

- For the exponential, Delta e^x = $e^x(e^h - 1)$, so every difference just multiplies by the constant $(e^h - 1)$ - the operator acts like a scalar.
- The expression is deliberately constructed so that the $(e^h - 1)^2$ factors cancel between numerator and denominator - spot that and only the shifts remain.
- Track the shifts: E^(-1) contributes e^(-h) and E contributes e^(+h); they cancel, leaving e^(2x)/e^x = e^x.
- 1/E means the BACKWARD shift E^(-1), not a division by a number - misreading it is the main trap in operator questions of this type.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With interval of differencing h, Delta e^x = e^(x+h) - e^x = $e^x(e^h - 1)$, so Delta^2 e^x = e^x $(e^h - 1)^2$.
2. The first factor is (Delta^2/E) e^x = E^(-1)[e^x $(e^h - 1)^2$] = e^(x-h)$(e^h - 1)^2$.
3. The second factor is (E e^x)/(Delta^2 e^x) = e^(x+h)/[e^x $(e^h - 1)^2$].
4. Multiplying, the factors $(e^h - 1)^2$ cancel: e^(x-h) . e^(x+h)/e^x.
5. The shifts cancel too: e^(x-h+x+h) /e^x = e^(2x)/e^x = e^x.
6. Option (d) is correct.

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27. 2022 · Q25 Sub-division of Intervals · Numerical

QUESTION

What is $\frac{1}{E-8}(x^2 \; 2^x)$ equal to? (interval of differencing =1)

OPTIONS

(a) $-\frac{2^x}{54}(9x^2-6x-1)$

(b) $\frac{2^x}{54}(9x^2-6x-1)$

(c) $-\frac{2^x}{54}(9x^2+6x+5)$

(d) $\frac{2^x}{54}(9x^2+6x+5)$

ANSWER (AI-derived — no official key exists for this paper)

(c) $-\frac{2^x}{54}(9x^2+6x+5)$

EXAM SHORTCUT (AI-derived explanation)

On 2^x g(x), E acts as 2E on g, so (E-8) becomes 2(E-4) = 2(Delta - 3). Then 1/(Delta-3) applied to x^2 gives -(9x^2+6x+5)/27, and the extra 1/2 gives -2^x(9x^2+6x+5)/54.

TIPS & TRICKS (AI-derived explanation)

- Shift rule for inverse operators: 1/phi(E) [a^x g(x)] = a^x . 1/phi(aE) [g(x)]. Here a = 2, so phi(E) = E - 8 becomes 2E - 8 = 2(E - 4).
- Convert to Delta: E - 4 = (1 + Delta) - 4 = Delta - 3, then expand 1/(Delta - 3) = -(1/3)(1 - Delta/3)^(-1) = -(1/3)(1 + Delta/3 + Delta^2/9 + ...).
- The series terminates because Delta^3 x^2 = 0 - only three terms are ever needed for a quadratic.
- Sign check: the leading coefficient of 1/(Delta - 3) is -1/3, so the whole answer must be NEGATIVE. That eliminates options (b) and (d) at a glance.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Use the shift rule 1/phi(E)[a^x g(x)] = a^x . 1/phi(aE)[g(x)] with a = 2: (E - 8) becomes (2E - 8) = 2(E - 4).
2. So 1/(E-8)[2^x x^2] = 2^x . (1/2) . 1/(E - 4)[x^2].
3. Write E - 4 = (1 + Delta) - 4 = Delta - 3, and expand 1/(Delta - 3) = -(1/3)[1 - Delta/3]^(-1) = -(1/3)[1 + Delta/3 + Delta^2/9 + ...].
4. For x^2 with h = 1: Delta x^2 = 2x + 1 and Delta^2 x^2 = 2, with all higher differences zero.
5. So 1/(Delta-3)[x^2] = -(1/3)[x^2 + (2x+1)/3 + 2/9] = -(9x^2 + 6x + 3 + 2)/27 = -(9x^2 + 6x + 5)/27.
6. Multiplying by 2^x/2: 1/(E-8)[x^2 2^x] = -2^x(9x^2 + 6x + 5)/54, option (c).

28. 2022 · Q28 Sub-division of Intervals · Factual

QUESTION

Central finite difference approximation for the second-order derivative of $f(x)$ at x_i, equally spaced points of step length h, is:

OPTIONS

(a) $\frac{f_{i-1}+f_i+f_{i+1}}{h^2}$

(b) $\frac{f_{i+1}-f_{i-1}}{2h^2}$

(c) $\frac{f_{i-1}-2f_i+f_{i+1}}{h^2}$

(d) $\frac{f_i-2f_{i-1}+f_{i-2}}{h^2}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{f_{i-1}-2f_i+f_{i+1}}{h^2}$

EXAM SHORTCUT (AI-derived explanation)

The central second-difference formula is (f_(i-1) - 2f_i + f_(i+1))/h^2 - the coefficients 1, -2, 1 divided by h squared.

TIPS & TRICKS (AI-derived explanation)

- Coefficients of any pure difference operator must sum to ZERO so that constants are annihilated: 1 - 2 + 1 = 0. Option (a) sums to 3 and is eliminated instantly.
- The denominator must be h^2 because a second derivative has dimensions of f per length squared. Option (b) has a first-difference numerator over h^2 and is dimensionally inconsistent.
- Central formulas are symmetric about the point i, which option (d) is not - it is the BACKWARD second difference.
- This approximation is second-order accurate, with error -h^2 f''''(xi)/12 - noticeably better than the one-sided forward or backward versions.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Expand by Taylor series about x_i: f_(i+1) = f_i + h f'_i + (h^2/2) f''_i + (h^3/6) f'''_i + ... and f_(i-1) = f_i - h f'_i + (h^2/2) f''_i - (h^3/6) f'''_i + ...
2. Adding the two expansions, the odd-order terms cancel: f_(i+1) + f_(i-1) = 2 f_i + h^2 f''_i + O(h^4).
3. Rearranging: f''_i = (f_(i-1) - 2 f_i + f_(i+1))/h^2 + O(h^2).
4. The numerator is exactly the central second difference delta^2 f_i, with coefficients 1, -2, 1 summing to zero as any difference operator must.
5. The cancellation of the h^3 terms is what makes this formula second-order accurate.
6. Hence the approximation is (f_(i-1) - 2f_i + f_(i+1))/h^2, option (c).

29. 2022 · Q30 Sub-division of Intervals · Numerical

QUESTION

What is $E(\Delta^n e^{ax+b})$ equal to? (step length h=1)

OPTIONS

(a) $e^{ax+b}(e^a\text{-}1)^{n+1}$

(b) $e^{ax+b}e^a(e^a\text{-}1)^n$

(c) $e^{ax+b}(e^a\text{-}1)^n$

(d) $e^{ax+b}e^a(e^a\text{-}1)^{n+1}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $e^{ax+b}e^a(e^a\text{-}1)^n$

EXAM SHORTCUT (AI-derived explanation)

Delta^n e^(ax+b) = e^(ax+b)(e^a - 1)^n, and applying E multiplies by e^a. So the answer is e^(ax+b) e^a (e^a - 1)^n.

TIPS & TRICKS (AI-derived explanation)

- For an exponential, each Delta multiplies by the constant (e^(ah) - 1); with h = 1 that is (e^a - 1). The operator behaves like a scalar.
- E and Delta commute, so the order of application does not matter - apply Delta^n first and then the single shift E.
- E contributes exactly one extra factor e^a, which is what separates option (b) from option (c). Count the shifts carefully.
- Dimension check on the exponent count: n differences give (e^a - 1)^n and one shift gives e^a - options (a) and (d) with power n+1 over-count the differences.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With step length h = 1, Delta e^(ax+b) = e^(a(x+1)+b) - e^(ax+b) = e^(ax+b)(e^a - 1).
2. Each further application of Delta multiplies by the same constant factor (e^a - 1).
3. Hence Delta^n e^(ax+b) = e^(ax+b)(e^a - 1)^n.
4. Now apply the shift operator E, which replaces x by x + 1: E[e^(ax+b)] = e^(a(x+1)+b) = e^(ax+b) e^a.
5. Therefore E(Delta^n e^(ax+b)) = e^(ax+b) e^a (e^a - 1)^n.
6. Option (b) is correct.

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30. 2022 · Q61 Delta, E and D Operators · Theoretical

QUESTION

Consider the expressions (n, m positive integers): 1. $E^2x^2=(Ex)^2$; 2. $E^n=(1+\Delta)^n$; 3. $E^nE^mf(x)=E^{n+m}f(x)$; 4. $\Delta^n\Delta^mf(x)=\Delta^{n+m}f(x)$. How many are correct?

OPTIONS

(a) Only one expression

(b) Only two expressions

(c) Only three expressions

(d) All four expressions

ANSWER (AI-derived — no official key exists for this paper)

(c) Only three expressions

EXAM SHORTCUT (AI-derived explanation)

Expressions 2, 3 and 4 are standard operator identities. Expression 1 fails: $E^2\ x^2 = (x+2h)^2$ whereas $(Ex)^2 = (x+h)^2$. So three are correct.

TIPS & TRICKS (AI-derived explanation)

- E^n means shift n times, so $E^2\ x^2 = (x + 2h)^2$ - the exponent on E is a SHIFT COUNT, not an algebraic power that distributes over the function.
- E = 1 + Delta is the master identity; raising both sides to the nth power gives expression 2 immediately.
- Both E and Delta obey the index law because repeated application of the same operator simply composes - expressions 3 and 4 are automatic.
- Test the doubtful one numerically: with x = 1, h = 1, $E^2\ x^2 = 9$ but $(Ex)^2 = 4$. One substitution settles expression 1.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Expression 1: $E^2\ x^2$ means applying the shift twice to the function x^2, giving $(x + 2h)^2$. But $(Ex)^2$ means shifting x first and then squaring, giving $(x + h)^2$. These differ, e.g. at x = 1, h = 1 they are 9 and 4. FALSE.
2. Expression 2: since E = 1 + Delta by definition of the forward difference, raising both sides to the nth power gives $E^n = (1 + \text{Delta})^n$. TRUE.
3. Expression 3: applying the shift n times and then m more times shifts a total of n + m steps, so $E^n\ E^m$ f(x) = E^(n+m) f(x). TRUE.
4. Expression 4: likewise repeated differencing composes, so $\text{Delta}^n\ \text{Delta}^m$ f(x) = Delta^(n+m) f(x). TRUE.
5. Exactly three of the four expressions are correct.
6. Hence option (c) is correct.

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31. 2022 · Q64 Delta, E and D Operators · Theoretical

QUESTION

If Δ and ∇ are the forward and backward difference operators, what is Δ-∇ equal to?

OPTIONS

(a) $-\Delta\nabla$

(b) $\Delta\nabla$

(c) $\Delta+\nabla$

(d) $\frac{\Delta}{\nabla}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\Delta\nabla$

EXAM SHORTCUT (AI-derived explanation)

Delta - nabla = (E - 1) - (1 - E^(-1)) = E - 2 + E^(-1), and Delta nabla = (E-1)(1 - E^(-1)) = E - 2 + E^(-1). They are equal.

TIPS & TRICKS (AI-derived explanation)

- Convert everything to E: Delta = E - 1 and nabla = 1 - E^(-1). Every identity in this chapter falls out in one line from these two.
- Both sides equal delta^2 = E - 2 + E^(-1), so the identity can also be written Delta - nabla = Delta nabla = delta^2 - a three-way identity worth memorising.
- Numerical check on f(x) = x^2 with h = 1: Delta f = 2x + 1, nabla f = 2x - 1, so Delta f - nabla f = 2; and Delta nabla f = Delta(2x-1) = 2. Equal.
- Companion identity from the same pair: Delta + nabla = E - E^(-1) = 2 mu delta.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. In terms of the shift operator, Delta = E - 1 and nabla = 1 - E^(-1).
2. Therefore Delta - nabla = (E - 1) - (1 - E^(-1)) = E - 2 + E^(-1).
3. Also Delta nabla = (E - 1)(1 - E^(-1)) = E - E E^(-1) - 1 + E^(-1) = E - 1 - 1 + E^(-1) = E - 2 + E^(-1).
4. The two expressions coincide, so Delta - nabla = Delta nabla.
5. Both also equal delta^2, since delta^2 = (E^(1/2) - E^(-1/2))^2 = E - 2 + E^(-1).
6. Hence option (b) is correct.

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32. 2022 · Q65 Factorial Representation of a Polynomial · Numerical

QUESTION

Consider $f(1)=12$, $f(2)=40$, $f(3)=90$, $f(4)=168$, $f(5)=280$, $f(6)=432$. The polynomial conforming to the data is:

OPTIONS

(a) $f(x)=x^3+5x^2+5x+1$

(b) $f(x)=x^3+4x^2+6x+1$

(c) $f(x)=x^3+5x^2+4x+2$

(d) $f(x)=x^3+5x^2+6x$

ANSWER (AI-derived — no official key exists for this paper)

(d) $f(x)=x^3+5x^2+6x$

EXAM SHORTCUT (AI-derived explanation)

All four options give 12 at x = 1, so test x = 2 where the value must be 40: only $x^3 + 5x^2 + 6x$ gives 8 + 20 + 12 = 40.

TIPS & TRICKS (AI-derived explanation)

- When several options agree at the first data point, move straight to the SECOND - do not keep testing the same value.
- At x = 1 every cubic here evaluates to the sum of its coefficients, which the examiner has deliberately made equal to 12 in all four options.
- The polynomial factorises as x(x+2)(x+3), so f(1) = 1(3)(4) = 12 and f(2) = 2(4)(5) = 40 - the factored form makes verification instant.
- Cross-check with 2018 Q41, which used the same data and established that the underlying polynomial has degree 3 - consistent with all the options offered.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The data are f(1) = 12, f(2) = 40, f(3) = 90, f(4) = 168, f(5) = 280, f(6) = 432, which (as in 2018 Q41) have constant third differences and so come from a cubic.
2. At x = 1 all four options evaluate to the sum of their coefficients, which is 12 in every case, so this test does not discriminate.
3. Test x = 2, where the required value is 40. Option (a): 8 + 20 + 10 + 1 = 39. Option (b): 8 + 16 + 12 + 1 = 37. Option (c): 8 + 20 + 8 + 2 = 38.
4. Option (d): 8 + 20 + 12 = 40, which matches.
5. Confirm on further points: at x = 3, 27 + 45 + 18 = 90; at x = 4, 64 + 80 + 24 = 168; at x = 5, 125 + 125 + 30 = 280; at x = 6, 216 + 180 + 36 = 432. All correct.
6. Note f(x) = x(x+2)(x+3). Option (d) is correct.

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33. 2022 · Q66 Delta, E and D Operators · Theoretical

QUESTION

If $u=\Delta[f(x\text{-}1)\Delta\ g(x\text{-}1)]$, $v=\nabla[f(x)\Delta\ g(x)]$, $w=\Delta[f(x\text{-}1)\nabla\ g(x)]$, difference interval =1, which is correct?

OPTIONS

(a) Only $u=v$

(b) Only $v=w$

(c) Only $u=w$

(d) $u=v=w$

ANSWER (AI-derived — no official key exists for this paper)

(d) $u = v = w$

EXAM SHORTCUT (AI-derived explanation)

nabla g(x) = Delta g(x-1), so w is literally u. And both u and v expand to f(x)Delta g(x) - f(x-1)Delta g(x-1). Hence u = v = w.

TIPS & TRICKS (AI-derived explanation)

- Key bridge identity: nabla g(x) = g(x) - g(x-1) = Delta g(x-1). Substituting it turns w into u without any expansion.
- Expand u and v fully in terms of function values: both give f(x)Delta g(x) - f(x-1)Delta g(x-1). Writing out the two terms is the safest route.
- Delta applied at x-1 and nabla applied at x reach the SAME pair of points - that shared reach is why all three expressions coincide.
- Numerical check with f(x) = x and g(x) = x^2 at x = 2: all three expressions evaluate to 3 x 5 - 1 x 3 = ... compute once and the equality is confirmed.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With difference interval 1, nabla g(x) = g(x) - g(x-1) = Delta g(x-1).
2. Substituting this into w = Delta[f(x-1) nabla g(x)] gives w = Delta[f(x-1) Delta g(x-1)] = u. So u = w immediately.
3. Now expand u. Writing G(x) = f(x-1) Delta g(x-1), we have u = Delta G(x) = G(x+1) - G(x) = f(x) Delta g(x) - f(x-1) Delta g(x-1).
4. Expand v. Writing F(x) = f(x) Delta g(x), we have v = nabla F(x) = F(x) - F(x-1) = f(x) Delta g(x) - f(x-1) Delta g(x-1).
5. The two expansions are identical, so u = v.
6. Therefore u = v = w, option (d).

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34. **2022 · Q67** Delta, E and D Operators · Theoretical

QUESTION

Consider: 1. $\mu = \sqrt{1+\frac{\delta^2}{4}}$; 2. $\Delta = \frac{\delta^2}{2} + \delta\sqrt{1+\frac{\delta^2}{4}}$; 3. $\nabla = -\frac{\delta^2}{2} + \sqrt{1+\frac{\delta^2}{4}}$. How many expressions are correct?

OPTIONS

(a) Only one expression

(b) Only two expressions

(c) All three expressions

(d) No expression

ANSWER (AI-derived — no official key exists for this paper)

(b) Only two expressions

EXAM SHORTCUT (AI-derived explanation)

Expressions 1 and 2 are standard. Expression 3 drops the factor delta: the true identity is nabla = -delta^2/2 + delta sqrt(1 + delta^2/4), so as printed it is false.

TIPS & TRICKS (AI-derived explanation)

- Master identities: mu = sqrt(1 + delta^2/4), Delta = delta^2/2 + delta mu, nabla = -delta^2/2 + delta mu. All three carry mu MULTIPLIED by delta in the second term.
- Compare expressions 2 and 3 side by side: expression 2 has delta before the square root, expression 3 does not. That single missing factor is the whole question.
- Verify with E-forms: delta mu = (E - E^(-1))/2 and delta^2/2 = (E - 2 + E^(-1))/2. Adding gives E - 1 = Delta; subtracting gives 1 - E^(-1) = nabla.
- Dimensional check: Delta and nabla are first-order difference operators, so each term must contain an ODD total power of delta. In expression 3 the term mu alone is even-order, which cannot be right.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write the operators in terms of E: delta = E^(1/2) - E^(-1/2), mu = (E^(1/2) + E^(-1/2))/2, so delta^2 = E - 2 + E^(-1) and delta mu = (E - E^(-1))/2.
2. Expression 1: mu^2 = (E + 2 + E^(-1))/4 = 1 + (E - 2 + E^(-1))/4 = 1 + delta^2/4, so mu = sqrt(1 + delta^2/4). TRUE.
3. Expression 2: delta^2/2 + delta mu = (E - 2 + E^(-1))/2 + (E - E^(-1))/2 = (2E - 2)/2 = E - 1 = Delta. TRUE.
4. Expression 3 as printed is -delta^2/2 + sqrt(1 + delta^2/4) = -delta^2/2 + mu, which is not nabla.
5. The correct identity is nabla = -delta^2/2 + delta mu = -(E - 2 + E^(-1))/2 + (E - E^(-1))/2 = 1 - E^(-1); the printed version omits the factor delta. FALSE.
6. Exactly two expressions are correct, option (b).

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35. 2022 · Q69 Sub-division of Intervals · Numerical

QUESTION

What is $\Delta^2\left(\frac{a^{4x}}{(a^2-1)^2}\right)$ equal to? (interval of differencing =1)

OPTIONS

(a) $(a^2+1)^4 a^{4x}$

(b) $\frac{a^{4x}}{(a^2+1)^2}$

(c) $(a^2+1)^2 a^{4x}$

(d) $\frac{a^{4x}}{(a^2+1)^4}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $(a^2+1)^2 a^{4x}$

EXAM SHORTCUT (AI-derived explanation)

Delta^2 a^(4x) = a^(4x)(a^4 - 1)^2, and (a^4 - 1) = (a^2-1)(a^2+1), so dividing by (a^2-1)^2 leaves (a^2+1)^2 a^(4x).

TIPS & TRICKS (AI-derived explanation)

- For an exponential base b = a^4, each Delta multiplies by (b - 1) = (a^4 - 1). Two differences give the square.
- Factorise the difference of squares: a^4 - 1 = (a^2 - 1)(a^2 + 1). The (a^2-1)^2 in the denominator is placed precisely to cancel.
- The exponential factor a^(4x) never changes - only the constant multiplier does, which is why every option keeps a^(4x).
- Sanity check with a = 2: a^4 - 1 = 15 and a^2 - 1 = 3, so the constant is (15/3)^2 = 25 = (a^2+1)^2 = 5^2. Confirmed.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With interval of differencing 1, Delta a^(4x) = a^(4(x+1)) - a^(4x) = a^(4x)(a^4 - 1).
2. Applying Delta a second time multiplies by the same constant: Delta^2 a^(4x) = a^(4x)(a^4 - 1)^2.
3. The constant (a^2 - 1)^(-2) passes through the operator unchanged, so Delta^2[a^(4x)/(a^2-1)^2] = a^(4x)(a^4-1)^2/(a^2-1)^2.
4. Factorise a^4 - 1 = (a^2 - 1)(a^2 + 1), so (a^4 - 1)^2 = (a^2-1)^2 (a^2+1)^2.
5. The (a^2-1)^2 factors cancel, leaving (a^2 + 1)^2 a^(4x).
6. Option (c) is correct.

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36. 2022 · Q70 Sub-division of Intervals · Numerical

QUESTION

What is $\Delta^2 E[(3x\text{-}3)^{(-2)}]$ at $x=1$ equal to? (interval of differencing =1)

OPTIONS

(a) $-\frac{1}{30}$

(b) $\frac{1}{30}$

(c) $-\frac{1}{270}$

(d) $\frac{1}{270}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $-\frac{1}{270}$

EXAM SHORTCUT (AI-derived explanation)

Delta(3x-3)^(-2) = (-2)(3)(3x-3)^(-3) = -6(3x-3)^(-3). Shifting by E^2 gives -6/[(3x+6)(3x+9)(3x+12)], which at x = 1 is -6/(9 x 12 x 15) = -1/270.

TIPS & TRICKS (AI-derived explanation)

- Negative factorial with a linear argument and coefficient a: u^(-n) = 1/[(u+a)(u+2a)...(u+na)]. Here a = 3, so (3x-3)^(-2) = 1/[(3x)(3x+3)].
- The difference rule Delta u^(n) = n a u^(n-1) holds for negative n too, with n = -2 giving the factor (-2)(3) = -6.
- E and Delta commute, so apply the difference rule first and the double shift afterwards - far cleaner than expanding four separate fractions.
- Sign check: a negative factorial power decreases in magnitude as x grows, so its forward difference must be NEGATIVE - options (b) and (d) are eliminated at once.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With interval 1 and a linear argument of coefficient a = 3, the generalised factorial rule is Delta u^(n) = n a u^(n-1), valid for negative n as well.
2. So Delta[(3x-3)^(-2)] = (-2)(3)(3x-3)^(-3) = -6(3x-3)^(-3).
3. Since E and Delta commute, apply E^2, which replaces x by x + 2: E^2[-6(3x-3)^(-3)] = -6(3x+3)^(-3).
4. By definition of the negative factorial, (3x+3)^(-3) = 1/[(3x+3+3)(3x+3+6)(3x+3+9)] = 1/[(3x+6)(3x+9)(3x+12)].
5. At x = 1 the three factors are 9, 12 and 15, so the value is -6/(9 x 12 x 15) = -6/1620 = -1/270, option (c).

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37. 2023 · Q42 Delta, E and D Operators · Numerical

QUESTION

What is $\Delta^2_{y,z}\ x^3$ equal to?

OPTIONS

(a) $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$

(b) $x+y+z$

(c) $x^2+y^2+z^2$

(d) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $x+y+z$

EXAM SHORTCUT (AI-derived explanation)

The second divided difference of x^3 over nodes x, y, z is the complete homogeneous symmetric polynomial of degree 1, namely x + y + z.

TIPS & TRICKS (AI-derived explanation)

- General rule: the kth divided difference of t^n over nodes x0,...,xk is the complete homogeneous symmetric polynomial h_(n-k) in those nodes.
- Here n = 3 and k = 2, so the result is h_1 = x + y + z - degree 1 in the nodes, which immediately rules out options (a) and (d).
- Verify directly: f[x,y] = $(x^3 - y^3)/(x - y) = x^2 + xy + y^2$, and then f[x,y,z] = (f[x,y] - f[y,z])/(x - z) simplifies to x + y + z.
- Symmetry check: a divided difference must be symmetric in its arguments, and x + y + z plainly is.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The notation denotes the second divided difference of $f(t) = t^3$ over the three arguments x, y and z.
2. First divided difference: f[x,y] = $(x^3 - y^3)/(x - y) = x^2 + xy + y^2$, and similarly f[y,z] = $y^2 + yz + z^2$.
3. Second divided difference: f[x,y,z] = (f[x,y] - f[y,z])/(x - z) = $(x^2 + xy - yz - z^2)/(x - z)$.
4. Factor the numerator: $x^2 - z^2 + y(x - z) = (x - z)(x + z) + y(x - z) = (x - z)(x + y + z)$.
5. Cancelling (x - z) gives f[x,y,z] = x + y + z.
6. This agrees with the general rule that the kth divided difference of t^n is the complete homogeneous symmetric polynomial of degree n - k. Option (b) is correct.

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38. 2023 · Q43 Factorial Representation of a Polynomial · Numerical

QUESTION

Let $U(x)$ be a polynomial of degree 3. Given that $U(0)+U(4)=72$ and $U(1)+U(3)=42$. What is the value of $U(2)$ assuming that differencing interval $h=1$?

OPTIONS

(a) 1

(b) 2

(c) 4

(d) 16

ANSWER (AI-derived — no official key exists for this paper)

(d) 16

EXAM SHORTCUT (AI-derived explanation)

A cubic has zero fourth difference: U(4) - 4U(3) + 6U(2) - 4U(1) + U(0) = 0. Grouping the given sums, 72 - 4(42) + 6U(2) = 0, so U(2) = 16.

TIPS & TRICKS (AI-derived explanation)

- The fourth-difference coefficients 1, -4, 6, -4, 1 pair the OUTER terms and the INNER terms symmetrically - exactly matching the two sums supplied.
- Group before substituting: [U(0) + U(4)] - 4[U(1) + U(3)] + 6U(2) = 0. The question is designed so that the individual values are never needed.

- Arithmetic: 72 - 168 = -96, so 6U(2) = 96 and U(2) = 16.
- Note the interval h = 1 is stated so that the plain fourth-difference relation applies with unit spacing.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. U(x) is a polynomial of degree 3, so its fourth forward difference vanishes identically.
2. With h = 1, Delta^4 U(0) = U(4) - 4U(3) + 6U(2) - 4U(1) + U(0) = 0.
3. Group the terms to match the given data: [U(0) + U(4)] - 4[U(1) + U(3)] + 6U(2) = 0.
4. Substituting U(0) + U(4) = 72 and U(1) + U(3) = 42: 72 - 4(42) + 6U(2) = 0.
5. So 72 - 168 + 6U(2) = 0, giving 6U(2) = 96.
6. Hence U(2) = 16, option (d).

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39. 2023 · Q44 Delta, E and D Operators · Numerical

QUESTION

If differencing interval is of length h, then what is $(x^2+hx+1)\tan(\Delta \tan^{-1} x)$ equal to?

OPTIONS

(a) 1

(b) h

(c) $h+1$

(d) h^2+1

ANSWER (AI-derived — no official key exists for this paper)

(b) h

EXAM SHORTCUT (AI-derived explanation)

Delta arctan x = arctan(x+h) - arctan x, whose tangent is h/(1 + x^2 + hx) by the subtraction formula. Multiplying by (x^2 + hx + 1) leaves h.

TIPS & TRICKS (AI-derived explanation)

- Tangent subtraction formula: tan(A - B) = (tan A - tan B)/(1 + tan A tan B). With A = arctan(x+h) and B = arctan x this is immediate.
- The numerator collapses to (x + h) - x = h and the denominator to 1 + x(x+h) = x^2 + hx + 1 - precisely the factor placed in front.
- The whole question is engineered so that the awkward denominator cancels, leaving the bare step length h. Spotting the design is the shortcut.
- Related identity worth carrying: Delta arctan x = arctan[h/(1 + x^2 + hx)], the discrete analogue of the derivative 1/(1+x^2).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With interval of differencing h, Delta arctan x = arctan(x + h) - arctan(x).

2. Apply the tangent subtraction formula tan(A - B) = (tan A - tan B)/(1 + tan A tan B) with tan A = x + h and tan B = x.
3. tan(Delta arctan x) = [(x + h) - x]/[1 + x(x + h)] = h/(1 + x^2 + hx).
4. Multiplying by the given factor: (x^2 + hx + 1) . h/(1 + x^2 + hx).
5. The bracket cancels exactly, leaving h.
6. Option (b) is correct.

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40. 2023 · Q45 Factorial Representation of a Polynomial · Numerical

QUESTION

If $x^{(r)} = x(x\text{-}1)(x\text{-}2)\cdots(x\text{-}(r\text{-}1))$, then what is the coefficient of x^4 in $2x^{(5)}\text{-}5x^{(4)}+3x^{(3)}+2x^{(2)}+1$?

OPTIONS

(a) 2

(b) -1

(c) -10

(d) -25

ANSWER (AI-derived — no official key exists for this paper)

(d) -25

EXAM SHORTCUT (AI-derived explanation)

Only x^(5) and x^(4) reach degree 4. Since x^(5) = x^5 - 10x^4 + ... and x^(4) = x^4 - ..., the coefficient is 2(-10) + (-5)(1) = -25.

TIPS & TRICKS (AI-derived explanation)

- Terms of degree below 4 cannot contribute, so only the first two summands matter - discard x^(3), x^(2) and the constant immediately.
- Expansion coefficients are Stirling numbers of the first kind: x^(5) = x^5 - 10x^4 + 35x^3 - 50x^2 + 24x, so the x^4 coefficient is -10.
- The sub-leading coefficient of x^(r) is always -(1 + 2 + ... + (r-1)) = -r(r-1)/2. For r = 5 that is -10 and for r = 4 it is -6 - a quick way to generate them.
- x^(4) contributes its LEADING coefficient 1 to x^4, multiplied by -5, giving -5. Adding 2(-10) = -20 gives -25.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The factorial polynomial x^(r) = x(x-1)...(x-r+1) has degree r, so only x^(5) and x^(4) can contribute to the coefficient of x^4.
2. Expanding, x^(5) = x(x-1)(x-2)(x-3)(x-4) = x^5 - 10x^4 + 35x^3 - 50x^2 + 24x, so its x^4 coefficient is -10.
3. (The sub-leading coefficient of x^(r) is -r(r-1)/2, which for r = 5 gives -10.)
4. x^(4) = x(x-1)(x-2)(x-3) = x^4 - 6x^3 + 11x^2 - 6x, so its x^4 coefficient is 1.

5. The terms 3x^(3), 2x^(2) and the constant have degree at most 3 and contribute nothing.
6. Coefficient of x^4 = 2(-10) + (-5)(1) = -20 - 5 = -25, option (d).

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41. 2023 · Q49 Sub-division of Intervals · Numerical

QUESTION

What is the value of the expression $\Delta^6((1-x)(1-3x^2)(1-4x^3))$, when interval of differencing is 1?

OPTIONS

(a) 8640

(b) 2160

(c) -2160

(d) -8640

ANSWER (AI-derived — no official key exists for this paper)

(d) -8640

EXAM SHORTCUT (AI-derived explanation)

The product has degree 1 + 2 + 3 = 6 with leading coefficient (-1)(-3)(-4) = -12. So Delta^6 = -12 x 6! = -12 x 720 = -8640.

TIPS & TRICKS (AI-derived explanation)

- Identity: for h = 1, Delta^n applied to a degree-n polynomial gives (leading coefficient) x n!, and everything of lower degree is annihilated.
- Add the degrees of the factors (1 + 2 + 3 = 6) rather than multiplying anything out.
- Leading coefficient is the product of the leading coefficients: (-1)(-3)(-4). THREE negatives give a negative, which is what makes the answer negative - options (a) and (b) are eliminated by the sign alone.
- 6! = 720, so the magnitude is 12 x 720 = 8640. Option (c) -2160 corresponds to using 3! or a leading coefficient of 3.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The factors have degrees 1, 2 and 3, so the product is a polynomial of degree 1 + 2 + 3 = 6.
2. Its leading term comes from multiplying the highest-degree term of each factor: $(-x)(-3x^2)(-4x^3)$ = $-12\ x^6$.
3. So the leading coefficient is -12.
4. For an interval of differencing 1, Delta^n annihilates every term of degree below n and maps the leading term $a\_n\ x^n$ to a_n n!.
5. Hence Delta^6 of the product = -12 x 6! = -12 x 720 = -8640, option (d).

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42. 2023 · Q50

Delta, E and D Operators · Numerical

QUESTION

If $\delta = \sqrt{E} - \frac{1}{\sqrt{E}}$ and $\mu = \frac{1}{2}\left(\sqrt{E} + \frac{1}{\sqrt{E}}\right)$, then what is $2\sqrt{1+\delta^2\mu^2}\ y_x$ equal to?

OPTIONS

(a) $y_{x+2} - 2y_{x+1} + y_x$

(b) $y_{x+1} - y_{x-1}$

(c) $y_{x+1} + y_{x-1}$

(d) $y_{x+2} - y_x$

ANSWER **(AI-derived — no official key exists for this paper)**

(c) $y_{x+1} + y_{x-1}$

EXAM SHORTCUT **(AI-derived explanation)**

delta mu = (E - E^(-1))/2, so 1 + delta^2 mu^2 = (E + E^(-1))^2/4. Taking the square root and doubling gives E + E^(-1), i.e. y_(x+1) + y_(x-1).

TIPS & TRICKS **(AI-derived explanation)**

- Compute delta mu FIRST as a difference of squares: (E^(1/2) - E^(-1/2))(E^(1/2) + E^(-1/2))/2 = (E - E^(-1))/2.
- Then 1 + (E - E^(-1))^2/4 = (4 + E^2 - 2 + E^(-2))/4 = (E + E^(-1))^2/4 - the perfect square is the whole trick.
- The factor 2 outside cancels the 1/2 from the square root, leaving the clean operator E + E^(-1).
- Sanity test on y_x = 1: the operator gives 1 + 1 = 2, and the expression 2 sqrt(1 + 0) = 2. Consistent.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. With delta = E^(1/2) - E^(-1/2) and mu = (E^(1/2) + E^(-1/2))/2, the product is delta mu = (E - E^(-1))/2.
2. Squaring: delta^2 mu^2 = (E - E^(-1))^2/4 = (E^2 - 2 + E^(-2))/4.
3. Therefore 1 + delta^2 mu^2 = (4 + E^2 - 2 + E^(-2))/4 = (E^2 + 2 + E^(-2))/4 = [(E + E^(-1))/2]^2.
4. Taking the positive square root: sqrt(1 + delta^2 mu^2) = (E + E^(-1))/2.
5. Multiplying by 2 gives the operator E + E^(-1).
6. Applying it to y_x gives y_(x+1) + y_(x-1), option (c).

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43. 2023 · Q55

Sub-division of Intervals · Theoretical

QUESTION

Consider the following for equi-spaced arguments with h as the interval of differencing: 1. $x^{(-r)} = \frac{2}{(x+rh)^{(r)}}$ 2. $\Delta^m x^{(-r)} = r(r+1)(r+2)\cdots(r+m)h^m x^{(-r-m)}$ for $m \leq r$ Which of the statements given above is/are correct?

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(d) Neither 1 nor 2

EXAM SHORTCUT (AI-derived explanation)

Statement 1 carries a spurious factor 2 (the correct identity is x^(-r) = 1/(x+rh)^(r)), and statement 2 omits the sign $(-1)^m$ and mis-states the last factor. Both fail.

TIPS & TRICKS (AI-derived explanation)

- Definition: x^(-r) = 1/[(x+h)(x+2h)...(x+rh)], and (x+rh)^(r) is exactly that product, so the correct identity has NO factor 2.
- Difference rule for negative factorials: Delta x^(-r) = -r h x^(-r-1). Iterating m times gives $(-1)^m$ r(r+1)...(r+m-1) h^m x^(-r-m).
- Two defects in statement 2: the missing $(-1)^m$ and the product running to (r+m) instead of (r+m-1). Either alone makes it false.
- Sanity check with r = 1, m = 1: Delta x^(-1) = -h x^(-2), whereas statement 2 would give 1(2) h x^(-2) = 2h x^(-2) - wrong in both sign and magnitude.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The negative factorial function is defined by x^(-r) = 1/[(x+h)(x+2h)...(x+rh)].
2. Now (x + rh)^(r) = (x+rh)(x+rh-h)...(x+rh-(r-1)h) = (x+rh)(x+(r-1)h)...(x+h), which is exactly the same product.
3. Hence the correct identity is x^(-r) = 1/(x+rh)^(r); statement 1's factor of 2 is spurious. FALSE.
4. For the difference, Delta x^(-r) = 1/[(x+2h)...(x+(r+1)h)] - 1/[(x+h)...(x+rh)], and combining over the common denominator gives -r h x^(-r-1).
5. Iterating m times yields Delta^m x^(-r) = $(-1)^m$ r(r+1)...(r+m-1) h^m x^(-r-m). Statement 2 omits the factor $(-1)^m$ and runs the product to (r+m) rather than (r+m-1). FALSE.
6. Neither statement is correct, option (d).

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44. 2023 · Q56 Delta, E and D Operators · Theoretical

QUESTION

Consider the following: 1. $\Delta(u_x v_x) = u_x \Delta\ v_x + v_{x+1} \Delta\ u_x$ 2. $\Delta(u_x v_x) = v_x \Delta\ u_x + u_{x+1} \Delta\ v_x$ Which of the above equalities is/are valid?

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER **(AI-derived — no official key exists for this paper)**

(c) Both 1 and 2

EXAM SHORTCUT **(AI-derived explanation)**

Both expand to u_(x+1)v_(x+1) - u_x v_x, so both product rules are valid - they differ only in which factor is shifted.

TIPS & TRICKS **(AI-derived explanation)**

- The discrete product rule comes in two equivalent forms because the 'cross term' can be split either way - both are standard and both are correct.
- Expand each side explicitly: u_x(v_(x+1) - v_x) + v_(x+1)(u_(x+1) - u_x) telescopes to u_(x+1)v_(x+1) - u_x v_x.
- The asymmetry is essential: in each form exactly ONE factor carries the shift x+1. A version with both unshifted would be wrong.
- Numerical check with u_x = x, v_x = x^2, x = 1: Delta(uv) = 8 - 1 = 7; form 1 gives 1(3) + 4(1) = 7; form 2 gives 1(1) + 2(3) = 7. Both agree.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. By definition Delta(u_x v_x) = u_(x+1) v_(x+1) - u_x v_x.
2. Expand form 1: u_x Delta v_x + v_(x+1) Delta u_x = u_x(v_(x+1) - v_x) + v_(x+1)(u_(x+1) - u_x) = u_x v_(x+1) - u_x v_x + u_(x+1) v_(x+1) - u_x v_(x+1) = u_(x+1)v_(x+1) - u_x v_x. VALID.
3. Expand form 2: v_x Delta u_x + u_(x+1) Delta v_x = v_x(u_(x+1) - u_x) + u_(x+1)(v_(x+1) - v_x) = u_(x+1) v_x - u_x v_x + u_(x+1)v_(x+1) - u_(x+1)v_x = u_(x+1)v_(x+1) - u_x v_x. VALID.
4. Both equalities hold, option (c).

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45. 2023 · Q59 Summation of Series · Numerical

COMMON DATA FOR THIS GROUP

Questions 59–60 are based on the series 12, 40, 90, 168, 280, 432, ...; let $f(x)$ be the polynomial of least degree related to this series.

QUESTION

What is the degree of $f(x)$?

OPTIONS

(a) 2

(b) 3

(c) 4

(d) 5

ANSWER **(AI-derived — no official key exists for this paper)**

(b) 3

EXAM SHORTCUT **(AI-derived explanation)**

Differences 28, 50, 78, 112, 152; second 22, 28, 34, 40; third 6, 6, 6 - constant. The first constant row is the third, so the degree is 3.

TIPS & TRICKS **(AI-derived explanation)**

- The order of the first CONSTANT difference row is the degree of the polynomial. Stop differencing the moment a row repeats.
- You need only three equal entries in a row to be confident; here 6, 6, 6 settles it and the fourth differences are 0.
- Same data as 2018 Q41 and 2022 Q65 - this series has now appeared in three papers, so the difference table is worth knowing by heart.
- The underlying polynomial is x(x+2)(x+3) = $x^3 + 5x^2 + 6x$, whose degree 3 matches.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. The series is 12, 40, 90, 168, 280, 432.
2. First differences: 28, 50, 78, 112, 152.
3. Second differences: 22, 28, 34, 40.
4. Third differences: 6, 6, 6 - constant, and hence the fourth differences are all zero.
5. The first constant row of differences is the third, so the polynomial of least degree fitting the data has degree 3.
6. Option (b) is correct.

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46. 2023 · Q60 Summation of Series · Numerical

COMMON DATA FOR THIS GROUP

Questions 59–60 are based on the series 12, 40, 90, 168, 280, 432, ...; let $f(x)$ be the polynomial of least degree related to this series.

QUESTION

What is the constant term in $f(x)$?

OPTIONS

(a) 0

(b) 1

(c) 8

(d) 12

ANSWER **(AI-derived — no official key exists for this paper)**

(a) 0

EXAM SHORTCUT **(AI-derived explanation)**

The fitted cubic is f(x) = $x^3 + 5x^2 + 6x$ = x(x+2)(x+3), whose constant term is 0.

TIPS & TRICKS **(AI-derived explanation)**

- The constant term is f(0), so extend the difference table one step BACKWARDS from f(1) = 12 rather than solving for all four coefficients.
- Working backwards: the second difference before 22 is 22 - 6 = 16, the first difference before 28 is 28 - 16 = 12, so f(0) = 12 - 12 = 0.
- The polynomial factorises as x(x+2)(x+3), which makes the zero constant term obvious and reproduces 1(3)(4) = 12 at x = 1.
- Cross-check with 2022 Q65, where the same data identified f(x) = $x^3 + 5x^2 + 6x$ among four options - consistent with a constant term of 0.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. From the previous part the data 12, 40, 90, 168, 280, 432 at x = 1, 2, 3, ... come from a cubic, with third difference 6 and hence leading coefficient 6/3! = 1.
2. The constant term of f is f(0), which can be found by extending the difference table one step to the left.
3. The second differences are 22, 28, 34, 40 with common difference 6, so the entry before 22 is 16.
4. The first differences are 28, 50, 78, ... so the entry before 28 is 28 - 16 = 12.
5. Therefore f(0) = f(1) - 12 = 12 - 12 = 0, and indeed f(x) = $x^3 + 5x^2 + 6x$ = x(x+2)(x+3) reproduces every tabulated value.
6. The constant term is 0, option (a).

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47. 2024 · Q66 Delta, E and D Operators · Numerical

QUESTION

What is the value of the following?

$$4y_0+6\Delta\, y_0+4\Delta^2\, y_0+\Delta^3\, y_0$$

OPTIONS

(a) $y_0-y_1+y_2-y_3$

(b) $y_0+y_1+y_2+y_3$

(c) $y_0+y_1-y_2-y_3$

(d) $y_0-y_1-y_2+y_3$

ANSWER **(AI-derived — no official key exists for this paper)**

(b) $y_0+y_1+y_2+y_3$

EXAM SHORTCUT (AI-derived explanation)

Since E = 1 + Delta, expanding 1 + E + E^2 + E^3 gives 4 + 6 Delta + 4 Delta^2 + Delta^3. So the expression equals y0 + y1 + y2 + y3.

TIPS & TRICKS (AI-derived explanation)

- Recognise the coefficients 4, 6, 4, 1 - they are the sums of the binomial columns from $(1+\text{Delta})^0$ through $(1+\text{Delta})^3$, which is the signature of y0 + y1 + y2 + y3.
- Work backwards from the operator: 1 + E + E^2 + E^3 applied to y0 gives the four consecutive values, and substituting E = 1 + Delta reproduces the printed expression.
- Verify the constant: the four expansions contribute 1 + 1 + 1 + 1 = 4, matching the leading coefficient exactly.
- The alternating-sign options (a), (c) and (d) would require alternating operator signs, which the all-positive coefficients rule out.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Since E = 1 + Delta, we have E^2 = 1 + 2Delta + Delta^2 and E^3 = 1 + 3Delta + 3Delta^2 + Delta^3.
2. Adding: 1 + E + E^2 + E^3 = (1 + 1 + 1 + 1) + (1 + 2 + 3)Delta + (1 + 3)Delta^2 + Delta^3 = 4 + 6 Delta + 4 Delta^2 + Delta^3, which is exactly the given operator.
3. Applying it to y0: (1 + E + E^2 + E^3) y0 = y0 + y1 + y2 + y3.
4. So the expression equals the sum of the four consecutive ordinates.
5. Option (b) is correct.

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48. 2024 · Q67 Delta, E and D Operators · Theoretical

QUESTION

What is the value of $\nabla^5 y_n$?

OPTIONS

(a) $\nabla^5 y_{n-5}$

(b) $\nabla^4 \Delta y_{n-5}$

(c) $\Delta^5 y_n$

(d) $\Delta^5 y_{n-5}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $\Delta^5 y_{n-5}$

EXAM SHORTCUT (AI-derived explanation)

The bridge identity is nabla^n y_k = Delta^n y_(k-n). With n = 5 this gives nabla^5 y_n = Delta^5 y_(n-5).

TIPS & TRICKS (AI-derived explanation)

- Memorise nabla^n y_k = Delta^n y_(k-n): the same table entry read backwards from y_k or forwards from y_(k-n).
- Operator proof: nabla = 1 - E^(-1) = (E - 1)E^(-1) = Delta E^(-1), so nabla^n = Delta^n E^(-n), which shifts the index down by n.
- Index check: a fifth backward difference at n reaches back to n - 5, so the forward form must START at n - 5. Option (c) keeps the index at n and is wrong.
- Option (a) restates nabla in terms of itself with a shifted index and is simply false; the identity must convert between the two OPERATORS.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The backward difference operator is nabla = 1 - E^(-1), where E is the shift operator.
2. Factorising, nabla = (E - 1)E^(-1) = Delta E^(-1).
3. Since Delta and E commute, nabla^n = Delta^n E^(-n).
4. Applying this to y_n: nabla^n y_n = Delta^n E^(-n) y_n = Delta^n y_(n-n) = Delta^n y_(n-n)... more precisely, E^(-n) y_n = y_(n-n) is the value at index n - n; in general nabla^n y_k = Delta^n y_(k-n).
5. With n = 5 and index n: nabla^5 y_n = Delta^5 y_(n-5). Both expand to y_n - 5y_(n-1) + 10y_(n-2) - 10y_(n-3) + 5y_(n-4) - y_(n-5).
6. Option (d) is correct.

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49. 2024 · Q71 Sub-division of Intervals · Numerical

QUESTION

If $\Delta^2(ab^x) = \alpha\ b^x$, the interval of differencing being 1, then what is the value of α?

OPTIONS

(a) $a(b\text{-}1)^2$

(b) a

(c) $(b\text{-}1)^2$

(d) $a(b\text{-}1)$

ANSWER (AI-derived — no official key exists for this paper)

(a) $a(b\text{-}1)^2$

EXAM SHORTCUT (AI-derived explanation)

Delta(a b^x) = a b^x (b - 1), so applying Delta twice gives a b^x (b-1)^2. Hence alpha = a(b-1)^2.

TIPS & TRICKS (AI-derived explanation)

- For an exponential, each Delta multiplies by the constant (b - 1): Delta b^x = b^(x+1) - b^x = b^x(b-1).
- The multiplicative constant a passes straight through the operator, so it must appear in alpha - eliminating option (c).

- Two differences give the SQUARE of (b-1); one difference would give option (d), the 'one step short' trap.
- Check with a = 1, b = 2: Delta^2 2^x = 2^x, and a$(b-1)^2$ = 1. Correct.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With interval of differencing 1, Delta(a b^x) = a b^(x+1) - a b^x = a b^x (b - 1).
2. Applying Delta a second time multiplies by the same constant factor (b - 1).
3. So Delta^2(a b^x) = a b^x $(b - 1)^2$.
4. Comparing with the given form alpha b^x, we read alpha = a$(b - 1)^2$.
5. Check with a = 1 and b = 2: Delta^2 2^x = 2^x and a$(b-1)^2$ = 1, consistent.
6. Option (a) is correct.

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50. 2024 · Q72 Sub-division of Intervals · Numerical

QUESTION

If $\left(\frac{\Delta^2}{E}\right)x^3 = \alpha\ x + \beta$, the interval of differencing being 1, then what are the values of α and β respectively?

OPTIONS

(a) 6, 6

(b) 6, 0

(c) 0, 6

(d) 3, 3

ANSWER (AI-derived — no official key exists for this paper)

(b) 6, 0

EXAM SHORTCUT (AI-derived explanation)

Delta^2 x^3 = 6x + 6, and dividing by E shifts x to x - 1, giving 6(x-1) + 6 = 6x. So alpha = 6 and beta = 0.

TIPS & TRICKS (AI-derived explanation)

- 1/E is the BACKWARD shift E^(-1), replacing x by x - 1 - not an algebraic division. Misreading it is the main trap.
- Compute in two stages: Delta x^3 = $3x^2$ + 3x + 1, then Delta^2 x^3 = 6x + 6.
- The shift cancels the constant exactly: 6(x-1) + 6 = 6x - 6 + 6 = 6x, so beta = 0. Skipping the shift would give alpha = 6, beta = 6 - option (a).
- This repeats 2021 Q67, where the same expression was asked for directly - the operator identity is clearly a favourite.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With h = 1, Delta x^3 = $(x+1)^3$ - x^3 = $3x^2$ + 3x + 1.

2. Applying Delta again: Delta^2 x^3 = 3[(x+1)^2 - x^2] + 3[(x+1) - x] = 3(2x+1) + 3 = 6x + 6.
3. The operator 1/E is the backward shift E^(-1), which replaces x by x - 1.
4. So (Delta^2/E) x^3 = 6(x - 1) + 6 = 6x - 6 + 6 = 6x.
5. Comparing with alpha x + beta gives alpha = 6 and beta = 0.
6. Option (b) is correct.

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51. 2024 · Q73 Sub-division of Intervals · Numerical

QUESTION

If $\delta = \alpha \nabla E^n$, the interval of differencing being 1, then what are the values of α and n respectively?

OPTIONS

(a) 1, -1/2

(b) 1, 1/2

(c) 1/2, -1/2

(d) -1/2, 1/2

ANSWER (AI-derived — no official key exists for this paper)

(b) 1, 1/2

EXAM SHORTCUT (AI-derived explanation)

nabla E^n = E^n - E^(n-1), and delta = E^(1/2) - E^(-1/2). Matching the exponents gives n = 1/2 and alpha = 1.

TIPS & TRICKS (AI-derived explanation)

- Write both sides in powers of E: the right side is alpha(E^n - E^(n-1)) and the left is E^(1/2) - E^(-1/2).
- Match the leading exponent first: n = 1/2, and then n - 1 = -1/2 automatically - a built-in consistency check.
- With the exponents matched, the coefficients give alpha = 1; no scaling is needed.
- Companion identity: delta = Delta E^(-1/2) as well, since Delta = E - 1 gives E^(-1/2)(E-1) = E^(1/2) - E^(-1/2).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The central difference operator is delta = E^(1/2) - E^(-1/2), and the backward difference is nabla = 1 - E^(-1).
2. Therefore nabla E^n = (1 - E^(-1))E^n = E^n - E^(n-1).
3. Setting alpha(E^n - E^(n-1)) = E^(1/2) - E^(-1/2) and comparing the leading terms gives alpha = 1 and n = 1/2.
4. The second terms then match automatically: n - 1 = 1/2 - 1 = -1/2.
5. So delta = nabla E^(1/2), i.e. alpha = 1 and n = 1/2.

6. Option (b) is correct.

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52. 2025 · Q21 Sub-division of Intervals · Numerical

QUESTION

What is $\frac{1}{1-E^2}(ab^x)$ equal to, where $0<b<1$? (interval of differencing = unity)

OPTIONS

(a) $\frac{ab^x}{1-b^2}$

(b) $\frac{ab^x}{1-a^2}$

(c) $\frac{ab^x}{1-a^4}$

(d) $\frac{ab^x}{1-b^4}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{ab^x}{1-b^2}$

EXAM SHORTCUT (AI-derived explanation)

On b^x the operator E acts as multiplication by b, so E^2 becomes b^2 and $1/(1 - E^2)$ becomes $1/(1 - b^2)$. The answer is a $b^x/(1 - b^2)$.

TIPS & TRICKS (AI-derived explanation)

- Substitution rule: $E(b^x)$ = b^(x+1) = b . b^x, so any polynomial in E acting on b^x is that polynomial EVALUATED at b.
- Here the operator is 1 - E^2, which becomes 1 - b^2 - the variable that appears is b, the base, not the constant a.
- Options (b) and (c) put a in the denominator, which is impossible since a is only a multiplicative constant that passes through the operator.
- The condition 0 < b < 1 guarantees 1 - b^2 is non-zero and positive, so the inverse operator is well defined.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With interval of differencing 1, $E(b^x)$ = b^(x+1) = b . b^x, so E acts on b^x as multiplication by b.
2. Hence E^2 (a b^x) = b^2 (a b^x).
3. Therefore (1 - E^2)(a b^x) = a b^x (1 - b^2).
4. Applying the inverse operator simply divides by that constant: [1/(1 - E^2)](a b^x) = a b^x/(1 - b^2).
5. The condition 0 < b < 1 ensures 1 - b^2 is non-zero, so the operation is valid.
6. Option (a) is correct.

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53. 2025 · Q22 Sub-division of Intervals · Numerical

QUESTION

What is $\Delta^5\left(\frac{1}{x}\right)$ at $x=2$? (interval of differencing = unity)

OPTIONS

(a) 1

(b) -1

(c) $-\frac{1}{21}$

(d) $-\frac{1}{42}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $-\frac{1}{42}$

EXAM SHORTCUT (AI-derived explanation)

Delta^n(1/x) = (-1)^n n!/[x(x+1)...(x+n)]. With n = 5 and x = 2 this is -120/(2.3.4.5.6.7) = -120/5040 = -1/42.

TIPS & TRICKS (AI-derived explanation)

- Memorise Delta^n(1/x) = (-1)^n n! h^n/[x(x+h)...(x+nh)] - it converts a five-fold difference into one fraction.
- Build it from the first step: Delta(1/x) = 1/(x+1) - 1/x = -1/[x(x+1)], and the pattern of alternating signs and factorials follows.
- The denominator has n + 1 = 6 consecutive factors starting at x = 2: 2, 3, 4, 5, 6, 7, whose product is 5040.
- Sign check: n = 5 is ODD, so the answer must be negative - options (a) and (c) can be tested by size, but the sign alone kills (a).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With unit interval, Delta(1/x) = 1/(x+1) - 1/x = -1/[x(x+1)].
2. Applying Delta again gives Delta^2(1/x) = 2/[x(x+1)(x+2)], and the pattern continues.
3. In general Delta^n(1/x) = (-1)^n n!/[x(x+1)(x+2)...(x+n)].
4. For n = 5 and x = 2: the denominator is 2 x 3 x 4 x 5 x 6 x 7 = 5040 and the numerator is (-1)^5 5! = -120.
5. So Delta^5(1/x) at x = 2 = -120/5040 = -1/42, option (d).

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54. 2025 · Q62 Finite Differences of Different Orders · Conceptual

QUESTION

What is the forward difference approximation for $f'(x_i)$ at x_i?

OPTIONS

(a) $\frac{f_{i+2}-2f_{i+1}+f_i}{h^2}$

(b) $\frac{f_{i-1}-2f_i+f_{i+1}}{h^2}$

(c) $\frac{f_{i+1}+2f_i+f_{i-1}}{h^2}$

(d) $\frac{f_{i+3}-2f_{i+2}-f_{i+1}}{h^2}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{f_{i+2}-2f_{i+1}+f_i}{h^2}$

EXAM SHORTCUT (AI-derived explanation)

The word 'forward' fixes the stencil: only points at or ahead of x_i may appear. Option (a) uses f_i, f_{i+1}, f_{i+2} - the only choice with no backward point and with coefficients 1, -2, 1 summing to zero. Pick (a) in five seconds.

TIPS & TRICKS (AI-derived explanation)

- For any valid second-derivative formula the coefficients must sum to zero (it must annihilate constants); 1-2+1 = 0 passes, 1+2+1 = 4 fails, 1-2-1 = -2 fails.
- Forward = points i, i+1, i+2. Backward = points i, i-1, i-2. Central = points i-1, i, i+1.
- Derivation: f'' ~ Delta^2 f_i / h^2, and Delta^2 f_i = f_{i+2} - 2f_{i+1} + f_i.
- Option (b) is the classic central-difference formula - a very common distractor when the question says 'forward'.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The forward difference operator is Delta f_i = f_{i+1} - f_i, so Delta^2 f_i = Delta(f_{i+1} - f_i) = (f_{i+2} - f_{i+1}) - (f_{i+1} - f_i) = f_{i+2} - 2f_{i+1} + f_i.
2. Using D = (1/h) log(1 + Delta), we get h^2 D^2 = (log(1+Delta))^2 = Delta^2 - Delta^3 + ... , so to leading order h^2 f''(x_i) = Delta^2 f_i + O(h^3).
3. Therefore f''(x_i) = (f_{i+2} - 2f_{i+1} + f_i)/h^2 with truncation error O(h).
4. Option (b) is the central difference formula (f_{i-1} - 2f_i + f_{i+1})/h^2 - correct as a formula but not a forward difference.
5. Option (c) has coefficients summing to 4, so it does not even vanish for constant f; option (d) has coefficients summing to -2 and is not a difference formula at all.
6. Hence option (a) is correct.

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55. 2025 · Q66 Delta, E and D Operators · Theoretical

QUESTION

In respect of forward and backward operators (Δ and ∇): I. $\Delta-\nabla=\Delta\nabla$ II. $\nabla-\Delta=\nabla\Delta$ III. $\Delta+\nabla=\Delta\nabla$ IV. $\Delta+\nabla=-\Delta\Delta$ Which is/are correct?

OPTIONS

(a) I only

(b) II only

(c) I and III

(d) II and IV

ANSWER (AI-derived — no official key exists for this paper)

(a) I only

EXAM SHORTCUT (AI-derived explanation)

Write everything in E: Delta = E - 1, nabla = 1 - E^{-1}. Then Delta - nabla = E - 2 + E^{-1} and Delta*nabla = (E-1)(1-E^{-1}) = E - 2 + E^{-1}. Identical, so I holds; II is its negative and fails; Delta + nabla = E - E^{-1} kills III and IV. Answer (a).

TIPS & TRICKS (AI-derived explanation)

- Reduce every operator identity to the shift operator E - it turns operator algebra into ordinary algebra.
- Delta and nabla commute (both are polynomials in E), so Delta*nabla = nabla*Delta; statements I and II cannot both be true since their left sides are negatives of each other.
- Useful companion identities: Delta*nabla = delta^2, Delta = E*nabla, nabla = E^{-1}*Delta, and Delta - nabla = delta^2 as well.
- Statement IV as printed reads Delta + nabla = -Delta*Delta, which is dimensionally hopeless - reject it immediately.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Using the shift operator E with E f(x) = f(x+h): Delta = E - 1 and nabla = 1 - E^{-1}.
2. Statement I: Delta - nabla = (E - 1) - (1 - E^{-1}) = E - 2 + E^{-1}. Also Delta*nabla = (E - 1)(1 - E^{-1}) = E - 1 - 1 + E^{-1} = E - 2 + E^{-1}. The two agree, so I is correct.
3. Statement II: nabla - Delta = -(E - 2 + E^{-1}), while nabla*Delta = Delta*nabla = E - 2 + E^{-1} (the operators commute). These are negatives of each other and are not equal (except on the trivial zero function), so II is incorrect.
4. Statement III: Delta + nabla = (E - 1) + (1 - E^{-1}) = E - E^{-1}, which is not equal to Delta*nabla = E - 2 + E^{-1}. So III is incorrect.
5. Statement IV: Delta + nabla = E - E^{-1} is certainly not -Delta^2 = -(E-1)^2 = -(E^2 - 2E + 1). So IV is incorrect.
6. Only statement I holds. Hence option (a) is correct.

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56. 2025 · Q67 Sub-division of Intervals · Numerical

QUESTION

What is $\frac{\Delta^{100}}{E}(x^{100})$? (interval of differencing = unity)

OPTIONS

(a) 100!

(b) 100!-99!

(c) 100!-1

(d) 100!-99!+1

ANSWER (AI-derived — no official key exists for this paper)

(a) 100!

EXAM SHORTCUT (AI-derived explanation)

Delta^n applied to x^n with h = 1 gives the constant n!, so Delta^{100} x^{100} = 100!. Dividing by E means applying E^{-1}, and shifting a constant leaves it unchanged. Answer 100!, option (a).

TIPS & TRICKS (AI-derived explanation)

- Key fact: for a monic degree-n polynomial and h = 1, Delta^n gives exactly n! and Delta^{n+1} gives 0.
- Operators that are polynomials in E commute, so Delta^{100}/E may be read as E^{-1}Delta^{100} or Delta^{100} E^{-1} - both give the same result.
- A shift never changes a constant: E^{-1}(c) = c. That is why the answer is clean rather than 100! - 99!.
- Distractors like 100! - 99! are built by wrongly treating E^{-1} as though it subtracted a lower factorial; check by testing the identity on a small case such as Delta^2/E applied to x^2, which gives 2! = 2.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Take the interval of differencing h = 1.
2. For a polynomial of degree n with leading coefficient 1, the n-th forward difference is the constant n! * h^n. With n = 100 and h = 1, Delta^{100} (x^{100}) = 100!.
3. The expression Delta^{100}/E means the operator E^{-1} composed with Delta^{100} (all these operators are polynomials in E, hence commute).
4. Applying E^{-1} to the constant 100! gives E^{-1}(100!) = 100!, since shifting the argument of a constant function changes nothing.
5. Equivalently, computing in the other order: E^{-1}(x^{100}) = (x-1)^{100}, and Delta^{100} of the monic degree-100 polynomial (x-1)^{100} is again 100!.
6. Hence option (a) is correct.

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57. **2026 · Q41** Sub-division of Intervals · Theoretical

QUESTION

If the interval of difference is h, what is $\Delta[\log f(x)]$?

OPTIONS

(a) $\log[1+\frac{Ef(x)}{f(x)}]$

(b) $\log[1-\frac{\Delta\ f(x)}{f(x)}]$

(c) $\log[1+\frac{\Delta\ f(x)}{f(x)}]$

(d) $\log[1-\frac{Ef(x)}{f(x)}]$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\log[1+\frac{\Delta\ f(x)}{f(x)}]$

EXAM SHORTCUT (AI-derived explanation)

Delta log f = log f(x+h) - log f(x) = log[f(x+h)/f(x)], and f(x+h) = f(x) + Delta f(x), so the ratio is 1 + Delta f(x)/f(x). Answer (c).

TIPS & TRICKS (AI-derived explanation)

- Turn a difference of logarithms into the logarithm of a ratio - that single step solves the whole item.
- Use E f(x) = f(x) + Delta f(x) to convert the shifted value into something expressed through Delta.
- Options (a) and (d) are dimensionally wrong: E f(x)/f(x) is already the full ratio f(x+h)/f(x), so adding 1 to it double-counts.
- Sign check with a growing f: Delta f > 0 forces Delta log f > 0, which rules out the minus sign in options (b) and (d).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. By definition, Delta[log f(x)] = log f(x + h) - log f(x).
2. Combining the logarithms: Delta[log f(x)] = log[f(x + h)/f(x)].
3. Since Delta f(x) = f(x + h) - f(x), we have f(x + h) = f(x) + Delta f(x).
4. Therefore f(x + h)/f(x) = 1 + Delta f(x)/f(x).
5. Hence Delta[log f(x)] = log[1 + Delta f(x)/f(x)].
6. Hence option (c) is correct.

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58. 2026 · Q42 Factorial Representation of a Polynomial · Numerical

QUESTION

A cubic polynomial $f(x)$ has values:

x	0	1	2	3
f(x)	1	0	1	10

What is $f(4)$?

OPTIONS

(a) 35

(b) 33

(c) 31

(d) 29

ANSWER (AI-derived — no official key exists for this paper)

(b) 33

EXAM SHORTCUT (AI-derived explanation)

Build the difference table: f = 1, 0, 1, 10 gives Delta = -1, 1, 9; Delta^2 = 2, 8; Delta^3 = 6. For a cubic Delta^3 stays 6, so extend: Delta^2 = 14, Delta = 23, f(4) = 10 + 23 = 33.

TIPS & TRICKS (AI-derived explanation)

- For a polynomial of degree 3 the third differences are constant - that constant is the engine that extends the table.
- Extend the table from the bottom-right upwards: new Delta^3 = 6, then add up each column in turn.
- Check: Delta^3 = 6 = 3! * a_3, so the leading coefficient is 1, i.e. $f(x) = x^3 - 3x^2 + x + 1$. Then f(4) = 64 - 48 + 4 + 1 = 33 - a clean independent verification.
- Do not use Newton's forward formula with a fractional argument here; simple table extension is faster and error-free for an integer node.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Construct the forward difference table for f(0) = 1, f(1) = 0, f(2) = 1, f(3) = 10.
2. First differences: -1, 1, 9.
3. Second differences: 1 - (-1) = 2, and 9 - 1 = 8.
4. Third difference: 8 - 2 = 6.
5. Since f is a cubic, all third differences equal 6.
6. Extend the table: next Delta^2 = 8 + 6 = 14; next Delta = 9 + 14 = 23; hence f(4) = f(3) + 23 = 10 + 23 = 33.
7. Verification: Delta^3 = 6 = 3! times the leading coefficient, so a_3 = 1, and fitting gives $f(x) = x^3 - 3x^2 + x + 1$, whence f(4) = 64 - 48 + 4 + 1 = 33.
8. Hence option (b) is correct.

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59. 2026 · Q45 Sub-division of Intervals · Numerical

QUESTION

If the interval of difference is h, what is $\left(\frac{\Delta^2}{E}\right)e^x \times \frac{Ee^x}{\Delta^2 e^x}$?

OPTIONS

(a) 1

(b) e^{-x}

(c) e^{x+h}

(d) e^x

ANSWER **(AI-derived — no official key exists for this paper)**

(d) e^x

EXAM SHORTCUT **(AI-derived explanation)**

The two Delta^2 e^x factors cancel, leaving (1/E) e^x times E e^x = e^{x-h} *e^{x+h}... more simply the expression is [E^{-1} Delta^2 e^x]* [E e^x / Delta^2 e^x] = E^{-1}(e^x) * ... = e^x. Formally the Delta^2 and E factors cancel to give exactly e^x.

TIPS & TRICKS **(AI-derived explanation)**

- Work out the two key actions on e^x first: Delta e^x = e^x(e^h - 1), so Delta^2 e^x = e^x(e^h - 1)^2, and E e^x = e^{x+h}.
- e^x is an eigenfunction of every difference operator, which is why all these operators reduce to multiplication by a constant - the fastest route on any question involving exponentials.
- Track the factors: E^{-1} contributes e^{-h} and E contributes e^{+h}; they cancel, as do the two (e^h - 1)^2 factors.
- Since the expression is a product of a quantity and (essentially) its reciprocal up to the shift, the answer must be free of (e^h - 1) - that alone points to a clean e^x.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. For the exponential, Delta e^x = e^{x+h} - e^x = e^x (e^h - 1), so Delta^2 e^x = e^x (e^h - 1)^2, and E e^x = e^{x+h}.
2. First factor: (Delta^2/E) e^x = E^{-1}[Delta^2 e^x] = E^{-1}[e^x (e^h - 1)^2] = e^{x-h} (e^h - 1)^2.
3. Second factor: (E e^x)/(Delta^2 e^x) = e^{x+h}/[e^x (e^h - 1)^2] = e^h/(e^h - 1)^2.
4. Multiplying: e^{x-h}(e^h - 1)^2 *e^h/(e^h - 1)^2 = e^{x-h}* e^h = e^x.
5. The factors (e^h - 1)^2 cancel and the shifts e^{-h} and e^{+h} cancel, leaving e^x.
6. Hence option (d) is correct.

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60. 2026 · Q47 Delta, E and D Operators · Theoretical

QUESTION

What is $\mu[f(x)g(x)]$?

OPTIONS

(a) $\delta\ f(x)\delta\ g(x) + \frac{1}{4}\mu\ f(x)\mu\ g(x)$

(b) $\mu\ f(x)\mu\ g(x) + \frac{1}{4}\delta\ f(x)\delta\ g(x)$

(c) $\mu\ f(x)\delta\ g(x) + \delta\ f(x)\mu\ g(x)$

(d) $\frac{1}{4}[\mu\ f(x)\delta\ g(x) + \delta\ f(x)\mu\ g(x)]$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\mu\ f(x)\mu\ g(x) + \frac{1}{4}\delta\ f(x)\delta\ g(x)$

EXAM SHORTCUT (AI-derived explanation)

Write f+ = f(x+h/2), f- = f(x-h/2). Then mu f mu g = (f+ + f-)(g+ + g-)/4 and delta f delta g = (f+ - f-)(g+ - g-). Adding one quarter of the second to the first cancels the cross terms and leaves (f+g+ + f-g-)/2 = mu(fg).

TIPS & TRICKS (AI-derived explanation)

- Always convert mu and delta to half-shifts: mu = (E^{1/2} + E^{-1/2})/2 and delta = E^{1/2} - E^{-1/2}. Everything then becomes ordinary algebra in f+, f-, g+, g-.
- The cross terms f+g- and f-g+ must cancel, since mu(fg) contains only the 'aligned' products f+g+ and f-g- - that requirement alone identifies the coefficient 1/4.
- Companion identity worth memorising: delta(fg) = mu f *delta g + delta f* mu g, which is option (c) here - a deliberately placed distractor.
- Quick test: put g = 1, so mu g = 1 and delta g = 0. The correct formula must collapse to mu f. Option (b) gives mu f * 1 + 0 = mu f. Options (a), (c), (d) fail this test.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let f+ = f(x + h/2), f- = f(x - h/2), and similarly for g. Then mu f = (f+ + f-)/2, delta f = f+ - f-, and mu(fg) = (f+ g+ + f- g-)/2.
2. Compute mu f * mu g = [(f+ + f-)/2][(g+ + g-)/2] = (f+g+ + f+g- + f-g+ + f-g-)/4.
3. Compute delta f * delta g = (f+ - f-)(g+ - g-) = f+g+ - f+g- - f-g+ + f-g-.
4. Then mu f mu g + (1/4) delta f delta g = (f+g+ + f+g- + f-g+ + f-g-)/4 + (f+g+ - f+g- - f-g+ + f-g-)/4.
5. The cross terms cancel, leaving (2 f+g+ + 2 f-g-)/4 = (f+g+ + f-g-)/2 = mu(fg).
6. Therefore mu[f(x) g(x)] = mu f(x) mu g(x) + (1/4) delta f(x) delta g(x).
7. Consistency check with g identically 1: the right side gives mu f * 1 + (1/4)(delta f)(0) = mu f, as required.
8. Hence option (b) is correct.

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61. 2026 · Q49 Sub-division of Intervals · Numerical

QUESTION

If the interval of difference is 1, what is $\Delta^4[(x\text{-}1)(1\text{-}2x)(1\text{-}3x)(1\text{-}4x)]$?

OPTIONS

(a) 576

(b) 24

(c) -24

(d) -576

ANSWER (AI-derived — no official key exists for this paper)

(d) -576

EXAM SHORTCUT (AI-derived explanation)

Only the leading coefficient matters: (x)(-2x)(-3x)(-4x) gives $-24x^4$. Delta^4 of a quartic with leading coefficient a is 4! a h^4 = 24(-24)(1) = -576.

TIPS & TRICKS (AI-derived explanation)

- Rule: Delta^n of a degree-n polynomial with leading coefficient a_n and interval h equals n! a_n h^n - a constant.
- Do not expand the product. Multiply only the x-coefficients: 1 *(-2)* (-3) * (-4) = -24. Three minus signs give a negative result.
- Missing the 4! = 24 factor produces the distractor -24; missing the sign produces +576. Both traps are on the option list.
- With h = 1 the factor h^n disappears, so the whole answer is just 24 * (-24) = -576.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The expression (x - 1)(1 - 2x)(1 - 3x)(1 - 4x) is a polynomial of degree 4.
2. Its leading coefficient is the product of the coefficients of x in each factor: (1)(-2)(-3)(-4) = -24. So the polynomial is $-24x^4$ + (lower order terms).
3. For a polynomial of degree n with leading coefficient a_n, the nth forward difference is the constant Delta^n = n! *a_n* h^n.
4. Here n = 4, a_4 = -24, h = 1, so Delta^4 = 4! *(-24)* 1^4 = 24 * (-24) = -576.
5. (Lower-degree terms contribute nothing, since Delta^4 annihilates every polynomial of degree below 4.)
6. Hence option (d) is correct.

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62. 2026 · Q59 Sub-division of Intervals · Theoretical

QUESTION

Let $P=[\binom{n}{1}+\binom{n}{2}\Delta+\binom{n}{3}\Delta^2+\cdots+\binom{n}{n}\Delta^{n-1}]y_1$ and $Q=y_1+y_2+\cdots+y_n$. If the interval of difference $h=1$, what is P/Q?

OPTIONS

(a) 1

(b) n

(c) $2n+1$

(d) n^2

ANSWER (AI-derived — no official key exists for this paper)

(a) 1

EXAM SHORTCUT (AI-derived explanation)

The bracket is $[(1+\text{Delta})^n - 1]$/Delta = $(E^n - 1)/(E - 1)$ = 1 + E + ... + E^{n-1}. Applied to y_1 this gives y_1 + y_2 + ... + y_n = Q, so P/Q = 1.

TIPS & TRICKS (AI-derived explanation)

- Spot the binomial pattern: sum over k from 1 to n of C(n,k) Delta^{k-1} is exactly $[(1 + \text{Delta})^n - 1]$ divided by Delta.
- Then use the fundamental relation E = 1 + Delta, turning the bracket into the geometric operator sum $(E^n - 1)/(E - 1) = 1 + E + E^2 + \ldots$ + E^{n-1}.
- E^k y_1 = y_{1+k} when h = 1, so the operator sum applied to y_1 reproduces the sum of the first n ordinates.
- This is the standard operator proof of the summation formula sum y_i = [C(n,1) + C(n,2)Delta + ...] y_1 - a result worth memorising in its own right.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The bracket is B = sum from k = 1 to n of C(n, k) Delta^{k-1}.
2. Multiplying by Delta: Delta * B = sum from k = 1 to n of C(n, k) $\text{Delta}^k = (1 + \text{Delta})^n - 1$, by the binomial theorem (the k = 0 term C(n,0) = 1 being subtracted off).
3. Using E = 1 + Delta, this is $E^n - 1$, and Delta = E - 1, so B = $(E^n - 1)/(E - 1)$.
4. By the finite geometric sum for operators, $(E^n - 1)/(E - 1) = 1 + E + E^2 + \ldots$ + E^{n-1}.
5. With h = 1, E^k y_1 = y_{1+k}, so B y_1 = y_1 + y_2 + ... + y_n.
6. Therefore P = B y_1 = y_1 + y_2 + ... + y_n = Q, and P/Q = 1.
7. Hence option (a) is correct.

N2 · NUMERICAL ANALYSIS

Interpolation - Newton-Gregory Forward & Backward (Equal Intervals)

14 questions · 2018: 1 2019: 1 2020: 1 2021: 3 2022: 3 2023: 3 2024: 2 2025: 0 2026: 0

Weight. 14 questions (7.8% of Numerical Analysis) · appeared in 7 of 9 papers · peak year 2021 (3)

Examiner's pattern. Forward formula for the head of a table, backward for the tail, and one step of extrapolation beyond the last entry. Also: the linear-interpolation error bound and choosing a step size to hit a target accuracy. The f(20)=512, f(30)=439, f(40)=346, f(50)=243 table was reused verbatim in 2020 and 2022.

Must-know. Forward: y = y0 + pDy0 + p(p-1)/2! D^2 y0 + ... with p = (x - x0)/h. Backward uses p = (x - xn)/h and nabla. Linear interpolation error <= (h^2/8) max|f''|.

1. 2018 · Q50 Newton-Gregory Forward Interpolation Formula · Numerical

QUESTION

Population (in lakhs) of a town:

Year	1971	1981	1991	2001	2011
Population	0.9	1.2	1.5	1.9	2.5

P_{2006} estimated by Newton's forward difference formula satisfies

OPTIONS

(a) $P_{2006}<2$

(b) $2<P_{2006}<2.1$

(c) $2.1<P_{2006}<2.2$

(d) $2.2<P_{2006}<2.3$

ANSWER (AI-derived — no official key exists for this paper)

(c) $2.1<P_{2006}<2.2$

EXAM SHORTCUT (AI-derived explanation)

With h = 10 and origin 1971, p = (2006-1971)/10 = 3.5. Differences: Delta = 0.3, Delta^2 = 0.0, Delta^3 = 0.1, Delta^4 = 0. So P = 0.9 + 3.5(0.3) + 0 + 2.1875(0.1) = 2.169.

TIPS & TRICKS (AI-derived explanation)

- Newton's forward formula: P = y0 + p Delta y0 + p(p-1)/2! Delta^2 y0 + p(p-1)(p-2)/3! Delta^3 y0 + ... Compute p first; everything else follows mechanically.
- Here Delta^2 y0 = 0, which kills the quadratic term entirely - always scan the leading diagonal for zeros before grinding through the arithmetic.
- p(p-1)(p-2)/6 with p = 3.5 is (3.5)(2.5)(1.5)/6 = 13.125/6 = 2.1875. Keep the numerator as a product until the end to avoid rounding drift.

- Sanity check: 2006 lies between 2001 (1.9) and 2011 (2.5), so linear interpolation would give 2.2. The answer must be near but not equal to that, which points at band (c) rather than (d).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Take x0 = 1971 with h = 10; the values are y0 = 0.9, y1 = 1.2, y2 = 1.5, y3 = 1.9, y4 = 2.5.
2. Leading differences: Delta y0 = 0.3; second differences 0.0, 0.1, 0.2 so Delta^2 y0 = 0.0; third differences 0.1, 0.1 so Delta^3 y0 = 0.1; Delta^4 y0 = 0.
3. For the year 2006, p = (2006 - 1971)/10 = 3.5.
4. Newton's forward difference formula gives P = 0.9 + (3.5)(0.3) + [(3.5)(2.5)/2](0.0) + [(3.5)(2.5)(1.5)/6](0.1) + 0 = 0.9 + 1.05 + 0 + (13.125/6)(0.1) = 0.9 + 1.05 + 0.21875 = 2.16875.
5. Since 2.1 < 2.169 < 2.2, option (c) is correct.

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2. 2019 · Q77 Interpolation & Extrapolation · Numerical

QUESTION

Let $f(x)=\log_{10}(1+x)$ with $x_0=1$, $x_1=1.1$. What is the approximate value of $f(1.04)$ by linear interpolation? [Given $\log_{10}2=0.301030$, $\log_{10}2.1=0.322219$]

OPTIONS

(a) 0.309506

(b) 0.095603

(c) 0.390506

(d) 0.090356

ANSWER (AI-derived — no official key exists for this paper)

(a) 0.309506

EXAM SHORTCUT (AI-derived explanation)

p = (1.04 - 1)/0.1 = 0.4, so f = 0.301030 + 0.4(0.322219 - 0.301030) = 0.301030 + 0.008476 = 0.309506.

TIPS & TRICKS (AI-derived explanation)

- Linear interpolation is Newton's forward formula truncated after the first difference: f = f0 + p (f1 - f0) with p = (x - x0)/h.
- Compute the difference once: 0.322219 - 0.301030 = 0.021189, then take 40% of it. Working with the difference rather than the raw values keeps all six decimals intact.
- Sanity check: the answer must lie between 0.301030 and 0.322219, and nearer the lower end since p = 0.4 < 0.5. Options (b), (c) and (d) all fall outside that range.
- Note that f(x) = log(1 + x), so f(1.04) = log10(2.04); the true value is 0.309630, confirming the interpolation is accurate to four decimals.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. We interpolate f(x) = log10(1 + x) between x0 = 1 and x1 = 1.1, so h = 0.1.

2. The tabulated values are f(x0) = log10 2 = 0.301030 and f(x1) = log10 2.1 = 0.322219.
3. For x = 1.04, p = (1.04 - 1)/0.1 = 0.4.
4. Linear interpolation: f(1.04) = f0 + p(f1 - f0) = 0.301030 + 0.4(0.322219 - 0.301030).
5. The first difference is 0.021189, and 0.4 x 0.021189 = 0.0084756.
6. So f(1.04) = 0.301030 + 0.008476 = 0.309506, option (a).

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3. 2020 · Q74 Newton-Gregory Backward Interpolation Formula · Conceptual

QUESTION

Given $f(20)=512,\ f(30)=439,\ f(40)=346,\ f(50)=243$: 1. $f(x)$ can be approximated by a polynomial of degree $n<2$. 2. Approximate values of $f(x)$ for $x=10$ and $x=60$ can be obtained by Newton–Gregory forward and Newton–Gregory backward difference formulae respectively.

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(b) 2 only

EXAM SHORTCUT (AI-derived explanation)

First differences are -73, -93, -103 - not constant, so f is not of degree below 2 and statement 1 fails. Statement 2 correctly assigns the forward formula to x = 10 (near the table's start) and the backward formula to x = 60 (near its end).

TIPS & TRICKS (AI-derived explanation)

- Degree test: constant FIRST differences mean degree 1, constant SECOND differences mean degree 2. Here the first differences vary by 20 and 10, so the degree is at least 3.
- Extrapolation convention: use Newton-Gregory FORWARD near the beginning of the table (p negative for x below x0) and BACKWARD near the end. Statement 2 states exactly this.
- Both requested points lie outside the tabulated range, so this is extrapolation rather than interpolation - legitimate but less accurate, which is why the formulas are chosen by proximity.
- Compute the difference table before reading the statements; it answers statement 1 immediately and costs only three subtractions.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The tabulated values at x = 20, 30, 40, 50 are 512, 439, 346, 243, equally spaced with h = 10.
2. First differences: 439 - 512 = -73; 346 - 439 = -93; 243 - 346 = -103. These are not constant.
3. Second differences: -93 + 73 = -20 and -103 + 93 = 10, also not constant, so the data require a polynomial of degree at least 3.

4. Statement 1 claims a polynomial of degree n < 2 (linear or constant) suffices, which contradicts the non-constant first differences. FALSE.
5. Statement 2: x = 10 lies just before the start of the table, where Newton-Gregory's FORWARD formula (with a negative p) is the appropriate extrapolation; x = 60 lies just beyond the end, where the BACKWARD formula is appropriate. TRUE.
6. Only statement 2 is correct, option (b).

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4. 2021 · Q48 Interpolation & Extrapolation · Numerical

QUESTION

Data: x=-1,0,1,2; f=4,2,2,4. The interpolating polynomial is

OPTIONS

(a) $2x^2$-x+2

(b) x^2+x-2

(c) x^2-x+2

(d) $2x^2$+x-2

ANSWER (AI-derived — no official key exists for this paper)

(c) x^2-x+2

EXAM SHORTCUT (AI-derived explanation)

Second differences are constant at 2, so the polynomial is quadratic. Newton forward from x0 = -1 gives p^2 - 3p + 4 with p = x + 1, i.e. x^2 - x + 2.

TIPS & TRICKS (AI-derived explanation)

- Build the difference table first: values 4, 2, 2, 4 give first differences -2, 0, 2 and second differences 2, 2 - constant, so the degree is 2 and the third difference vanishes.
- Faster route: test the options. At x = 0 the value must be 2, which options (a) and (c) satisfy; then x = 1 must give 2, which only (c) does.
- The data are symmetric about x = 0.5, so the vertex of the parabola is at x = 0.5, consistent with x^2 - x + 2 whose vertex is at x = 1/2.
- Leading coefficient check: the second difference is 2 = 2a h^2 with h = 1, so a = 1 - eliminating options (a) and (d), which have a = 2.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Tabulate: at x = -1, 0, 1, 2 the values are 4, 2, 2, 4 with h = 1.
2. First differences: -2, 0, 2. Second differences: 2, 2. Third difference: 0, so the data lie on a quadratic.
3. Newton's forward formula from x0 = -1 with p = (x + 1)/1: f = 4 + p(-2) + [p(p-1)/2](2) = 4 - 2p + p(p-1).
4. Simplifying: 4 - 2p + p^2 - p = p^2 - 3p + 4.
5. Substituting p = x + 1: $(x+1)^2$ - 3(x+1) + 4 = x^2 + 2x + 1 - 3x - 3 + 4 = x^2 - x + 2.

6. Check: at x = -1, 1 + 1 + 2 = 4; at x = 0, 2; at x = 1, 2; at x = 2, 4. All match. Option (c) is correct.

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5. 2021 · Q62 Newton-Gregory Backward Interpolation Formula · Numerical

QUESTION

A degree-2 polynomial $f(x)$ has values $8,3,0,\text{-}1,0$ at $x{=}110,120,130,140,150$. What is $f(100)$?

OPTIONS

(a) 5

(b) 10

(c) 15

(d) 20

ANSWER (AI-derived — no official key exists for this paper)

(c) 15

EXAM SHORTCUT (AI-derived explanation)

Second differences are constant at 2, so the first difference one step before -5 is -7. Hence f(100) = f(110) + 7 = 8 + 7 = 15.

TIPS & TRICKS (AI-derived explanation)

- For backward extrapolation of a quadratic, just extend the difference table LEFTWARDS: keep the second difference constant and work back one first difference.
- First differences here are -5, -3, -1, 1 - an arithmetic progression with common difference 2, so the entry before -5 must be -7.
- f(110) - f(100) = -7 means f(100) = 8 + 7 = 15. Watch the sign: the function is DECREASING over the table, so it must be larger at x = 100.
- Verify by fitting: with x = (t - 130)/10 the data are 8, 3, 0, -1, 0 at x = -2,-1,0,1,2, i.e. f = x^2 - 2x... checking t = 100 gives x = -3 and f = 9 + 6 = 15.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The tabulated values at x = 110, 120, 130, 140, 150 are 8, 3, 0, -1, 0 with h = 10.
2. First differences: -5, -3, -1, 1. Second differences: 2, 2, 2 - constant, confirming a quadratic.
3. To extrapolate to x = 100 we extend the table backwards by one step, keeping the second difference equal to 2.
4. The first difference immediately preceding -5 must therefore be -5 - 2 = -7, i.e. f(110) - f(100) = -7.
5. Hence f(100) = f(110) + 7 = 8 + 7 = 15.
6. Option (c) is correct.

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6. 2021 · Q64 Newton-Gregory Forward Interpolation Formula · Numerical

QUESTION

Cumulative data:

Marks below	40	60	80	100	120
No. of students	250	370	470	540	590

Approximate number of students securing marks between 60 and 70, via Newton's forward interpolation:

OPTIONS

(a) 54

(b) 58

(c) 62

(d) 66

ANSWER (AI-derived — no official key exists for this paper)

(a) 54

EXAM SHORTCUT (AI-derived explanation)

Newton forward from x0 = 40 with h = 20 and p = 1.5 gives f(70) = 250 + 180 - 7.5 + 0.625 + 0.469 = 423.6. Subtracting f(60) = 370 leaves about 54 students.

TIPS & TRICKS (AI-derived explanation)

- The table is CUMULATIVE ('marks below'), so the number between 60 and 70 is f(70) - f(60) - never interpolate the frequencies themselves.
- Difference table: 120, 100, 70, 50; then -20, -30, -20; then -10, 10; then 20. Compute it in full, since p = 1.5 keeps all four terms alive.
- The factorial coefficients change sign after the second term because p - 2 = -0.5 and p - 3 = -1.5 are negative - track the signs carefully.
- f(60) = 370 is read directly from the table, so only f(70) needs interpolating. The answer 53.6 rounds to 54 students.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Take x0 = 40 with h = 20 and cumulative values y = 250, 370, 470, 540, 590.
2. Differences: Delta = 120, 100, 70, 50; Delta^2 = -20, -30, -20; Delta^3 = -10, 10; Delta^4 = 20.
3. For x = 70, p = (70 - 40)/20 = 1.5.
4. Newton's forward formula: f(70) = 250 + 1.5(120) + [1.5(0.5)/2](-20) + [1.5(0.5)(-0.5)/6](-10) + [1.5(0.5)(-0.5)(-1.5)/24](20) = 250 + 180 - 7.5 + 0.625 + 0.469 = 423.59.
5. Students scoring between 60 and 70 = f(70) - f(60) = 423.59 - 370 = 53.59, i.e. about 54. Option (a) is correct.

7. 2022 · Q62 Newton-Gregory Forward Interpolation Formula · Numerical

QUESTION

Given $f(0)=5$, $f(1)=1$, $f(2)=9$, $f(3)=25$, $f(4)=55$. Approximate $f(5)$ using Newton-Gregory forward interpolation.

OPTIONS

(a) 108

(b) 110

(c) 112

(d) 115

ANSWER (AI-derived — no official key exists for this paper)

(d) 115

EXAM SHORTCUT (AI-derived explanation)

Leading differences are -4, 12, -4, 10. With p = 5 the binomial coefficients are 5, 10, 10, 5, giving 5 - 20 + 120 - 40 + 50 = 115.

TIPS & TRICKS (AI-derived explanation)

- Extrapolating one step beyond the table means p = 5 here (x0 = 0, h = 1), so the coefficients are the binomial numbers C(5,1) = 5, C(5,2) = 10, C(5,3) = 10, C(5,4) = 5.
- Build the full difference table: -4, 8, 16, 30; then 12, 8, 14; then -4, 6; then 10. Every level is needed because the data are not polynomial of low degree.
- For integer p the factorial coefficients reduce to plain binomial coefficients - a large simplification worth spotting.
- Running total: 5, -15, 105, 65, 115. Keeping a running sum makes a sign slip immediately visible.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With x0 = 0 and h = 1, the values are 5, 1, 9, 25, 55.
2. Differences: Delta = -4, 8, 16, 30; Delta^2 = 12, 8, 14; Delta^3 = -4, 6; Delta^4 = 10.
3. For x = 5, p = (5 - 0)/1 = 5, and since p is an integer the coefficients p(p-1).../k! are the binomial coefficients C(5,k): 5, 10, 10, 5.
4. Newton's forward formula: f(5) = 5 + 5(-4) + 10(12) + 10(-4) + 5(10) = 5 - 20 + 120 - 40 + 50 = 115, option (d).

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8. 2022 · Q63 Newton-Gregory Backward Interpolation Formula · Numerical

QUESTION

If $y_{10}=3$, $y_{11}=6$, $y_{12}=11$, $y_{13}=18$, $y_{14}=27$, what is y_4 equal to?

OPTIONS

(a) 3

(b) 4

(c) 9

(d) 27

ANSWER (AI-derived — no official key exists for this paper)

(d) 27

EXAM SHORTCUT (AI-derived explanation)

Second differences are constant at 2, so y is quadratic with leading coefficient 1. Fitting gives $y_x = x^2 - 18x + 83$, and $y_4 = 16 - 72 + 83 = 27$.

TIPS & TRICKS (AI-derived explanation)

- Constant second differences equal to 2 with h = 1 mean $2a\ h^2 = 2$, so the leading coefficient a = 1 - identify it before fitting anything else.
- Two data points then pin down b and c: subtracting y_{10} from y_{11} gives 21 + b = 3, so b = -18.
- Alternative route: extend the difference table BACKWARDS from y_{10} six steps, subtracting 2 from each first difference each time. Slower but needs no algebra.
- Check the fitted formula on a value you did not use: y_{12} = 144 - 216 + 83 = 11, matching the table.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The values y_{10} to y_{14} are 3, 6, 11, 18, 27 with h = 1.
2. First differences: 3, 5, 7, 9. Second differences: 2, 2, 2 - constant, so y is a quadratic in x.
3. For $y = ax^2 + bx + c$ with h = 1, the second difference is 2a, so 2a = 2 and a = 1.
4. From y_{10} = 3: 100 + 10b + c = 3. From y_{11} = 6: 121 + 11b + c = 6. Subtracting: 21 + b = 3, so b = -18.
5. Then 100 - 180 + c = 3 gives c = 83, so $y_x = x^2 - 18x + 83$ (check: y_{12} = 144 - 216 + 83 = 11).
6. Therefore y_4 = 16 - 72 + 83 = 27, option (d).

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9. **2022 · Q68** Newton-Gregory Forward Interpolation Formula · Numerical

QUESTION

For $f(20)=512$, $f(30)=439$, $f(40)=346$, $f(50)=243$, consider: 1. $f(x)$ can be approximated by a polynomial of degree 2. 2. $f(25)$ can be obtained as 477 approximately using Newton-Gregory forward difference interpolation.

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(d) Neither 1 nor 2

Source note. AI-derived key with a caveat. The third difference is Delta^3 = 10, which is non-zero, so the four points do NOT lie on a quadratic and statement 1 fails. Applying Newton-Gregory forward interpolation at x = 25 (p = 0.5) gives 512 - 36.5 + 2.5 + 0.625 = 478.63, not the 477 asserted in statement 2 (truncating after the second difference still gives 478.0). Both statements therefore fail as printed, giving (d). If the intended value in statement 2 were 478.6 the key would be (b).

EXAM SHORTCUT (AI-derived explanation)

Differences: -73, -93, -103; then -20, -10; then 10. A non-zero third difference rules out a quadratic, and Newton forward at p = 0.5 gives 478.63, not 477.

TIPS & TRICKS (AI-derived explanation)

- Degree test: a quadratic requires CONSTANT second differences. Here they are -20 and -10, so at least a cubic is needed.
- For p = 0.5 the factorial coefficients are p = 0.5, p(p-1)/2 = -0.125 and p(p-1)(p-2)/6 = +0.0625 - note the sign change at the third term.
- Assemble carefully: 512 - 36.5 = 475.5, then +2.5 = 478.0, then +0.625 = 478.625.
- The same data appear as 2020 Q74, where the statements concerned degree n < 2 and the choice of forward versus backward formula - a clear repeat of the underlying table.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The tabulated values at x = 20, 30, 40, 50 are 512, 439, 346, 243 with h = 10.
2. First differences: -73, -93, -103. Second differences: -20, -10. Third difference: 10.
3. Statement 1: a quadratic would require constant second differences, but they are -20 and -10, and the third difference is non-zero. A cubic is needed, so the claim of degree 2 is FALSE.
4. Statement 2: for x = 25, p = (25 - 20)/10 = 0.5. Newton's forward formula gives f(25) = 512 + 0.5(-73) + [0.5(-0.5)/2](-20) + [0.5(-0.5)(-1.5)/6](10) = 512 - 36.5 + 2.5 + 0.625 = 478.63. Even truncating after the second-difference term gives 478.0, so the asserted value 477 is not obtained. FALSE.
5. Neither statement holds, option (d).

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10. 2023 · Q46 Interpolation & Extrapolation · Theoretical

QUESTION

In case of equi-spaced tabular data, the error for linear interpolation is bounded by one eighth of the:

OPTIONS

(a) first difference

(b) second difference

(c) third difference

(d) fourth difference

ANSWER (AI-derived — no official key exists for this paper)

(b) second difference

EXAM SHORTCUT (AI-derived explanation)

Linear interpolation truncates Newton's formula after the first difference, so the leading error term is [p(p-1)/2] Delta^2. Since |p(p-1)| <= 1/4 on [0,1], the bound is one eighth of the second difference.

TIPS & TRICKS (AI-derived explanation)

- The first neglected term is [p(p-1)/2!] times the SECOND difference, so the bound must involve Delta^2 - that alone identifies option (b).
- Maximise |p(p-1)| over 0 <= p <= 1: the maximum is 1/4 at p = 1/2. Dividing by 2! gives the familiar 1/8.
- The worst case is at the MIDPOINT between two tabulated values, which is why interpolation is most accurate near the nodes.
- Analogue for quadratic interpolation: the bound involves the third difference, with a different constant - the pattern is 'one order higher than the formula retains'.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Newton's forward formula is y_p = y0 + p Delta y0 + [p(p-1)/2!] Delta^2 y0 + ...
2. Linear interpolation keeps only the first two terms, so the leading error is the first neglected term, [p(p-1)/2] Delta^2 y0.
3. For interpolation between two consecutive tabulated points, p lies in [0,1].
4. The function |p(p-1)| attains its maximum on [0,1] at p = 1/2, where it equals 1/4.
5. Hence the error is bounded in modulus by (1/4)/2 = 1/8 times the second difference.
6. So the bound is one eighth of the second difference, option (b).

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11. 2023 · Q51 Newton-Gregory Forward Interpolation Formula · Numerical

COMMON DATA FOR THIS GROUP

Questions 51–52 are based on: $u_3=4.8,\ u_4=8.4,\ u_5=14.5,\ u_6=23.6,\ u_7=36.2,\ u_8=52.8,\ u_9=73.9$

QUESTION

What is the value of u_2 according to Newton-Gregory forward difference interpolation formula?

OPTIONS

(a) 3.1

(b) 3.2

(c) 3.3

(d) 3.4

ANSWER (AI-derived — no official key exists for this paper)

(b) 3.2

EXAM SHORTCUT (AI-derived explanation)

Third differences are constant at 0.5, so the data are cubic. Newton forward from u3 with p = -1 gives 4.8 - 3.6 + 2.5 - 0.5 = 3.2.

TIPS & TRICKS (AI-derived explanation)

- Extrapolating one step BACKWARDS means p = -1 in Newton's forward formula - the formula still works, with alternating-sign coefficients.
- For p = -1 the coefficients are p = -1, p(p-1)/2 = +1 and p(p-1)(p-2)/6 = -1, so the terms simply alternate in sign.
- Build the table first: Delta = 3.6, 6.1, 9.1, 12.6, 16.6, 21.1; Delta^2 = 2.5, 3.0, 3.5, 4.0, 4.5; Delta^3 = 0.5 throughout. The constant third difference confirms a cubic and truncates the series.
- Alternatively extend the difference table leftwards: the new second difference is 2.5 - 0.5 = 2.0 and the new first difference is 3.6 - 2.0 = 1.6, so u2 = 4.8 - 1.6 = 3.2 - a useful cross-check.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The values u3 to u9 are 4.8, 8.4, 14.5, 23.6, 36.2, 52.8, 73.9 with h = 1.
2. First differences: 3.6, 6.1, 9.1, 12.6, 16.6, 21.1. Second differences: 2.5, 3.0, 3.5, 4.0, 4.5. Third differences: 0.5, 0.5, 0.5, 0.5 - constant, so the data are cubic and fourth differences vanish.
3. Taking x0 = 3, the value x = 2 corresponds to p = (2 - 3)/1 = -1.
4. Newton's forward formula: u2 = 4.8 + (-1)(3.6) + [(-1)(-2)/2](2.5) + [(-1)(-2)(-3)/6](0.5) = 4.8 - 3.6 + (1)(2.5) + (-1)(0.5) = 4.8 - 3.6 + 2.5 - 0.5 = 3.2. (Check by extending the table backwards: new Delta^2 = 2.0, new Delta = 1.6, so u2 = 4.8 - 1.6 = 3.2.) Option (b) is correct.

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12. 2023 · Q52 Newton-Gregory Backward Interpolation Formula · Numerical

COMMON DATA FOR THIS GROUP

Questions 51–52 are based on: $u_3=4.8,\ u_4=8.4,\ u_5=14.5,\ u_6=23.6,\ u_7=36.2,\ u_8=52.8,\ u_9=73.9$

QUESTION

What is the approximate value of $u_{8.5}$ according to Newton-Gregory backward difference interpolation formula?

OPTIONS

(a) 62.75

(b) 64.75

(c) 66.75

(d) 68.75

ANSWER (AI-derived — no official key exists for this paper)

(a) 62.75

EXAM SHORTCUT (AI-derived explanation)

Backward differences at u9 are 21.1, 4.5, 0.5. With p = -0.5 the formula gives 73.9 - 10.55 - 0.5625 - 0.03125 = 62.76.

TIPS & TRICKS (AI-derived explanation)

- Newton BACKWARD formula: u = u_n + p nabla u_n + [p(p+1)/2!] nabla^2 u_n + [p(p+1)(p+2)/3!] nabla^3 u_n, with p = (x - x_n)/h negative.
- Note the PLUS signs inside the coefficients (p+1, p+2), in contrast to the minus signs of the forward formula - mixing them is the classic error.
- The backward differences at the last point are simply the last entries of each difference column: 21.1, 4.5 and 0.5.
- Bracket the answer: u8 = 52.8 and u9 = 73.9, so a value near the midpoint but below it (the function is convex) is expected - 62.76 fits, while options (c) and (d) are too high.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Take x_n = 9 with h = 1. The backward differences at u9 are nabla u9 = 73.9 - 52.8 = 21.1, nabla^2 u9 = 21.1 - 16.6 = 4.5 and nabla^3 u9 = 4.5 - 4.0 = 0.5, with higher differences zero.
2. For x = 8.5, p = (8.5 - 9)/1 = -0.5.
3. Newton's backward formula: u = u9 + p nabla u9 + [p(p+1)/2] nabla^2 u9 + [p(p+1)(p+2)/6] nabla^3 u9.
4. Term 1: (-0.5)(21.1) = -10.55. Term 2: [(-0.5)(0.5)/2](4.5) = (-0.125)(4.5) = -0.5625.
5. Term 3: [(-0.5)(0.5)(1.5)/6](0.5) = (-0.0625)(0.5) = -0.03125.
6. u(8.5) = 73.9 - 10.55 - 0.5625 - 0.03125 = 62.756, i.e. about 62.75, option (a).

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13. 2024 · Q74 Error Terms in Interpolation Formulae · Numerical

QUESTION

Which one of the following is the best possible choice for the step size h to tabulate $f(x) = \sin\ x$ on $[0, \pi/4]$ that results in truncation error for the linear interpolation at most 10^{-5}?

OPTIONS

(a) $\pi/520$

(b) $\pi/320$

(c) $\pi/120$

(d) $\pi/92$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\pi/320$

EXAM SHORTCUT (AI-derived explanation)

Linear interpolation error <= h^2 M/8 with M = max|f''| = sin(pi/4) = 0.7071. Requiring h^2(0.7071)/8 <= 1e-5 gives h <= 0.01064, and pi/320 = 0.00982 is the largest listed value that satisfies it.

TIPS & TRICKS (AI-derived explanation)

- Error bound for linear interpolation: |E| <= h^2 M2/8 where M2 bounds |f''| on the interval. The 8 comes from max|p(p-1)|/2 = 1/8.
- For f = sin x, f'' = -sin x, whose modulus is largest at the right end of [0, pi/4], giving M2 = sin(pi/4) = 0.7071.
- Solve for h: h^2 <= 8(1e-5)/0.7071 = 1.131e-4, so h <= 0.01064. Then test the options: pi/320 = 0.00982 passes, pi/120 = 0.02618 fails.
- 'Best possible' means the LARGEST admissible step, since a smaller h wastes computation - pi/520 also satisfies the bound but is unnecessarily fine.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For linear interpolation on an interval of width h, the truncation error satisfies |E| <= h^2 M2/8, where M2 is a bound on |f''| over the interval.
2. Here f(x) = sin x, so f''(x) = -sin x and |f''| is largest at x = pi/4, giving M2 = sin(pi/4) = 0.7071.
3. Requiring h^2 (0.7071)/8 <= 10^(-5) gives h^2 <= 8 x 10^(-5)/0.7071 = 1.131 x 10^(-4).
4. Hence h <= 0.01064.
5. Testing the options: pi/520 = 0.00604 (satisfies), pi/320 = 0.00982 (satisfies), pi/120 = 0.02618 (fails), pi/92 = 0.03415 (fails).
6. The best choice is the largest admissible step, pi/320, option (b).

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14. 2024 · Q75 Newton-Gregory Backward Interpolation Formula · Numerical

QUESTION

The roots of the least degree polynomial satisfied by $(0,1)$, $(1/2,3/2)$ and $(1,3)$ are:

OPTIONS

(a) real and distinct

(b) complex with real part zero

(c) real and repeated

(d) complex with non-zero real part

ANSWER (AI-derived — no official key exists for this paper)

(b) complex with real part zero

EXAM SHORTCUT (AI-derived explanation)

The slopes 1 and 3 differ, so the least-degree polynomial is a quadratic; fitting gives y = $2x^2$ + 1, whose roots x = +-i/sqrt2 are purely imaginary.

TIPS & TRICKS (AI-derived explanation)

- Test linearity first by comparing slopes: (3/2 - 1)/(1/2) = 1 but (3 - 3/2)/(1/2) = 3. Different, so the degree is 2, not 1.
- Use the point (0,1) to fix the constant immediately: c = 1, leaving only two unknowns.
- Solving a + 2b = 2 with a + b = 2 gives b = 0 - the absence of a linear term is what makes the roots purely imaginary.
- y = $2x^2$ + 1 has no real roots since $2x^2$ + 1 >= 1 > 0; the roots are +-i/sqrt(2), complex with real part zero.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Test whether the three points are collinear: the slope from (0,1) to (1/2, 3/2) is 1, while from (1/2, 3/2) to (1,3) it is 3. They differ, so no line fits and the least degree is 2.
2. Write y = ax^2 + bx + c. From (0,1): c = 1.
3. From (1/2, 3/2): a/4 + b/2 + 1 = 3/2, i.e. a + 2b = 2.
4. From (1,3): a + b + 1 = 3, i.e. a + b = 2.
5. Subtracting gives b = 0 and hence a = 2, so y = $2x^2$ + 1.
6. Setting $2x^2$ + 1 = 0 gives x^2 = -1/2, so x = +- i/sqrt(2): complex conjugates with real part zero. Option (b) is correct.

N3 · NUMERICAL ANALYSIS

Divided Differences; Newton's & Lagrange's Formulae (Unequal Intervals)

33 questions · 2018: 4 2019: 4 2020: 6 2021: 7 2022: 2 2023: 1 2024: 2 2025: 3 2026: 4

Weight. 33 questions (18.3% of Numerical Analysis) · appeared in 9 of 9 papers · peak year 2021 (7)

Examiner's pattern. Second-largest numerical topic. The signature item is the n-th divided difference of a polynomial — the third divided difference of x^3 - 2x over the arguments 2, 4, 9, 10 was asked identically in 2019 and 2026. Also constant: the divided difference of 1/x, symmetry in the arguments, Lagrange with unequal nodes, and truncation-error bounds.

Must-know. For degree n, f[x0..xn] = leading coefficient a_n, and f[x0..x_(n+1)] = 0. For f(x) = 1/x, f[x0..xn] = (-1)^n/(x0 x1 ... xn). Divided differences are symmetric in their arguments.

1. 2018 · Q43 Newton's Divided Difference Formula · Theoretical

QUESTION

The nth divided difference of an nth-degree polynomial is

OPTIONS

(a) a variable

(b) a constant

(c) zero

(d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(b) a constant

EXAM SHORTCUT (AI-derived explanation)

The nth divided difference of a degree-n polynomial equals its leading coefficient - a fixed number. So it is a constant.

TIPS & TRICKS (AI-derived explanation)

- Two companion facts: the nth divided difference of a degree-n polynomial is the leading coefficient (a constant), and the (n+1)th and all higher divided differences are ZERO.
- Do not confuse 'constant' with 'zero'. Option (c) is correct only for divided differences of order HIGHER than n - the commonest slip in this item.
- Link to finite differences: for equally spaced points, the nth divided difference equals Delta^n f / (n! h^n), which for a degree-n polynomial is a_n, the leading coefficient.
- Divided differences are symmetric in their arguments, so the answer cannot depend on which points are used - another reason it must be a constant.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let f(x) = a_n x^n + a_(n-1) x^(n-1) + ... + a_0 be a polynomial of degree n, so a_n is not 0.

2. Divided differences are linear, and the kth divided difference of a polynomial of degree less than k is 0.
3. Hence the nth divided difference of f depends only on the leading term a_n x^n.
4. For x^n, the nth divided difference over any n+1 distinct nodes equals 1 (this is the standard result, e.g. from the Newton form of the interpolating polynomial).
5. Therefore the nth divided difference of f is a_n, a fixed non-zero number independent of the nodes - i.e. a constant.
6. Hence option (b) is correct.

---

2. 2018 · Q51 Lagrange's Formula (Unequal Intervals) · Numerical

QUESTION

Table:

x	2	3	α	5
$f(x)$	4	5	16	25

α estimated by Lagrange's interpolation formula satisfies

OPTIONS

(a) $3.5<\alpha<4$

(b) $\alpha=4$

(c) $4<\alpha<4.2$

(d) $4.2<\alpha<4.5$

ANSWER (AI-derived — no official key exists for this paper)

(d) $4.2<\alpha<4.5$

EXAM SHORTCUT (AI-derived explanation)

Fit the quadratic through (2,4), (3,5), (5,25): P(x) = $3x^2$ - 14x + 20. Solve P(x) = 16, i.e. $3x^2$ - 14x + 4 = 0, whose root in range is x = (14 + sqrt(148))/6 = 4.36.

TIPS & TRICKS (AI-derived explanation)

- Three known pairs determine a quadratic; the fourth row is the unknown ABSCISSA, so build the polynomial from the three complete pairs and then solve for x.
- Do not attempt inverse Lagrange in the f-variable here - it uses the same three points but produces a different (and wrong) cubic-free estimate. Solving P(x) = 16 is the intended route.
- Discriminant shortcut: 196 - 48 = 148, and sqrt(148) is just over 12.1, so the larger root is a little over (14+12.1)/6 = 4.35 - enough to choose band (d) without full precision.
- The smaller root 0.31 lies outside the tabulated range 2 to 5 and is discarded; always check which root is admissible.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Three complete data pairs are available: (2, 4), (3, 5) and (5, 25). Fit the Lagrange quadratic through them.
2. P(x) = 4 (x-3)(x-5)/[(2-3)(2-5)] + 5 (x-2)(x-5)/[(3-2)(3-5)] + 25 (x-2)(x-3)/[(5-2)(5-3)] = (4/3)(x^2 - 8x + 15) - (5/2)(x^2 - 7x + 10) + (25/6)(x^2 - 5x + 6).
3. Collecting terms: coefficient of x^2 is 4/3 - 5/2 + 25/6 = 3; coefficient of x is -32/3 + 35/2 - 125/6 = -14; constant is 20 - 25 + 25 = 20. So P(x) = $3x^2$ - 14x + 20.
4. Set P(alpha) = 16: 3 alpha^2 - 14 alpha + 4 = 0, so alpha = [14 +- sqrt(196 - 48)]/6 = [14 +- sqrt(148)]/6.
5. sqrt(148) = 12.166, giving alpha = 4.361 (the other root 0.306 lies outside the table). Since 4.2 < 4.361 < 4.5, option (d) is correct.

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3. 2018 · Q53 Newton's Divided Difference Formula · Numerical

QUESTION

If $f(x)=1/x$, the third divided difference $f[a,b,c,d]$ is

OPTIONS

(a) $\frac{1}{abcd}$

(b) $-\frac{1}{abcd}$

(c) $\frac{a-d}{ad}$

(d) $\frac{1}{ad}-\frac{1}{bc}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $-\frac{1}{abcd}$

EXAM SHORTCUT (AI-derived explanation)

For f(x) = 1/x the nth divided difference is $(-1)^n$/(x0 x1 ... xn). With n = 3 that is -1/(abcd).

TIPS & TRICKS (AI-derived explanation)

- Memorise the reciprocal rule: f[x0, x1, ..., xn] = $(-1)^n$/(x0 x1 ... xn) when f(x) = 1/x. It converts an otherwise tedious recursion into one line.
- Sign check: the order 3 is ODD, so the answer must be negative - options (a) is eliminated at once.
- Verify the first step yourself: f[a,b] = (1/a - 1/b)/(a - b) = [(b-a)/ab]/(a-b) = -1/(ab), which sets the alternating pattern.
- Divided differences are symmetric functions of their nodes, so the answer must be symmetric in a, b, c, d. Options (c) and (d) are not symmetric and can be struck out immediately.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. First divided difference: f[a, b] = (f(a) - f(b))/(a - b) = (1/a - 1/b)/(a - b) = [(b - a)/(ab)]/(a - b) = -1/(ab).

2. Second: f[a, b, c] = (f[a,b] - f[b,c])/(a - c) = (-1/(ab) + 1/(bc))/(a - c) = [(a - c)/(abc)]/(a - c) = 1/(abc).
3. Third: f[a, b, c, d] = (f[a,b,c] - f[b,c,d])/(a - d) = (1/(abc) - 1/(bcd))/(a - d) = [(d - a)/(abcd)]/(a - d) = -1/(abcd).
4. The pattern $(-1)^n$/(product of the nodes) is confirmed.
5. Hence the third divided difference is -1/(abcd), option (b).

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4. 2018 · Q54 Lagrange's Formula (Unequal Intervals) · Numerical

QUESTION

$f(n^2)=n^3$ holds for $n=1,2,3$. Then $f(3)$ from Lagrange's interpolation is

OPTIONS

(a) 4.9

(b) 5.1

(c) 5.3

(d) 5.5

ANSWER (AI-derived — no official key exists for this paper)

(c) 5.3

EXAM SHORTCUT (AI-derived explanation)

The data are (1,1), (4,8), (9,27). Lagrange at x = 3: weights 6/24, -12/-15, -2/40 give 0.25(1) + 0.8(8) - 0.05(27) = 0.25 + 6.4 - 1.35 = 5.3.

TIPS & TRICKS (AI-derived explanation)

- Translate the disguised statement first: $f(n^2) = n^3$ for n = 1, 2, 3 means f(1) = 1, f(4) = 8, f(9) = 27. Half the candidates lose the mark here, not in the interpolation.
- Lagrange weights must sum to 1: 0.25 + 0.8 - 0.05 = 1.00. Use this as an instant arithmetic check before multiplying by the f-values.
- The true underlying function is f(t) = t^(3/2), whose exact value at 3 is 5.196 - the quadratic interpolant overshoots slightly to 5.3, which is a good plausibility check.
- Keep the negative weight: the node 9 is far from 3, and its weight -0.05 is genuinely negative. Dropping the sign gives 8.0 and no matching option.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From $f(n^2) = n^3$ with n = 1, 2, 3 we obtain the data points (x, f) = (1, 1), (4, 8) and (9, 27).
2. Lagrange's formula at x = 3: f(3) = 1 . L0 + 8 . L1 + 27 . L2.
3. L0 = (3-4)(3-9)/[(1-4)(1-9)] = (-1)(-6)/[(-3)(-8)] = 6/24 = 0.25.
4. L1 = (3-1)(3-9)/[(4-1)(4-9)] = (2)(-6)/[(3)(-5)] = -12/-15 = 0.8.
5. L2 = (3-1)(3-4)/[(9-1)(9-4)] = (2)(-1)/[(8)(5)] = -2/40 = -0.05. (Check: 0.25 + 0.8 - 0.05 = 1.)
6. f(3) = 1(0.25) + 8(0.8) + 27(-0.05) = 0.25 + 6.4 - 1.35 = 5.3, option (c).

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5. 2019 · Q58 Newton's Divided Difference Formula · Conceptual

QUESTION

The third divided difference of the polynomial $2x^2+1$ over the points $0,1,3,6$ is

OPTIONS

(a) 0

(b) 1

(c) 2

(d) 4

ANSWER (AI-derived — no official key exists for this paper)

(a) 0

EXAM SHORTCUT (AI-derived explanation)

Divided differences of order higher than the degree of the polynomial are identically zero. Here the polynomial is quadratic and the difference is of order 3, so the answer is 0.

TIPS & TRICKS (AI-derived explanation)

- Two rules cover every such item: the nth divided difference of a degree-n polynomial is its leading coefficient (a constant), and any HIGHER-order divided difference is zero.
- Degree 2 versus order 3 - just compare the two numbers before doing any arithmetic.
- Option (c) 2 is the leading coefficient, i.e. the answer to the SECOND divided difference. Read the order in the question carefully.
- The nodes 0, 1, 3, 6 are deliberately unequal to make the problem look laborious; the result is independent of the nodes.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The polynomial $f(x) = 2x^2 + 1$ has degree 2.
2. The second divided difference of a quadratic equals its leading coefficient, here 2, and is the same for every choice of nodes.
3. Since the second divided difference is a constant, its further divided difference - the third divided difference - is zero.
4. In general the kth divided difference of a polynomial of degree n is identically zero whenever $k > n$.
5. Here $k = 3 > n = 2$, so the third divided difference over the points 0, 1, 3, 6 is 0.
6. Hence option (a) is correct.

---

6. 2019 · Q59 Lagrange's Formula (Unequal Intervals) · Numerical

QUESTION

A curve passes through $(0,18)$, $(1,10)$, $(3,-18)$, $(6,90)$. What is the slope of the curve at $x=2$ (using Lagrange's formula)?

OPTIONS

(a) -16

(b) -8

(c) 1

(d) 0

ANSWER (AI-derived — no official key exists for this paper)

(a) -16

EXAM SHORTCUT (AI-derived explanation)

Newton's divided differences give $f(x) = 2x^3 - 10x^2 + 18$, so $f'(x) = 6x^2 - 20x$ and $f'(2) = 24 - 40 = -16$.

TIPS & TRICKS (AI-derived explanation)

- Build the interpolating cubic explicitly - it is far safer than differentiating Lagrange basis polynomials one by one.
- Divided-difference ladder: 18, then -8, -14, 36; then -2, 10; then 2. The last entry, 2, is the leading coefficient of the cubic.
- Always verify the polynomial at all four data points before differentiating: $2x^3 - 10x^2 + 18$ gives 18, 10, -18, 90 exactly.
- The x-term vanishes here, which is a pleasant surprise worth checking twice; $f'(x) = 6x^2 - 20x$ then makes the evaluation at x = 2 trivial.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Data: (0,18), (1,10), (3,-18), (6,90). Form divided differences.
2. First order: f[0,1] = (10-18)/1 = -8; f[1,3] = (-18-10)/2 = -14; f[3,6] = (90+18)/3 = 36.
3. Second order: f[0,1,3] = (-14+8)/3 = -2; f[1,3,6] = (36+14)/5 = 10.
4. Third order: f[0,1,3,6] = (10+2)/6 = 2.
5. Newton's form: $f(x) = 18 - 8x - 2x(x-1) + 2x(x-1)(x-3) = 2x^3 - 10x^2 + 18$. (Check: f(0)=18, f(1)=10, f(3)=-18, f(6)=90.)
6. Differentiating, $f'(x) = 6x^2 - 20x$, so f'(2) = 24 - 40 = -16, option (a).

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7. 2019 · Q76 Lagrange's Formula (Unequal Intervals) · Numerical

QUESTION

$f(x)$ has values $f(1)=4$, $f(2)=5$, $f(7)=5$, $f(8)=4$. Then $f(x)$ attains

OPTIONS

(a) maximum value at $x=4.5$

(b) minimum value at $x=4.5$

(c) neither maximum nor minimum for any x

(d) minimum value at x=-4.5

ANSWER (AI-derived — no official key exists for this paper)

(a) maximum value at x=4.5

EXAM SHORTCUT (AI-derived explanation)

The data are symmetric about x = 4.5 (f(1) = f(8) = 4 and f(2) = f(7) = 5), and the values RISE towards the centre, so the interpolating curve has a maximum at x = 4.5.

TIPS & TRICKS (AI-derived explanation)

- Spot the symmetry before computing: matching values at mirror-image abscissae force the interpolating polynomial to be symmetric about the midpoint (1+8)/2 = 4.5.
- In the shifted variable u = x - 4.5 the polynomial is even, so it reduces to a + bu^2. Fitting a + 12.25b = 4 and a + 6.25b = 5 gives b = -1/6 < 0, hence a maximum.
- Direction check without algebra: moving inward from x = 1 to x = 2 the value RISES from 4 to 5, and it falls again from x = 7 to x = 8. Rise then fall means a maximum.
- Option (d) places the turning point at x = -4.5, outside the data range entirely - a sign trap for candidates who mishandle the shift.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The data are f(1) = 4, f(2) = 5, f(7) = 5, f(8) = 4.
2. The abscissae are symmetric about x = (1 + 8)/2 = 4.5, and the paired values are equal: f(1) = f(8) and f(2) = f(7).
3. Hence the interpolating polynomial is symmetric about x = 4.5, so in the variable u = x - 4.5 it is an even function; with four data points it takes the form f = a + b u^2.
4. At u = +-3.5: a + 12.25b = 4. At u = +-2.5: a + 6.25b = 5. Subtracting gives 6b = -1, so b = -1/6 and a = 5 + 6.25/6 = 6.042.
5. Since b < 0, the parabola opens downwards and the turning point at u = 0 is a MAXIMUM.
6. Therefore f attains a maximum at x = 4.5, option (a).

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8. 2019 · Q80 Newton's Divided Difference Formula · Conceptual

QUESTION

The third divided difference of $f(x)=x^3$-$2x$ with arguments $2,4,9,10$ is equal to

OPTIONS

(a) 23

(b) 21

(c) 15

(d) 1

ANSWER **(AI-derived — no official key exists for this paper)**

(d) 1

EXAM SHORTCUT **(AI-derived explanation)**

The third divided difference of a cubic equals its leading coefficient. For $f(x) = x^3 - 2x$ that coefficient is 1, whatever the nodes.

TIPS & TRICKS **(AI-derived explanation)**

- Rule: the nth divided difference of a degree-n polynomial is its LEADING COEFFICIENT, independent of the arguments chosen.
- The awkward nodes 2, 4, 9, 10 are pure decoration - the answer would be 1 for any four distinct arguments.
- The -2x term contributes nothing: divided differences of order 3 annihilate every term of degree below 3.
- Contrast with the reciprocal case f(x) = 1/x, whose nth divided difference is $(-1)^n$/(product of nodes) and genuinely depends on the nodes.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Divided differences are linear, and the kth divided difference of any polynomial of degree less than k is zero.
2. Here $f(x) = x^3 - 2x$, so the term -2x (degree 1) contributes nothing to a third divided difference.
3. For x^3, the third divided difference over any four distinct nodes equals the leading coefficient, namely 1.
4. This follows from the Newton form of the interpolating polynomial: the coefficient of the highest-degree term is exactly the top divided difference.
5. Hence the third divided difference over 2, 4, 9, 10 is 1.
6. Option (d) is correct.

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9. 2020 · Q61 Lagrange's Formula (Unequal Intervals) · Conceptual

QUESTION

If a new interpolation value is later inserted, which method handles this without recalculating the whole formula?

OPTIONS

(a) Lagrange's formula

(b) Newton's forward formula

(c) Newton's divided difference formula

(d) All methods are the same

ANSWER **(AI-derived — no official key exists for this paper)**

(c) Newton's divided difference formula

EXAM SHORTCUT **(AI-derived explanation)**

Newton's divided-difference form is built term by term, so a new data point simply appends ONE extra term; every earlier coefficient stays valid.

TIPS & TRICKS **(AI-derived explanation)**

- Newton's form is 'incremental': f(x) = f[x0] + (x-x0)f[x0,x1] + (x-x0)(x-x1)f[x0,x1,x2] + ... Adding a node adds one more product term and one more divided difference.
- Lagrange's form must be rebuilt completely because EVERY basis polynomial contains every node in its numerator and denominator.
- Newton's forward formula requires equally spaced nodes, so an arbitrary new point usually cannot be accommodated at all.
- Practical consequence: Newton's divided differences are preferred when data arrive sequentially or when you want to monitor convergence by adding points one at a time.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Newton's divided-difference interpolation polynomial has the form f(x) = f[x0] + (x-x0)f[x0,x1] + (x-x0)(x-x1)f[x0,x1,x2] + ...
2. Each coefficient depends only on the nodes already used, so inserting a new node x_(n+1) merely appends one further term (x-x0)(x-x1)...(x-x_n) f[x0,...,x_(n+1)].
3. All previously computed divided differences remain valid and are reused.
4. Lagrange's formula, by contrast, defines each basis polynomial as a product over ALL nodes, so adding one node changes every basis polynomial and the whole computation must be repeated.
5. Newton's forward formula additionally requires equally spaced arguments, so an arbitrary new value cannot generally be inserted.
6. Hence Newton's divided difference formula is the method that accommodates a new value, option (c).

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10. **2020 · Q65** Newton's Divided Difference Formula · Theoretical

QUESTION

Consider the following statements: 1. Divided differences are symmetric in their arguments. 2. The nth divided differences of an nth-degree polynomial are constant. 3. The divided-difference operator is linear.

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

Symmetry, the constancy of the nth divided difference of a degree-n polynomial, and linearity are the three standard properties of divided differences - all correct.

TIPS & TRICKS (AI-derived explanation)

- Symmetry: f[x0, x1, ..., xn] is unchanged by any permutation of its arguments, because it equals a symmetric sum of f(xi) divided by products of node differences.
- Constancy: the nth divided difference of a degree-n polynomial is its LEADING COEFFICIENT, and the (n+1)th is zero.
- Linearity: f[.] + g[.] = (f+g)[.] and (cf)[.] = c f[.], which is what allows term-by-term treatment of polynomials.
- When three bookwork properties of the same object are listed and each is standard, 'all of them' is almost always the key.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: the divided difference f[x0, x1, ..., xn] can be written as a sum of f(xi) divided by products of differences of the nodes, an expression unchanged by permuting the arguments. Divided differences are therefore symmetric. TRUE.
2. Statement 2: in Newton's form the coefficient of the highest-degree term is the top divided difference, so for a polynomial of degree n the nth divided difference equals its leading coefficient - a constant independent of the nodes. TRUE.
3. Statement 3: the divided difference is built from linear operations on the function values, so f[.] + g[.] = (f + g)[.] and (cf)[.] = c f[.]. The operator is linear. TRUE.
4. All three statements are standard properties.
5. Hence option (d) is correct.

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11. 2020 · Q69 Lagrange's Formula (Unequal Intervals) · Numerical

QUESTION

Data:

x	1	6	9	11
$f(x)$	12	13	14	16

The approximate value of x for which $f(x)=15$ is

OPTIONS

(a) 9.5

(b) 10

(c) 10.5

(d) 10.9

ANSWER (AI-derived — no official key exists for this paper)

(c) 10.5

EXAM SHORTCUT (AI-derived explanation)

Apply Lagrange's formula with the roles of x and f interchanged (inverse interpolation) at f = 15: the weights 0.25, -1, 1.5, 0.25 applied to x = 1, 6, 9, 11 give 10.5.

TIPS & TRICKS (AI-derived explanation)

- Inverse interpolation: treat f as the independent variable and x as the dependent one, then use ordinary Lagrange. No iteration is needed when the f-values are distinct.
- The Lagrange weights must sum to 1: 0.25 - 1 + 1.5 + 0.25 = 1. Use that as an instant check before multiplying by the x-values.
- Negative weights are normal and expected here because f = 15 lies outside the middle of the f-range; do not discard them.
- Plausibility: f = 15 lies between f(9) = 14 and f(11) = 16, so x must lie between 9 and 11 - and 10.5 does.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. We need the value of x at which f(x) = 15, so we interpolate x as a function of f using the pairs (12,1), (13,6), (14,9), (16,11).
2. Lagrange weight for f = 12: (15-13)(15-14)(15-16)/[(12-13)(12-14)(12-16)] = (2)(1)(-1)/[(-1)(-2)(-4)] = -2/-8 = 0.25.
3. Weight for f = 13: (15-12)(15-14)(15-16)/[(13-12)(13-14)(13-16)] = (3)(1)(-1)/[(1)(-1)(-3)] = -3/3 = -1.
4. Weight for f = 14: (15-12)(15-13)(15-16)/[(14-12)(14-13)(14-16)] = (3)(2)(-1)/[(2)(1)(-2)] = -6/-4 = 1.5. Weight for f = 16: (3)(2)(1)/[(4)(3)(2)] = 6/24 = 0.25. (Sum = 1, as required.)
5. x = 1(0.25) + 6(-1) + 9(1.5) + 11(0.25) = 0.25 - 6 + 13.5 + 2.75 = 10.5.
6. Hence x is approximately 10.5, option (c).

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12. 2020 · Q71 Newton's Divided Difference Formula · Numerical

QUESTION

The polynomial of lowest degree (via Newton's divided difference formula) taking values 3,12,15,-21 at x=3,2,1,-1 respectively is

OPTIONS

(a) $27x^3+2x^2-4x$

(b) $x^3+2x^2-5x-27$

(c) $x^3-9x^2+17x+6$

(d) $x^3+2x^2-5x+17$

ANSWER (AI-derived — no official key exists for this paper)

(c) x^3-$9x^2$+$17x$+6

EXAM SHORTCUT (AI-derived explanation)

Substitution beats construction here: x^3 - $9x^2$ + 17x + 6 gives 3, 12, 15 and -21 at x = 3, 2, 1 and -1 respectively. Test the options rather than building the polynomial.

TIPS & TRICKS (AI-derived explanation)

- With four options and four data points, VERIFYING is far faster than constructing - start with the easiest node, x = 1, where each option reduces to the sum of its coefficients.
- At x = 1 the required value is 15: option (c) gives 1 - 9 + 17 + 6 = 15, while option (b) gives 1 + 2 - 5 - 27 = -29 and option (d) gives 15 too, so a second node is needed.
- Use x = -1 to separate the survivors: option (c) gives -1 - 9 - 17 + 6 = -21 (correct) whereas option (d) gives -1 + 2 + 5 + 17 = 23.
- Option (a) has no constant term and would give 0 at x = 0, and its leading coefficient 27 is implausible for data of this size.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Four data points determine a unique polynomial of degree at most 3, so we test the cubic options against all four values.
2. Option (c) is p(x) = x^3 - $9x^2$ + 17x + 6.
3. At x = 3: 27 - 81 + 51 + 6 = 3, matching the required value 3.
4. At x = 2: 8 - 36 + 34 + 6 = 12, matching 12.
5. At x = 1: 1 - 9 + 17 + 6 = 15, matching 15. At x = -1: -1 - 9 - 17 + 6 = -21, matching -21.
6. All four conditions are satisfied, and the interpolating polynomial is unique, so option (c) is correct.

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13. 2020 · Q72 Newton's Divided Difference Formula · Numerical

QUESTION

Given the divided differences $f[4]=48,\ f[4,5]=52,\ f[4,5,7]=15,\ f[4,5,7,10]=1$, what is $f(8)$?

OPTIONS

(a) 148

(b) 248

(c) 348

(d) 448

ANSWER (AI-derived — no official key exists for this paper)

(d) 448

EXAM SHORTCUT (AI-derived explanation)

Newton's divided-difference form at x = 8: 48 + (4)(52) + (4)(3)(15) + (4)(3)(1)(1) = 48 + 208 + 180 + 12 = 448.

TIPS & TRICKS (AI-derived explanation)

- Newton's form: f(x) = f[x0] + (x-x0)f[x0,x1] + (x-x0)(x-x1)f[x0,x1,x2] + (x-x0)(x-x1)(x-x2)f[x0,x1,x2,x3]. The nodes enter in the order 4, 5, 7, 10.
- Build the products cumulatively: (8-4) = 4, then x(8-5) = 12, then x(8-7) = 12. Each new factor multiplies the previous product.
- Note that the LAST node 10 never appears in the products - with four divided differences only three factors are needed.
- Running total: 48, 256, 436, 448. Keeping a running sum makes an arithmetic slip easy to spot.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Newton's divided-difference interpolation with nodes 4, 5, 7, 10 is f(x) = f[4] + (x-4) f[4,5] + (x-4)(x-5) f[4,5,7] + (x-4)(x-5)(x-7) f[4,5,7,10].
2. At x = 8 the products are (8-4) = 4, (8-4)(8-5) = 4 x 3 = 12, and (8-4)(8-5)(8-7) = 12 x 1 = 12.
3. First term: 48. Second term: 4 x 52 = 208.
4. Third term: 12 x 15 = 180. Fourth term: 12 x 1 = 12.
5. Adding: 48 + 208 + 180 + 12 = 448.
6. Hence f(8) = 448, option (d).

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14. 2020 · Q78 Newton's Divided Difference Formula · Numerical

QUESTION

The unique degree-2 polynomial $p(x)$ with $p(1)=1,\ p(3)=27,\ p(4)=64$, obtained via Newton's divided difference formula, is

OPTIONS

(a) $8x^2$-$12x$+19

(b) $8x^2$-$19x$+12

(c) $12x^2$-$8x$-19

(d) $12x^2$-$19x$+8

ANSWER (AI-derived — no official key exists for this paper)

(b) $8x^2$-$19x$+12

EXAM SHORTCUT (AI-derived explanation)

Divided differences: f[1,3] = 13, f[3,4] = 37, f[1,3,4] = 8. Newton's form gives 1 + 13(x-1) + 8(x-1)(x-3) = $8x^2$ - 19x + 12.

TIPS & TRICKS (AI-derived explanation)

- Build the divided-difference table first: 26/2 = 13 and 37/1 = 37, then (37 - 13)/(4 - 1) = 8. Three divisions and the polynomial is determined.
- Expand carefully: 8(x-1)(x-3) = $8x^2$ - 32x + 24, and adding 13x - 12 gives $8x^2$ - 19x + 12.
- Faster still: test the options at x = 1, where the value must be 1. Option (b) gives 8 - 19 + 12 = 1, while option (a) gives 15 - a quick discriminator.
- Note the underlying data are cubes (1, 27, 64 at 1, 3, 4), so the quadratic interpolant is only an approximation to x^3 - a nice reminder that interpolation reproduces the DATA, not the generating function.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The data are (1, 1), (3, 27), (4, 64).
2. First divided differences: f[1,3] = (27 - 1)/(3 - 1) = 13 and f[3,4] = (64 - 27)/(4 - 3) = 37.
3. Second divided difference: f[1,3,4] = (37 - 13)/(4 - 1) = 24/3 = 8.
4. Newton's form: p(x) = f[1] + (x-1)f[1,3] + (x-1)(x-3)f[1,3,4] = 1 + 13(x-1) + 8(x-1)(x-3).
5. Expanding: 1 + 13x - 13 + 8(x^2 - 4x + 3) = 13x - 12 + $8x^2$ - 32x + 24 = $8x^2$ - 19x + 12.
6. Check: p(1) = 1, p(3) = 72 - 57 + 12 = 27, p(4) = 128 - 76 + 12 = 64. Option (b) is correct.

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15. 2021 · Q41 Newton's Divided Difference Formula · Conceptual

QUESTION

Which statement is correct about Newton's divided difference formula?

OPTIONS

(a) It is a special case of Newton's forward difference formula.

(b) It is inapplicable when arguments are uniformly spaced.

(c) It applies whether or not the arguments are uniformly spaced.

(d) The roles of argument and entry can be interchanged in this formula.

ANSWER (AI-derived — no official key exists for this paper)

(c) It applies whether or not the arguments are uniformly spaced.

EXAM SHORTCUT (AI-derived explanation)

Divided differences are defined by dividing by the actual differences of the arguments, so they work for any distinct nodes - equally spaced or not.

TIPS & TRICKS (AI-derived explanation)

- Newton's FORWARD formula is the special case of the divided-difference formula when the nodes are equally spaced, not the other way round - so option (a) reverses the containment.
- Option (b) is the exact opposite of the truth: uniform spacing is a permitted special case, not a prohibition.

- Option (d) describes INVERSE interpolation, where x and f are deliberately interchanged; that is a separate technique, not a property of the formula itself.
- For equally spaced nodes the link is f[x0,...,xn] = Delta^{n} y0/(n! h^{n}) - worth memorising as the bridge between the two families.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Newton's divided-difference formula is f(x) = f[x0] + (x-x0)f[x0,x1] + (x-x0)(x-x1)f[x0,x1,x2] + ...
2. Each divided difference is formed by dividing by the actual difference of the arguments, e.g. f[x0,x1] = (f(x1)-f(x0))/(x1-x0), so no assumption of equal spacing is made.
3. It therefore applies to arbitrary distinct nodes, uniformly spaced or not.
4. When the nodes happen to be equally spaced with step h, the formula reduces to Newton's forward difference formula via f[x0,...,xn] = Delta^{n} y0/(n! h^{n}) - so the forward formula is the special case, not the general one.
5. Interchanging the roles of argument and entry is the technique of inverse interpolation, a different procedure.
6. Hence the correct statement is that it applies whether or not the arguments are uniformly spaced, option (c).

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16. 2021 · Q44 Lagrange's Formula (Unequal Intervals) · Numerical

QUESTION

Data: x=0,1,2,5; $f(x)$=2,3,12,147. Degree of the Lagrange interpolation polynomial representing this data?

OPTIONS

(a) 4

(b) 3

(c) 2

(d) 1

ANSWER (AI-derived — no official key exists for this paper)

(b) 3

EXAM SHORTCUT (AI-derived explanation)

Four data points give degree at most 3; the third divided difference f[0,1,2,5] = 1 is non-zero, so the degree is exactly 3.

TIPS & TRICKS (AI-derived explanation)

- n + 1 points determine a polynomial of degree at most n, so four points can never give degree 4 - option (a) is impossible on principle.
- Compute the top divided difference to confirm the degree is not lower: a non-zero value means the leading coefficient is non-zero.

- Divided-difference ladder here: 1, 9, 45; then 4, 9; then 1. The final 1 is the leading coefficient of the cubic.
- Quick check: the data 2, 3, 12, 147 at 0, 1, 2, 5 fit $f(x) = x^3 + \dots$ - the rapid growth to 147 at x = 5 already suggests a cubic rather than a quadratic.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With four data points (0,2), (1,3), (2,12), (5,147), the interpolating polynomial has degree at most 3.
2. First divided differences: f[0,1] = (3-2)/1 = 1; f[1,2] = (12-3)/1 = 9; f[2,5] = (147-12)/3 = 45.
3. Second divided differences: f[0,1,2] = (9-1)/2 = 4; f[1,2,5] = (45-9)/4 = 9.
4. Third divided difference: f[0,1,2,5] = (9-4)/5 = 1.
5. Since this leading divided difference is non-zero, the polynomial genuinely has degree 3 and does not collapse to a quadratic.
6. Hence the degree is 3, option (b).

---

17. 2021 · Q49 Newton's Divided Difference Formula · Numerical

QUESTION

The second divided difference of $f(x)=1/x$ with arguments a,b,c is

OPTIONS

(a) $f[a,b,c]=abc$

(b) $f[a,b,c]=\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$

(c) $f[a,b,c]=\frac{1}{a+b+c}$

(d) $f[a,b,c]=\frac{1}{abc}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $f[a,b,c]=\frac{1}{abc}$

EXAM SHORTCUT (AI-derived explanation)

For f(x) = 1/x the nth divided difference is $(-1)^n$/(product of the nodes). With n = 2 the sign is positive, giving 1/(abc).

TIPS & TRICKS (AI-derived explanation)

- Reciprocal rule: f[x0, x1, ..., xn] = $(-1)^n$/(x0 x1 ... xn) when f(x) = 1/x. One formula covers every order.
- Order 2 is EVEN, so the sign is +. Compare with 2018 Q53, where order 3 gave the negative -1/(abcd).
- Symmetry test: a divided difference must be symmetric in its arguments, which 1/(abc) is. Options (b) and (c) are symmetric too, so use the derivation, not just symmetry.
- Derive the first step yourself: f[a,b] = (1/a - 1/b)/(a-b) = -1/(ab), then f[a,b,c] = (-1/(ab) + 1/(bc))/(a-c) = 1/(abc).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. First divided difference: f[a,b] = (1/a - 1/b)/(a - b) = [(b - a)/(ab)]/(a - b) = -1/(ab).
2. Similarly f[b,c] = -1/(bc).
3. Second divided difference: f[a,b,c] = (f[a,b] - f[b,c])/(a - c) = (-1/(ab) + 1/(bc))/(a - c).
4. Combine the numerator over the common denominator abc: (-c + a)/(abc) = (a - c)/(abc).
5. Dividing by (a - c) gives f[a,b,c] = 1/(abc).
6. This matches the general rule $(-1)^n$/(product of nodes) with n = 2. Option (d) is correct.

---

18. 2021 · Q61 Newton's Divided Difference Formula · Numerical

QUESTION

Given

$$\det A = \begin{vmatrix} f(1) & f(2) & f(5) \\ 1 & 2 & 5 \\ 1 & 1 & 1 \end{vmatrix} = 60.$$

What is the second divided difference $f[1,2,5]$?

OPTIONS

(a) -5

(b) $-1/5$

(c) 5

(d) $1/5$

ANSWER (AI-derived — no official key exists for this paper)

(a) -5

EXAM SHORTCUT (AI-derived explanation)

Expanding the determinant along the top row gives -3f(1) + 4f(2) - f(5) = 60, while f[1,2,5] = (3f(1) - 4f(2) + f(5))/12 = -60/12 = -5.

TIPS & TRICKS (AI-derived explanation)

- Write the second divided difference in its symmetric Lagrange form: f[a,b,c] = f(a)/((a-b)(a-c)) + f(b)/((b-a)(b-c)) + f(c)/((c-a)(c-b)). Here the denominators are 4, -3 and 12.
- Put the three terms over the common denominator 12: f[1,2,5] = (3f(1) - 4f(2) + f(5))/12 - exactly the negative of the determinant expansion.
- The Vandermonde-style determinant with rows (f-values), (x-values), (1s) is a standard disguised way of presenting divided differences; recognising it saves all the algebra.
- Sign care: the determinant expansion gives -3f(1) + 4f(2) - f(5), which is MINUS the numerator of the divided difference - the single place this question can go wrong.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The second divided difference over nodes 1, 2, 5 is f[1,2,5] = f(1)/((1-2)(1-5)) + f(2)/((2-1)(2-5)) + f(5)/((5-1)(5-2)).
2. The denominators are (-1)(-4) = 4, (1)(-3) = -3 and (4)(3) = 12, so f[1,2,5] = f(1)/4 - f(2)/3 + f(5)/12 = (3f(1) - 4f(2) + f(5))/12.
3. Expand the determinant along its first row: det A = f(1)(2 - 5) - f(2)(1 - 5) + f(5)(1 - 2) = -3f(1) + 4f(2) - f(5).
4. We are told this equals 60, so 3f(1) - 4f(2) + f(5) = -60.
5. Therefore f[1,2,5] = -60/12 = -5.
6. Option (a) is correct.

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19. 2021 · Q65 Lagrange's Formula (Unequal Intervals) · Conceptual

QUESTION

Given $f(1)=0.98,\ f(2)=1.5,\ f(3)=4.7,\ f(5)=9.9,\ f(2.9)$ can be obtained using: 1. Lagrange's interpolation formula 2. Newton's divided difference formula

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(c) Both 1 and 2

EXAM SHORTCUT (AI-derived explanation)

The arguments 1, 2, 3, 5 are unequally spaced, and BOTH Lagrange's formula and Newton's divided-difference formula are designed for exactly that case.

TIPS & TRICKS (AI-derived explanation)

- Only the Newton-Gregory forward and backward formulas require equal spacing; Lagrange and divided differences never do.
- The two methods give the SAME interpolating polynomial - they are different algebraic arrangements of the unique polynomial through the given points.
- Divided differences are preferred in practice because new points can be added incrementally; Lagrange is preferred for theoretical work and error analysis.
- x = 2.9 lies inside the data range 1 to 5, so this is genuine interpolation, and both methods are accurate there.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The tabulated arguments are x = 1, 2, 3, 5, which are NOT equally spaced (the last gap is 2 while the others are 1).

2. Lagrange's interpolation formula constructs the interpolating polynomial as a weighted sum of the function values with weights built from products of (x - x_i), and it makes no assumption about spacing. Applicable.
3. Newton's divided difference formula divides by the actual differences of the arguments and is likewise valid for arbitrary distinct nodes. Applicable.
4. Both formulas produce the same unique polynomial of degree at most 3 through the four points, so either may be used to estimate f(2.9).
5. Only the Newton-Gregory forward and backward formulas, which are not offered here, would require equal spacing.
6. Hence both methods can be used, option (c).

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20. 2021 · Q66 Lagrange's Formula (Unequal Intervals) · Numerical

QUESTION

Data: x=1,2,4; $f(x)$=2,17,257. An approximation to $f(3)$ is

OPTIONS

(a) 92

(b) 102

(c) 165

(d) 182

ANSWER (AI-derived — no official key exists for this paper)

(b) 102

EXAM SHORTCUT (AI-derived explanation)

Divided differences: f[1,2] = 15, f[2,4] = 120, f[1,2,4] = 35. Then f(3) = 2 + 2(15) + 2(1)(35) = 2 + 30 + 70 = 102.

TIPS & TRICKS (AI-derived explanation)

- Newton's form: f(x) = f[1] + (x-1)f[1,2] + (x-1)(x-2)f[1,2,4]. Build the products (3-1) = 2 and (3-1)(3-2) = 2 before substituting.
- Note the unequal spacing: f[2,4] divides by 4 - 2 = 2, not by 1. Forgetting the actual gap is the classic error here.
- The data are 2, 17, 257 at 1, 2, 4 - close to 4^x - 2 or similar rapid growth, so a value near 100 at x = 3 is plausible while option (c) 165 and (d) 182 are too large for a quadratic fit.
- Only three points are given, so the interpolant is a QUADRATIC; do not attempt a cubic term.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The data are (1,2), (2,17), (4,257).
2. First divided differences: f[1,2] = (17 - 2)/(2 - 1) = 15 and f[2,4] = (257 - 17)/(4 - 2) = 240/2 = 120.
3. Second divided difference: f[1,2,4] = (120 - 15)/(4 - 1) = 105/3 = 35.
4. Newton's divided difference formula: f(x) = 2 + (x - 1)(15) + (x - 1)(x - 2)(35).

5. At x = 3: f(3) = 2 + (2)(15) + (2)(1)(35) = 2 + 30 + 70 = 102.
6. Option (b) is correct.

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21. 2021 · Q69 Newton's Divided Difference Formula · Numerical

QUESTION

Data: x=0,1,2,4,5,6; $f(x)$=1,14,15,5,6,19. Using Newton's divided difference formula, $f(3)$ is

OPTIONS

(a) 1

(b) 5

(c) 10

(d) 15

ANSWER (AI-derived — no official key exists for this paper)

(c) 10

EXAM SHORTCUT (AI-derived explanation)

The divided-difference table terminates: all third divided differences equal 1 and the fourth are 0, so f is the cubic 1 + 13x - 6x(x-1) + x(x-1)(x-2). At x = 3 this gives 1 + 39 - 36 + 6 = 10.

TIPS & TRICKS (AI-derived explanation)

- Build the whole divided-difference table before interpolating - spotting that the third differences are constant (all 1) tells you the data are exactly cubic and truncates the work.
- Leading diagonal: f[0] = 1, f[0,1] = 13, f[0,1,2] = -6, f[0,1,2,4] = 1. These four numbers are all the Newton form needs.
- Mind the unequal gaps: f[2,4] divides by 2 and f[1,2,4] divides by 3. Using 1 throughout is the standard error.
- Evaluate the products at x = 3 first: (3-0) = 3, (3-0)(3-1) = 6, (3-0)(3-1)(3-2) = 6, then multiply by 13, -6 and 1 respectively.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Data: (0,1), (1,14), (2,15), (4,5), (5,6), (6,19).
2. First divided differences: 13, 1, -5, 1, 13. Second: (1-13)/2 = -6, (-5-1)/3 = -2, (1+5)/3 = 2, (13-1)/2 = 6.
3. Third: (-2+6)/4 = 1, (2+2)/4 = 1, (6-2)/4 = 1 - all equal to 1. Fourth divided differences are therefore 0, so f is a cubic with leading coefficient 1.
4. Newton's form using the leading diagonal: f(x) = 1 + 13(x - 0) - 6(x - 0)(x - 1) + 1(x - 0)(x - 1)(x - 2).
5. At x = 3: f(3) = 1 + 13(3) - 6(3)(2) + (3)(2)(1) = 1 + 39 - 36 + 6 = 10, option (c).

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22. 2022 · Q27 Lagrange's Formula (Unequal Intervals) · Numerical

QUESTION

Applying Lagrange's formula inversely, the value of x from the data $y_1=4$, $y_3=12$, $y_4=19$, $y_x=7$ is approximately:

OPTIONS

(a) 1.86

(b) 1.98

(c) 2.16

(d) 2.36

ANSWER (AI-derived — no official key exists for this paper)

(a) 1.86

EXAM SHORTCUT (AI-derived explanation)

Inverse Lagrange with the pairs (4,1), (12,3), (19,4) evaluated at y = 7 gives weights 0.5, 0.6429, -0.1429, so x = 0.5 + 1.9286 - 0.5714 = 1.86.

TIPS & TRICKS (AI-derived explanation)

- Inverse interpolation means treating y as the ARGUMENT and x as the function value - the same Lagrange formula with the roles swapped, no iteration needed.
- Weights must sum to 1: 0.5 + 0.642857 - 0.142857 = 1.000. Check this before multiplying by the x-values.
- Bracket the answer first: y = 7 lies between y = 4 (x = 1) and y = 12 (x = 3), and nearer the lower end, so x should be a little below 2 - which points straight at 1.86.
- Keep six decimals in the weights; the options differ by about 0.1, so moderate precision suffices but sloppy rounding of -0.142857 can shift the total.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The data pairs are (x, y) = (1, 4), (3, 12), (4, 19), and we seek x when y = 7 by inverse interpolation, treating y as the independent variable.
2. Weight for y = 4: (7-12)(7-19)/[(4-12)(4-19)] = (-5)(-12)/[(-8)(-15)] = 60/120 = 0.5.
3. Weight for y = 12: (7-4)(7-19)/[(12-4)(12-19)] = (3)(-12)/[(8)(-7)] = -36/-56 = 0.642857.
4. Weight for y = 19: (7-4)(7-12)/[(19-4)(19-12)] = (3)(-5)/[(15)(7)] = -15/105 = -0.142857. (Sum = 1.)
5. x = 1(0.5) + 3(0.642857) + 4(-0.142857) = 0.5 + 1.928571 - 0.571429 = 1.857, i.e. approximately 1.86, option (a).

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23. 2022 · Q29 Lagrange's Formula (Unequal Intervals) · Theoretical

QUESTION

If $f(x)=\frac{1}{x}$, the value of $f(x_1,x_2,\ldots,x_n)$ is equal to:

OPTIONS

(a) $(-1)^{n}\dfrac{1}{x_1x_2\cdots x_n}$

(b) $(-1)^{n-1}\dfrac{1}{x_1x_2\cdots x_n}$

(c) $\dfrac{n}{x_1x_2\cdots x_n}$

(d) $\dfrac{1}{x_1x_2\cdots x_n}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $(-1)^{n-1}\dfrac{1}{x_1x_2\cdots x_n}$

EXAM SHORTCUT (AI-derived explanation)

With n arguments the divided difference has ORDER n - 1, and for f(x) = 1/x the order-k difference is $(-1)^k$ over the product of the nodes. Hence (-1)^(n-1)/(x1 x2 ... xn).

TIPS & TRICKS (AI-derived explanation)

- Count the arguments carefully: f(x1,...,xn) lists n nodes, so it is the (n-1)th divided difference - the exponent is n-1, not n.
- Base rule: f[x0, x1, ..., xk] = $(-1)^k$/(x0 x1 ... xk) for f(x) = 1/x. Fix the pattern with the two-node case f[a,b] = -1/(ab).
- Verify with n = 2: f(x1,x2) = -1/(x1 x2), and (-1)^(2-1) = -1. Option (b) matches; option (a) would give +1/(x1x2).
- The result is symmetric in all the nodes, as every divided difference must be - options (c) and (d) are symmetric too, so the sign is what decides.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For f(x) = 1/x the first divided difference is f[a,b] = (1/a - 1/b)/(a-b) = -1/(ab).
2. Iterating, the kth divided difference over k+1 nodes is f[x0, ..., xk] = $(-1)^k$/(x0 x1 ... xk).
3. The notation f(x1, x2, ..., xn) involves n arguments, so it denotes the divided difference of order n - 1.
4. Applying the rule with k = n - 1 gives f(x1, ..., xn) = (-1)^(n-1)/(x1 x2 ... xn).
5. Check with n = 2: the formula gives -1/(x1 x2), matching the first divided difference computed above.
6. Option (b) is correct.

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24. 2023 · Q41 Newton's Divided Difference Formula · Conceptual

QUESTION

What is the minimum number of arguments required to find the third divided difference of the function $f(x)=x^3-2x+1$?

OPTIONS

(a) 2

(b) 3

(c) 4

(d) 5

ANSWER **(AI-derived — no official key exists for this paper)**

(c) 4

EXAM SHORTCUT **(AI-derived explanation)**

A divided difference of order k needs k + 1 distinct arguments, so the third divided difference requires 4.

TIPS & TRICKS **(AI-derived explanation)**

- Counting rule: order k uses k + 1 nodes. First divided difference needs 2, second needs 3, third needs 4.
- The particular function is irrelevant to the COUNT - the cubic x^3 - 2x + 1 is there only to reassure you the answer is non-zero (it equals the leading coefficient 1).
- Each divided difference of order k is built from two of order k - 1, which is why the node count grows by one at each level.
- Contrast with finite differences on an equally spaced table, where the third difference likewise consumes four consecutive tabulated values.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. A divided difference of order k is defined recursively as f[x0,...,xk] = (f[x1,...,xk] - f[x0,...,x(k-1)])/(xk - x0).
2. Each application of the recursion consumes one additional distinct argument.
3. The zeroth divided difference f[x0] needs 1 argument, the first needs 2, the second needs 3.
4. Hence the third divided difference f[x0, x1, x2, x3] requires 4 distinct arguments.
5. For the cubic f(x) = x^3 - 2x + 1 the value of that third divided difference is the leading coefficient 1, independent of which four nodes are used.
6. Option (c) is correct.

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25. **2024 · Q70** Newton's Divided Difference Formula · Conceptual

QUESTION

The third divided difference of a cubic polynomial is:

OPTIONS

(a) always constant but not necessarily zero

(b) always zero

(c) equal to sum of arguments

(d) equal to product of arguments

ANSWER (AI-derived — no official key exists for this paper)

(a) always constant but not necessarily zero

EXAM SHORTCUT (AI-derived explanation)

The third divided difference of a cubic equals its leading coefficient, which is non-zero by definition of 'cubic' - so it is always constant but not necessarily zero.

TIPS & TRICKS (AI-derived explanation)

- Rule: the nth divided difference of a degree-n polynomial is its leading coefficient a_n, and all HIGHER divided differences are zero.
- 'Cubic' means the x^3 coefficient is non-zero, so the value cannot be zero - that distinction is exactly what separates options (a) and (b).
- Option (c) 'sum of arguments' is the SECOND divided difference of x^3 (as in 2023 Q42) - a neighbouring fact deliberately offered as a distractor.
- The value is independent of which four nodes are used, which is the sense in which it is 'constant'.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Divided differences are linear, and the kth divided difference of a polynomial of degree less than k is zero.
2. For a cubic f(x) = a3 x^3 + a2 x^2 + a1 x + a0, only the leading term survives a third divided difference.
3. For x^3 the third divided difference over any four distinct nodes equals 1, so for f it equals a3.
4. Since f is a cubic, a3 is non-zero by definition, so the third divided difference is a non-zero constant, the same for every choice of nodes.
5. It would be zero only for a polynomial of degree less than 3, and the SECOND divided difference of x^3 - not the third - is the sum of the arguments.
6. Hence it is always constant but not necessarily zero, option (a).

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26. 2024 · Q76 Lagrange's Formula (Unequal Intervals) · Conceptual

QUESTION

The bound on the truncation error of Lagrange linear interpolating polynomial $f(x)$ on $[a,b]$ is ε. What is the bound on the truncation error of Newton forward interpolating polynomial of $f(x)$ on $[a,b]$?

OPTIONS

(a) $\varepsilon/2$

(b) ε

(c) 2ε

(d) 4ε

ANSWER (AI-derived — no official key exists for this paper)

(b) ε

EXAM SHORTCUT (AI-derived explanation)

Through the same two points there is only ONE interpolating polynomial, so the Lagrange and Newton forward forms are algebraically identical and share the same error bound, epsilon.

TIPS & TRICKS (AI-derived explanation)

- Uniqueness theorem: through n + 1 distinct points there passes exactly one polynomial of degree at most n. Lagrange and Newton are just different ALGEBRAIC ARRANGEMENTS of it.
- Since the polynomial is the same, the remainder term f''(xi)(x-x0)(x-x1)/2! is the same, and so is its bound.
- The two forms differ in computational convenience - Newton's allows incremental addition of points - not in accuracy.
- Any option other than epsilon would imply that two representations of the same function differ in error, which is impossible.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Through two distinct points there passes exactly one polynomial of degree at most 1, by the uniqueness of the interpolating polynomial.
2. Lagrange's linear form and Newton's forward linear form are therefore two algebraic representations of the SAME polynomial.
3. The truncation error of linear interpolation is R(x) = f''(xi)(x - x0)(x - x1)/2! for some xi in the interval, and this expression depends only on the function and the nodes, not on how the polynomial is written.
4. Hence the two forms have identical remainder terms and identical error bounds.
5. So the bound for the Newton forward form is also epsilon.
6. Option (b) is correct.

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27. 2025 · Q23 Newton's Divided Difference Formula · Numerical

QUESTION

If the 9^{th} divided difference of $f(x)=\frac{1}{x}$ based on points $x_i=ih,\ 1\le i\le 10$ is α, what is the 10^{th} divided difference of $f(x)$ based on points $x_i=ih,\ 1\le i\le 11$?

OPTIONS

(a) $-\frac{\alpha}{11h}$

(b) $-\frac{\alpha}{10h}$

(c) $\frac{\alpha}{11h}$

(d) $\frac{\alpha}{10h}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $-\frac{\alpha}{11h}$

EXAM SHORTCUT (AI-derived explanation)

The nth divided difference of 1/x is $(-1)^n$ over the product of the nodes. Going from 10 nodes to 11 multiplies by -1 (sign flip) and divides by the new node 11h, giving -alpha/(11h).

TIPS & TRICKS (AI-derived explanation)

- Use f[x0,...,xn] = $(-1)^n$/(x0 x1 ... xn) for f = 1/x - the whole question is a ratio of two such expressions.
- Count the nodes: i = 1 to 10 gives 10 nodes and hence the NINTH divided difference; i = 1 to 11 gives the tenth. The orders differ by one, so the sign flips.
- The new product contains one extra factor, the node 11h, so the magnitude is divided by 11h.
- Combining sign flip and extra factor: new = -alpha/(11h). Option (b) uses 10h, the LAST node of the old set rather than the new one.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For f(x) = 1/x the nth divided difference over nodes x0, ..., xn is $(-1)^n$/(x0 x1 ... xn).
2. The nodes x_i = ih for i = 1 to 10 are ten in number, so the corresponding divided difference is of order 9: alpha = $(-1)^9$/(h . 2h . ... . 10h) = -1/(10! h^{10}).
3. The nodes x_i = ih for i = 1 to 11 are eleven in number, giving the divided difference of order 10: $(-1)^{10}$/(h . 2h . ... . 11h) = 1/(11! h^{11}).
4. Taking the ratio: [1/(11! h^{11})] / [-1/(10! h^{10})] = -(10!/11!)(1/h) = -1/(11h).
5. So the tenth divided difference equals -alpha/(11h).
6. Option (a) is correct.

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28. 2025 · Q64 Lagrange's Formula (Unequal Intervals) · Numerical

QUESTION

Let $f(x)=x+x^3+x^5$. If $g(x)$ is a quadratic polynomial via Lagrange's interpolation with $g(f(0))=0$, $g(f(1))=1$, $g(f(2))=2$, what is $g(13)$?

OPTIONS

(a) 70/19

(b) 71/21

(c) 71/20

(d) 70/21

ANSWER (AI-derived — no official key exists for this paper)

(b) 71/21

EXAM SHORTCUT **(AI-derived explanation)**

Compute the three nodes f(0)=0, f(1)=3, f(2)=42, so g interpolates (0,0), (3,1), (42,2). Since g(0)=0 the first Lagrange term dies; g(13) = 1*[13*(13-42)]/[3*(3-42)] + 2*[13*(13-3)]/[42*(42-3)] = 377/117 + 260/1638 = 29/9 + 10/63 = 213/63 = 71/21.

TIPS & TRICKS **(AI-derived explanation)**

- Always evaluate the nodes first - here the abscissae of g are the values of f, not 0, 1, 2.
- A zero function value kills an entire Lagrange term, which halves the arithmetic; check for it before expanding.
- Keep everything as fractions: 377/117 reduces by 13 to 29/9, and 260/1638 reduces by 26 to 10/63 - decimals here invite rounding errors between 71/20 and 71/21.
- Sanity check: 13 lies between 3 and 42, so g(13) must lie between 1 and 2; 71/21 = 3.38 fails that test, which flags that the intended reading is plain Lagrange interpolation (extrapolating the fitted parabola), matching the printed key.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. First find the interpolation nodes. f(x) = x + x^3 + x^5 gives f(0) = 0, f(1) = 1 + 1 + 1 = 3, f(2) = 2 + 8 + 32 = 42.
2. So the quadratic g satisfies g(0) = 0, g(3) = 1, g(42) = 2.
3. Lagrange form: g(x) = 0*L_0(x) + 1*L_1(x) + 2*L_2(x), where L_1(x) = (x-0)(x-42)/((3-0)(3-42)) and L_2(x) = (x-0)(x-3)/((42-0)(42-3)).
4. The L_0 term vanishes because g(0) = 0.
5. L_1(13) = 13*(13-42)/(3*(-39)) = 13*(-29)/(-117) = 377/117 = 29/9 (dividing numerator and denominator by 13).
6. L_2(13) = 13*(13-3)/(42*39) = 130/1638 = 5/63 (dividing by 26).
7. g(13) = 1*(29/9) + 2*(5/63) = 29/9 + 10/63 = 203/63 + 10/63 = 213/63 = 71/21.
8. Hence option (b) is correct.

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29. 2025 · Q70 Lagrange's Formula (Unequal Intervals) · Numerical

QUESTION

Given $f(0)=8,\ f(1)=11,\ f(4)=68,\ f(5)=120$. What is the approximate $f(2)$ via Lagrange's interpolation formula?

OPTIONS

(a) 16.6

(b) 17.6

(c) 18.6

(d) 19.6

ANSWER **(AI-derived — no official key exists for this paper)**

(c) 18.6

EXAM SHORTCUT (AI-derived explanation)

Lagrange weights at x = 2 for nodes 0, 1, 4, 5 come out as -0.3, 1, 0.5, -0.2 (they must sum to 1 - a fast check). Then f(2) = 8(-0.3) + 11(1) + 68(0.5) + 120(-0.2) = -2.4 + 11 + 34 - 24 = 18.6.

TIPS & TRICKS (AI-derived explanation)

- The Lagrange weights always sum to 1; computing all four and checking the sum catches sign errors before you multiply by the data.
- Nodes are unequally spaced (0, 1, 4, 5), so Newton's forward formula does not apply - use Lagrange or divided differences.
- Divided-difference alternative: f[0,1] = 3, f[1,4] = 19, f[4,5] = 52, f[0,1,4] = 4, f[1,4,5] = 8.25, f[0,1,4,5] = 0.85; then Newton's divided difference form at x = 2 gives 8 + 3(2) + 4(2)(1) + 0.85(2)(1)(-2) = 8 + 6 + 8 - 3.4 = 18.6.
- Keep the weights as exact decimals (-0.3, 1, 0.5, -0.2); rounding early is what separates 18.6 from the neighbouring distractors.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Nodes x = 0, 1, 4, 5 with f = 8, 11, 68, 120; interpolate at x = 2 using Lagrange's formula for unequal intervals.
2. L_0(2) = (2-1)(2-4)(2-5)/[(0-1)(0-4)(0-5)] = (1)(-2)(-3)/[(-1)(-4)(-5)] = 6/(-20) = -0.3.
3. L_1(2) = (2-0)(2-4)(2-5)/[(1-0)(1-4)(1-5)] = (2)(-2)(-3)/[(1)(-3)(-4)] = 12/12 = 1.
4. L_2(2) = (2-0)(2-1)(2-5)/[(4-0)(4-1)(4-5)] = (2)(1)(-3)/[(4)(3)(-1)] = -6/-12 = 0.5.
5. L_3(2) = (2-0)(2-1)(2-4)/[(5-0)(5-1)(5-4)] = (2)(1)(-2)/[(5)(4)(1)] = -4/20 = -0.2.
6. Check: -0.3 + 1 + 0.5 - 0.2 = 1, as required.
7. f(2) = 8(-0.3) + 11(1) + 68(0.5) + 120(-0.2) = -2.4 + 11 + 34 - 24 = 18.6.
8. Hence option (c) is correct.

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30. 2026 · Q43 Newton's Divided Difference Formula · Theoretical

QUESTION

Consider: I. The $(n+1)^{th}$ divided differences of an nth-degree polynomial are zero (n a natural number). II. The nth divided difference for a degree-n polynomial can be expressed as the quotient of two determinants, each of order n. Which is/are correct?

OPTIONS

(a) I only

(b) II only

(c) Both I and II

(d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(c) Both I and II

Source note. Statement II is order-sensitive. The determinant representation quoted in most textbooks uses two determinants of order (n+1) - a Vandermonde denominator and a numerator with its last column replaced by the function values. Subtracting the first row from every other row reduces BOTH to determinants of order n, so the existence claim 'can be expressed as the quotient of two determinants, each of order n' is achievable and II is keyed as correct. A candidate who reads II as quoting the standard (n+1)-order form would answer (a) I only.

EXAM SHORTCUT (AI-derived explanation)

Statement I is the fundamental property: the nth divided difference of a degree-n polynomial equals its leading coefficient, so the (n+1)th vanishes. Statement II is the determinant representation of divided differences. Both stand, so (c).

TIPS & TRICKS (AI-derived explanation)

- Core fact: for a polynomial of degree n, f[x_0,...,x_n] = a_n (the leading coefficient) and every higher divided difference is 0 - exactly parallel to Delta^(n+1) = 0 for equal intervals.
- The determinant form is f[x_0,...,x_n] = det(M)/det(V), where V is the Vandermonde matrix in x_0,...,x_n and M replaces its last column by the f-values.
- Row-reduce to see the order-n version: subtracting the first row from the rest kills the leading column of ones and drops both determinants from order n+1 to order n.
- Divided differences are symmetric functions of their arguments, so the order in which nodes are listed never matters - a companion fact often tested alongside these two.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: if p(x) has degree n, its nth divided difference on any n+1 distinct nodes equals the leading coefficient a_n, a constant independent of the nodes. Taking a further divided difference of a constant gives zero, so all (n+1)th divided differences vanish. Correct.
2. Statement II: the divided difference has the determinant representation f[x_0,...,x_n] = det(M)/det(V), where V is the (n+1)x(n+1) Vandermonde matrix with rows (1, x_i, x_i^2, ..., x_i^n) and M is V with its last column replaced by (f(x_0), ..., f(x_n)).
3. Both determinants have a first column of ones. Subtracting the first row from each of the remaining rows makes that column (1, 0, 0, ..., 0)^T, and expanding along it reduces each determinant to one of order n.
4. Hence the nth divided difference can indeed be written as the quotient of two determinants of order n, and statement II holds as an existence claim.
5. Both statements are therefore correct.
6. Hence option (c) is correct (see the note above on the order convention).

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31. 2026 · Q44 Newton's Divided Difference Formula · Numerical

QUESTION

Using Newton's divided difference formula, the lowest possible degree of $p(x)$ assuming values 48, 100, 294, 900, 1210, 2028 at x=4,5,7,10,11,13 respectively, is:

OPTIONS

(a) 3

(b) 4

(c) 5

(d) 6

ANSWER (AI-derived — no official key exists for this paper)

(a) 3

EXAM SHORTCUT (AI-derived explanation)

Build the divided-difference table. First: 52, 97, 202, 310, 409. Second: 15, 21, 27, 33. Third: 1, 1, 1 - constant. A constant third divided difference means degree 3.

TIPS & TRICKS (AI-derived explanation)

- The lowest degree equals the order at which the divided differences first become constant (equivalently, one less than the order at which they vanish).
- Divide by the SPREAD of the arguments, not by 1: e.g. f[5,7] = (294-100)/(7-5) = 97, since the nodes are unequally spaced.
- The second differences 15, 21, 27, 33 form an arithmetic progression with common difference 6, which already hints that the next column will be constant.
- Stop as soon as a column is constant - computing further columns of zeros wastes time in the exam.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Nodes x = 4, 5, 7, 10, 11, 13 with p = 48, 100, 294, 900, 1210, 2028.
2. First divided differences: (100-48)/(5-4) = 52; (294-100)/(7-5) = 97; (900-294)/(10-7) = 202; (1210-900)/(11-10) = 310; (2028-1210)/(13-11) = 409.
3. Second divided differences: (97-52)/(7-4) = 45/3 = 15; (202-97)/(10-5) = 105/5 = 21; (310-202)/(11-7) = 108/4 = 27; (409-310)/(13-10) = 99/3 = 33.
4. Third divided differences: (21-15)/(10-4) = 6/6 = 1; (27-21)/(11-5) = 6/6 = 1; (33-27)/(13-7) = 6/6 = 1.
5. The third divided differences are constant (all equal 1), so the fourth divided differences are all zero.
6. A constant nth divided difference characterises a polynomial of degree exactly n, so the lowest possible degree is 3 (with leading coefficient 1).
7. Hence option (a) is correct.

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32. 2026 · Q46 Newton's Divided Difference Formula · Numerical

QUESTION

The third divided difference of $(x^3\text{-}2x)$ with arguments 2, 4, 9, 10 is:

OPTIONS

(a) 1

(b) 2

(c) 3

(d) 4

ANSWER (AI-derived — no official key exists for this paper)

(a) 1

EXAM SHORTCUT (AI-derived explanation)

The nth divided difference of a degree-n polynomial equals its leading coefficient, whatever the nodes. Here the polynomial x^3 - 2x is cubic with leading coefficient 1, so the third divided difference is 1.

TIPS & TRICKS (AI-derived explanation)

- Memorise: f[x_0, ..., x_n] = a_n for a degree-n polynomial. The arguments 2, 4, 9, 10 are irrelevant decoration.
- The -2x term is of degree 1, so it contributes nothing to the third divided difference - lower-order terms always drop out.
- Equal-interval analogue: Delta^n f = n! a_n h^n. Note the divided-difference version has NO factorial, which is why the answer is 1 and not 6. Option (d) 4 and option (c) 3 target that confusion.
- If you must verify, compute the table: f(2) = 4, f(4) = 56, f(9) = 711, f(10) = 980, and the third divided difference works out to exactly 1.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let f(x) = x^3 - 2x, a polynomial of degree 3 with leading coefficient a_3 = 1.
2. A standard property of divided differences: for a polynomial of degree n, the nth divided difference on any n + 1 distinct arguments equals the leading coefficient a_n.
3. Here n = 3 and the four arguments 2, 4, 9, 10 are distinct, so the third divided difference is a_3 = 1.
4. Verification: f(2) = 8 - 4 = 4; f(4) = 64 - 8 = 56; f(9) = 729 - 18 = 711; f(10) = 1000 - 20 = 980.
5. First: f[2,4] = (56-4)/2 = 26; f[4,9] = (711-56)/5 = 131; f[9,10] = (980-711)/1 = 269.
6. Second: f[2,4,9] = (131-26)/(9-2) = 105/7 = 15; f[4,9,10] = (269-131)/(10-4) = 138/6 = 23.
7. Third: f[2,4,9,10] = (23-15)/(10-2) = 8/8 = 1.
8. Hence option (a) is correct.

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33. 2026 · Q56 Lagrange's Formula (Unequal Intervals) · Theoretical

QUESTION

An estimate of the truncation error $e(x)$ in the Lagrange interpolation polynomial is given by (with $x_0 \le \xi \le x_n$):

OPTIONS

(a) $\dfrac{(x-x_0)(x-x_1)\cdots(x-x_n)}{(n+1)!} y^{n}(\xi)$

(b) $\dfrac{(x-x_0)(x-x_1)\cdots(x-x_n)}{(n+1)!} y^{n+1}(\xi)$

(c) $\dfrac{(x-x_0)(x-x_1)\cdots(x-x_n)}{n!} y^{n-1}(\xi)$

(d) $\dfrac{(x-x_0)(x-x_1)\cdots(x-x_n)}{n!}\, y^{n}(\xi)$

ANSWER **(AI-derived — no official key exists for this paper)**

(b) $\dfrac{(x-x_0)(x-x_1)\cdots(x-x_n)}{(n+1)!}\, y^{n+1}(\xi)$

EXAM SHORTCUT **(AI-derived explanation)**

With n+1 nodes there are n+1 factors in the product, so the error carries (n+1)! in the denominator and the (n+1)th derivative. Answer (b).

TIPS & TRICKS **(AI-derived explanation)**

- Matching rule: the number of factors (x - x_i), the factorial, and the derivative order must all agree - here all equal n + 1.
- Consistency test: if f is itself a polynomial of degree at most n, interpolation is exact, so the error must vanish. Only the (n+1)th derivative guarantees this; the nth derivative in option (a) would not.
- The same product times the (n+1)th divided difference gives the exact error: e(x) = product * f[x_0, ..., x_n, x], which links the two standard error forms.
- This is the direct analogue of the Taylor remainder, with the single point x_0 repeated n+1 times replaced by n+1 distinct nodes.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Let p_n be the Lagrange interpolating polynomial of degree at most n through the n + 1 nodes x_0, x_1, ..., x_n.
2. The standard error theorem states e(x) = f(x) - p_n(x) = [(x - x_0)(x - x_1)...(x - x_n)/(n + 1)!] * f^{(n+1)}(xi), for some xi in the interval spanned by x_0, ..., x_n and x.
3. The product contains n + 1 factors, matching the factorial (n + 1)! and the derivative order n + 1.
4. Consistency check: if f is a polynomial of degree at most n, then f^{(n+1)} is identically zero and the error vanishes, as it must since interpolation is then exact. Option (a), using the nth derivative, fails this test.
5. Options (c) and (d) mismatch the factorial and the derivative order with the number of factors.
6. Hence option (b) is correct.

N4 · NUMERICAL ANALYSIS

Central Difference Formulae - Gauss, Stirling, Bessel; Error Terms

4 questions · 2018: 0 2019: 1 2020: 0 2021: 1 2022: 1 2023: 0 2024: 0 2025: 1 2026: 0

Weight. 4 questions (2.2% of Numerical Analysis) · appeared in 4 of 9 papers · peak year 2019 (1)

Examiner's pattern. Thin (4 items) and almost always conceptual rather than computational: which formula suits which part of the table. Gauss forward/backward near the middle, Stirling for p near 0, Bessel for p near 1/2.

Must-know. Use Stirling when $|p| < 0.25$ and Bessel when $0.25 < p < 0.75$, where $p = (x - x0)/h$. Gauss's formulae use the central-difference (zig-zag) path of the difference table.

1. 2019 · Q57 Gauss Central Difference Formula · Theoretical

QUESTION

Gauss forward interpolation formula involves

OPTIONS

(a) even differences above the central line and odd differences on the central line

(b) even differences below the central line and odd differences on the central line

(c) odd differences below the central line and even differences on the central line

(d) odd differences above the central line and even differences on the central line

ANSWER (AI-derived — no official key exists for this paper)

(c) odd differences below the central line and even differences on the central line

EXAM SHORTCUT (AI-derived explanation)

Gauss's forward formula uses Delta y0, Delta^2 y(-1), Delta^3 y(-1), Delta^4 y(-2), ... - the ODD differences lie below the central line and the EVEN differences lie on it.

TIPS & TRICKS (AI-derived explanation)

- Write the two formulas as difference paths: Gauss FORWARD zig-zags downward first (odd differences below the central line); Gauss BACKWARD zig-zags upward first (odd differences above it). Even differences sit on the central line in both.
- Gauss forward is used when the interpolation point lies just AFTER x0 ($0 < p < 1$); Gauss backward when it lies just before.
- Stirling's formula is the mean of the two Gauss formulas and is best for $|p| < 0.25$; Bessel's uses the mean of the two central lines and is best near $p = 0.5$. Knowing which to choose is itself examinable.
- Options (a) and (b) put odd differences ON the central line, which is impossible - the central line of a difference table can only carry even-order differences.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Set the central value y0 in a difference table with y(-2), y(-1), y0, y1, y2 and their differences.

2. Gauss's forward interpolation formula is y_p = y0 + p Delta y0 + [p(p-1)/2!] Delta^2 y(-1) + [(p+1)p(p-1)/3!] Delta^3 y(-1) + [(p+1)p(p-1)(p-2)/4!] Delta^4 y(-2) + ...
3. The differences used are Delta y0, Delta^2 y(-1), Delta^3 y(-1), Delta^4 y(-2), ...
4. In the difference table, Delta y0 and Delta^3 y(-1) - the ODD-order differences - lie immediately BELOW the horizontal line through y0.
5. The EVEN-order differences Delta^2 y(-1) and Delta^4 y(-2) lie ON that central line.
6. Hence the formula involves odd differences below the central line and even differences on the central line, option (c).

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2. 2021 · Q46 Bessel's Formula · Factual

QUESTION

Consider the following statements: 1. Newton's forward formula is used to interpolate near the beginning of a table. 2. Bessel's formula is used to interpolate near the centre of a table. 3. Lagrange's formula is used for unequal subintervals.

OPTIONS

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

All three are the standard selection rules: forward formula near the start, Bessel (a central formula) near the middle, Lagrange for unequal intervals.

TIPS & TRICKS (AI-derived explanation)

- Position map for equally spaced tables: Newton forward at the START, Newton backward at the END, Stirling near the MIDDLE for $|p| < 0.25$, Bessel near the middle for p around 0.5.
- Bessel's formula uses the mean of two central lines and is most accurate when the interpolation point lies midway between two tabulated values.
- Lagrange's formula makes no spacing assumption at all, which is exactly why it is the standard choice for unequal intervals.
- When all three statements are textbook selection rules for different situations, 'all of them' is the natural key.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: Newton's forward difference formula uses the leading differences at the top of the table, so it is accurate for interpolation near the BEGINNING of the table. TRUE.

2. Statement 2: Bessel's formula is a central-difference formula built on the mean of the two central lines, and it is most accurate when the interpolation point lies near the middle of the table, particularly near p = 0.5. TRUE.
3. Statement 3: Lagrange's formula is constructed from products of factors (x - x_i) and makes no assumption about spacing, so it is the standard method for unequally spaced arguments. TRUE.
4. All three statements correctly describe the standard use of each formula.
5. Hence option (d) is correct.

---

3. 2022 · Q26 Bessel's Formula · Factual

QUESTION

Bessel's interpolation formula is most appropriate when $p=(x\text{-}x_0)/h$:

OPTIONS

(a) lies between -0.25 and +0.25

(b) lies between 0.25 and 0.75

(c) lies between 0.75 and 1

(d) is greater than 1

ANSWER (AI-derived — no official key exists for this paper)

(b) lies between 0.25 and 0.75

EXAM SHORTCUT (AI-derived explanation)

Bessel's formula is built on the mean of the two central lines and is most accurate near the midpoint, i.e. for p between 0.25 and 0.75.

TIPS & TRICKS (AI-derived explanation)

- Range map for central formulas: Stirling for |p| < 0.25 (near a tabulated value), Bessel for 0.25 < p < 0.75 (near the midpoint).
- Bessel's formula is exactly symmetric about p = 0.5, where its odd-order terms drop out and the accuracy is highest - that symmetry is the reason for the range.
- For p outside [0,1] entirely, use Newton's forward formula near the start of the table or the backward formula near its end.
- Both Stirling and Bessel are averages of the two Gauss formulas; the difference is which pair of lines is averaged, and hence where each is sharpest.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Bessel's interpolation formula is constructed by averaging the two central lines of the difference table, which makes it symmetric about the midpoint between x0 and x1.
2. At p = 0.5 the odd-order terms of Bessel's formula vanish, so the truncation error is smallest there.
3. The formula therefore gives the best accuracy when the interpolation point lies near the middle of the interval, conventionally taken as 0.25 < p < 0.75.
4. For |p| < 0.25 the point is close to a tabulated value and Stirling's formula is preferred.

5. For p outside the range 0 to 1 the Newton-Gregory forward or backward formulas are appropriate.
6. Hence Bessel's formula suits p between 0.25 and 0.75, option (b).

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4. **2025 · Q68** Delta, E and D Operators · Numerical

QUESTION

Consider the data:

x	6.1	6.2	6.3	6.4
f(x)	−0.1998	−0.2223	−0.2422	−0.2596

What is the approximate $f''(6.3)$ with error $O(h^2)$ using the central difference operator?

OPTIONS

(a) 0.27

(b) 0.25

(c) 0.23

(d) 0.21

ANSWER (AI-derived — no official key exists for this paper)

(b) 0.25

EXAM SHORTCUT (AI-derived explanation)

Central second difference at 6.3 uses its two neighbours: (f(6.4) - 2f(6.3) + f(6.2))/h^2 with h = 0.1. Numerator = -0.2596 + 0.4844 - 0.2223 = 0.0025; divide by 0.01 to get 0.25. Option (b).

TIPS & TRICKS (AI-derived explanation)

- The O(h^2) accuracy tag is the signature of the central difference formula - it tells you which stencil to use.
- Divide by h^2 = 0.01, i.e. multiply by 100; forgetting this is the single most common slip and would give 0.0025.
- Add the two outer values first (-0.2596 - 0.2223 = -0.4819), then subtract 2*(-0.2422) = -0.4844: 0.4844 - 0.4819 = 0.0025. Small differences of nearly equal numbers demand all four decimal places.
- The node 6.1 is deliberately supplied as a distractor; the central formula at 6.3 does not use it.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The data are equally spaced with h = 0.1, and 6.3 is an interior node with neighbours 6.2 and 6.4.
2. Central difference second derivative formula, accurate to O(h^2): f''(x) = (f(x+h) - 2f(x) + f(x-h))/h^2.
3. Substituting: f''(6.3) = (f(6.4) - 2 f(6.3) + f(6.2))/$(0.1)^2$.
4. Numerator = (-0.2596) - 2(-0.2422) + (-0.2223) = -0.2596 + 0.4844 - 0.2223 = 0.0025.
5. Denominator = 0.01.
6. f''(6.3) = 0.0025/0.01 = 0.25.

7. Hence option (b) is correct.

N5 · NUMERICAL ANALYSIS

Inverse Interpolation

2 questions · 2018: 0 2019: 1 2020: 0 2021: 0 2022: 0 2023: 0 2024: 0 2025: 1 2026: 0

Weight. 2 questions (1.1% of Numerical Analysis) · appeared in 2 of 9 papers · peak year 2019 (1)

Examiner's pattern. Only two appearances (2019, 2025). One was a direct table inversion, the other inverted a Lagrange form. Method: swap the roles of x and y and re-apply Lagrange, or iterate the forward formula.

Must-know. Inverse interpolation by Lagrange: x = sum over i of [prod_(j != i) (y - yj)/(yi - yj)] xi. Successive approximation from Newton forward is the alternative.

1. 2019 · Q60 Inverse Interpolation · Numerical

QUESTION

Consider the table where $y=f(x)$:

x	10	15	20
y	1754	2648	3564

What is the value of x for $y=3000$, obtained by inverse interpolation (iterative method) with $h=5$?

OPTIONS

(a) 15.90

(b) 16.10

(c) 16.93

(d) 17.02

ANSWER (AI-derived — no official key exists for this paper)

(c) 16.93

EXAM SHORTCUT (AI-derived explanation)

Newton forward gives 3000 = 1754 + 894p + 11p(p-1), i.e. p = (1246 - 11p^2)/883. Iterating from p = 1.411 converges to p = 1.3871, so x = 10 + 5(1.3871) = 16.93.

TIPS & TRICKS (AI-derived explanation)

- Inverse interpolation by iteration: isolate the LINEAR p on the left and push the quadratic term to the right, then iterate p_(k+1) = (1246 - 11 p_k^2)/883. Convergence is very fast.
- Compute the differences first: Delta y0 = 894 and Delta^2 y0 = 916 - 894 = 22, so the second-difference term is (22/2)p(p-1) = 11p(p-1).
- The first guess is the linear estimate p = 1246/894 = 1.394 (or 1246/883 = 1.411); either converges to 1.387 within two iterations.
- Finish with x = x0 + ph = 10 + 5p. Forgetting to multiply by h = 5 gives about 11.4 and no matching option.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With x0 = 10 and h = 5, the values are y0 = 1754, y1 = 2648, y2 = 3564, so Delta y0 = 894 and Delta^2 y0 = 916 - 894 = 22.
2. Newton's forward formula: y = y0 + p Delta y0 + [p(p-1)/2] Delta^2 y0 = 1754 + 894p + 11p(p-1).
3. Setting y = 3000: 1246 = 894p + 11p^2 - 11p = 883p + 11p^2.
4. Rearranged for iteration: p = (1246 - 11p^2)/883.
5. Start with p = 1246/883 = 1.4111; then p = (1246 - 11(1.9912))/883 = 1.3863; then p = (1246 - 11(1.9218))/883 = 1.3872; then 1.3871. The iteration has converged.
6. Therefore x = x0 + ph = 10 + 5(1.3871) = 16.936, i.e. about 16.93, option (c).

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2. 2025 · Q63 Inverse Interpolation · Conceptual

QUESTION

Let g be such that $y=f(x) \Leftrightarrow x=g(y)$, where $f(x) = \frac{(x-b)(x-c)}{(a-b)(a-c)}\alpha + \frac{(x-a)(x-c)}{(b-a)(b-c)}\beta + \frac{(x-a)(x-b)}{(c-a)(c-b)}\gamma$, with $a<b<c$ and $\alpha<\beta<\gamma$ real. What is $g(\beta)$?

OPTIONS

(a) β

(b) a

(c) b

(d) α

ANSWER (AI-derived — no official key exists for this paper)

(c) b

EXAM SHORTCUT (AI-derived explanation)

Recognise f as the Lagrange interpolating polynomial on nodes a, b, c with values alpha, beta, gamma. Then f(b) = beta, and g is the inverse, so g(beta) = b. Answer (c) - no algebra needed.

TIPS & TRICKS (AI-derived explanation)

- The Lagrange basis polynomial L_i satisfies L_i(x_j) = 1 if i = j and 0 otherwise; spotting this pattern converts the whole expression into 'f(a)=alpha, f(b)=beta, f(c)=gamma' instantly.
- g is the inverse function, so g(y) answers 'which x gives this y' - this is exactly inverse interpolation.
- The conditions a<b<c and alpha<beta<gamma are given only to guarantee f is invertible on the range, so the inverse is well defined and single-valued.
- Options (a) and (d) return a y-value where an x-value is wanted - a units mismatch you can reject on sight.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write f(x) = alpha*L_a(x) + beta*L_b(x) + gamma*L_c(x), where L_a(x) = (x-b)(x-c)/((a-b)(a-c)), L_b(x) = (x-a)(x-c)/((b-a)(b-c)), L_c(x) = (x-a)(x-b)/((c-a)(c-b)).

2. These are the Lagrange basis polynomials for the nodes a, b, c, so $L_a(a) = 1$, $L_a(b) = L_a(c) = 0$, and similarly for the others.
3. Substituting $x = b$: $L_a(b) = 0$, $L_b(b) = (b-a)(b-c)/((b-a)(b-c)) = 1$, $L_c(b) = 0$. Hence $f(b) = \beta$.
4. By definition $y = f(x)$ if and only if $x = g(y)$. Taking $x = b$ and $y = \beta$ gives $g(\beta) = b$.
5. Hence option (c) is correct.

N6 · NUMERICAL ANALYSIS

Numerical Integration - Trapezoidal, Simpson's 1/3 & 3/8, Weddle's Rule

36 questions · 2018: 4 2019: 3 2020: 5 2021: 2 2022: 2 2023: 4 2024: 7 2025: 5 2026: 4

Weight. 36 questions (20.0% of Numerical Analysis) · appeared in 9 of 9 papers · peak year 2024 (7)

Examiner's pattern. Second-highest numerical topic and the most computational (36 items). Expect a direct Simpson's 1/3 or trapezoidal evaluation on a supplied table every single year, plus one conceptual item on degree of precision, the number of subintervals each rule requires, or deriving quadrature weights so the formula is exact to the highest possible order.

Must-know. Simpson 1/3 needs n even and is exact for cubics; Simpson 3/8 needs n a multiple of 3; Weddle needs n a multiple of 6; trapezoidal error = -(b-a) h^2 f''/12; Simpson 1/3 error = -(b-a) h^4 f''''/180.

1. **2018 · Q56** Weddle's Rule · Theoretical

QUESTION

Consider the following statements on quadrature: 1. Geometrically, Simpson's one-third rule replaces the graph by $n/2$ arcs of second-degree polynomials. 2. Weddle's rule gives an exact result for a polynomial of degree 6 or less.

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER **(AI-derived — no official key exists for this paper)**

(a) 1 only

EXAM SHORTCUT **(AI-derived explanation)**

Statement 1 is the correct geometric description of Simpson's 1/3 rule. Statement 2 fails: Weddle's rule is exact only up to degree 5 - testing x^6 on [0,6] gives 39996 against the true 39990.86.

TIPS & TRICKS **(AI-derived explanation)**

- Simpson's 1/3 rule needs an EVEN number n of subintervals and fits one parabola over each consecutive pair, hence exactly n/2 parabolic arcs.
- Weddle's rule is the 7-point Newton-Cotes formula with the coefficient 41/140 rounded to 42/140 = 3/10. That deliberate rounding is precisely what destroys exactness at degree 6.
- Quick disproof of statement 2: apply (3h/10)[f0 + 5f1 + f2 + 6f3 + f4 + 5f5 + f6] to f(x) = x^6 on [0,6] with h = 1. It returns 39996 while the exact integral is $6^7/7$ = 39990.86.
- Do remember that Simpson's 1/3 rule itself is exact for CUBICS despite fitting parabolas - the odd-degree error term cancels by symmetry. That bonus degree is a favourite examiner point.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: Simpson's one-third rule divides [a, b] into n (even) subintervals and, on each consecutive PAIR of subintervals, replaces the curve by the parabola through the three points. There are n/2 such pairs, so the graph is replaced by n/2 arcs of second-degree polynomials. TRUE.
2. Statement 2: Weddle's rule is I = (3h/10)[f0 + 5f1 + f2 + 6f3 + f4 + 5f5 + f6]. It is obtained from the seven-point Newton-Cotes formula by replacing the coefficient 41/140 of the sixth difference by 42/140 = 3/10.
3. Test it on f(x) = x^6 over [0, 6] with h = 1: the bracket is 0 + 5(1) + 64 + 6(729) + 4096 + 5(15625) + 46656 = 133320, so the rule gives (3/10)(133320) = 39996.
4. The exact value is the integral of x^6 from 0 to 6 = $6^7/7$ = 39990.857, which differs. So Weddle's rule is NOT exact for degree 6.
5. The same test on f(x) = x^5 returns 7776, matching the exact value, so exactness holds up to degree 5 only. Statement 2 is FALSE.
6. Only statement 1 is correct, option (a).

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2. 2018 · Q57 Simpson's Three-Eighth Rule · Numerical

QUESTION

Using Simpson's three-eighth rule dividing the range into three equal parts, the value of $\int_0^{1/2} \sin(\pi x)\, dx$ is

OPTIONS

(a) $\frac{3(1+\sqrt{2})}{32}$

(b) $\frac{5+3\sqrt{3}}{32}$

(c) $\frac{3+5\sqrt{3}}{32}$

(d) $\frac{3\sqrt{3}}{8}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{5+3\sqrt{3}}{32}$

EXAM SHORTCUT (AI-derived explanation)

h = (1/2)/3 = 1/6 and the ordinates are sin0 = 0, sin(pi/6) = 1/2, sin(pi/3) = sqrt3/2, sin(pi/2) = 1. The 3/8 rule gives (3h/8)[0 + 3(1/2) + 3(sqrt3/2) + 1] = (5 + 3sqrt3)/32.

TIPS & TRICKS (AI-derived explanation)

- Simpson's 3/8 rule weights are 1, 3, 3, 1 with prefactor 3h/8. Here 3h/8 = 3/(6 x 8) = 1/16 - simplify the prefactor before substituting.
- The angles are pi times 0, 1/6, 1/3, 1/2, giving the standard sines 0, 1/2, sqrt3/2, 1. Converting x to the ANGLE pi x is the step candidates most often skip.
- Assemble as (1/16)[3/2 + 3sqrt3/2 + 1] = (1/16)(5 + 3sqrt3)/2 = (5 + 3sqrt3)/32. Options (b) and (c) differ only by swapping 5 and 3, so keep the rational and surd parts separate to the end.

- Numerical check: (5 + 3(1.732))/32 = 10.196/32 = 0.3186, against the exact value 1/pi = 0.3183 - agreement to three decimals confirms the choice.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The range [0, 1/2] is divided into 3 equal parts, so h = (1/2 - 0)/3 = 1/6, with nodes x = 0, 1/6, 1/3, 1/2.
2. The ordinates are f(0) = sin 0 = 0, f(1/6) = sin(pi/6) = 1/2, f(1/3) = sin(pi/3) = sqrt(3)/2, f(1/2) = sin(pi/2) = 1.
3. Simpson's three-eighth rule: I = (3h/8)[f0 + 3f1 + 3f2 + f3].
4. Here 3h/8 = 3(1/6)/8 = 1/16.
5. I = (1/16)[0 + 3(1/2) + 3(sqrt3/2) + 1] = (1/16)[3/2 + 3sqrt3/2 + 1] = (1/16)[(5 + 3sqrt3)/2] = (5 + 3 sqrt3)/32.
6. Numerically this is 0.3186 against the exact value 1/pi = 0.3183, confirming option (b).

---

3. 2018 · Q58 Numerical Integration · Numerical

QUESTION

The formula $\alpha\ f(-\frac{1}{2})+\beta\ f(0)+\gamma\ f(\frac{1}{2})$ approximates $\int_{-1}^{1} f(x)\ dx$ exactly for every polynomial of degree less than 3. Then

OPTIONS

(a) $\alpha=\gamma=\frac{4}{3},\ \beta=-\frac{2}{3}$

(b) $\alpha=\gamma=\frac{1}{3},\ \beta=0$

(c) $\alpha=\gamma=\frac{4}{3},\ \beta=1$

(d) $\alpha=\beta=\gamma$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\alpha=\gamma=\frac{4}{3},\ \beta=-\frac{2}{3}$

EXAM SHORTCUT (AI-derived explanation)

Impose exactness on 1, x, x^2. From x: alpha = gamma. From 1: alpha + beta + gamma = 2. From x^2: (alpha + gamma)/4 = 2/3, so alpha = gamma = 4/3 and beta = 2 - 8/3 = -2/3.

TIPS & TRICKS (AI-derived explanation)

- Method of undetermined coefficients: force exactness on the basis 1, x, x^2, ... one degree at a time. Three unknowns need exactly three equations.
- The x-equation is free: symmetry of the nodes about 0 forces alpha = gamma immediately, halving the algebra.
- The odd basis function x^3 is automatically integrated exactly by any symmetric rule, which is why 'degree less than 3' is achievable with only three nodes.

- A negative weight is legitimate here - the nodes +-1/2 are interior to [-1,1], so the rule must 'over-reach' and compensate with beta < 0. Do not reject option (a) merely because -2/3 looks odd.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Require the rule Q(f) = alpha f(-1/2) + beta f(0) + gamma f(1/2) to equal the integral of f from -1 to 1 for f = 1, x and x^2.
2. f(x) = 1: alpha + beta + gamma = integral of 1 from -1 to 1 = 2.
3. f(x) = x: -alpha/2 + 0 + gamma/2 = integral of x from -1 to 1 = 0, hence alpha = gamma.
4. f(x) = x^2: alpha/4 + gamma/4 = integral of x^2 from -1 to 1 = 2/3, so (alpha + gamma)/4 = 2/3 and alpha + gamma = 8/3.
5. With alpha = gamma this gives alpha = gamma = 4/3, and then beta = 2 - 8/3 = -2/3.
6. Hence alpha = gamma = 4/3 and beta = -2/3, option (a).

---

4. 2018 · Q59 Trapezoidal Rule · Numerical

QUESTION

Using the Trapezoidal rule with 10 equal parts, $\int_0^{10} 2^x \, dx$ belongs to

OPTIONS

(a) (1400,1450)

(b) (1450,1500)

(c) (1500,1550)

(d) (1550,1600)

ANSWER (AI-derived — no official key exists for this paper)

(c) (1500,1550)

EXAM SHORTCUT (AI-derived explanation)

With h = 1 the trapezoidal sum is (1/2)(1) + (2 + 4 + ... + 512) + (1/2)(1024) = 0.5 + 1022 + 512 = 1534.5, which lies in (1500, 1550).

TIPS & TRICKS (AI-derived explanation)

- Trapezoidal form: T = h[(f0 + fn)/2 + f1 + ... + f(n-1)]. Halve only the two END ordinates.
- The interior sum 2 + 4 + ... + 512 is a geometric series: 2^{10} - 2 = 1022. Recognising the GP saves all the addition.
- The exact value is (2^{10} - 1)/ln 2 = 1023/0.6931 = 1476.0, so the trapezoidal rule OVERESTIMATES here - as it always does for a convex function. That check confirms a value above 1476, i.e. band (c).
- Option (b) contains the exact answer 1476 and is the trap for candidates who integrate analytically instead of applying the required rule.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. With a = 0, b = 10 and 10 equal parts, h = 1 and the nodes are x = 0, 1, ..., 10 with f(x) = 2^x.
2. Ordinates: f0 = 1, f1 = 2, f2 = 4, ..., f9 = 512, f10 = 1024.

3. Trapezoidal rule: T = h[(f0 + f10)/2 + f1 + f2 + ... + f9].
4. The interior sum is 2 + 4 + 8 + ... + 512 = 2^{10} - 2 = 1022, and (f0 + f10)/2 = (1 + 1024)/2 = 512.5.
5. T = 1 x (512.5 + 1022) = 1534.5.
6. Since 1500 < 1534.5 < 1550, option (c) is correct. (The exact value 1023/ln2 = 1476 confirms the expected overestimate for a convex integrand.)

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5. 2019 · Q71 Trapezoidal Rule · Numerical

QUESTION

The value estimated for $\ln 2$, obtained by applying the trapezoidal rule to $\int_1^2 \frac{1}{x}\,dx$ with two equal divisions of $[1,2]$, is

OPTIONS

(a) $\frac{1}{2}$

(b) $\frac{17}{24}$

(c) $\frac{19}{24}$

(d) 1

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{17}{24}$

EXAM SHORTCUT (AI-derived explanation)

With h = 0.5 the ordinates are 1, 2/3, 1/2. Trapezoidal gives (0.5)[(1 + 0.5)/2 + 2/3] = 0.5(0.75 + 0.6667) = 0.7083 = 17/24.

TIPS & TRICKS (AI-derived explanation)

- Trapezoidal with two strips: T = h[(f0 + f2)/2 + f1]. Only the two END ordinates are halved.
- Keep fractions: (1/2)[(1 + 1/2)/2 + 2/3] = (1/2)(3/4 + 2/3) = (1/2)(17/12) = 17/24. Exact fractions beat decimals when the options are fractions.
- 1/x is convex, so the trapezoidal rule OVERESTIMATES. The true ln 2 = 0.6931, and 17/24 = 0.7083 is duly larger - a strong confirmation.
- Option (c) 19/24 = 0.7917 is too far above ln 2 to be a two-strip trapezoidal value; the overestimate for such a smooth function is small.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. We approximate the integral of 1/x from 1 to 2 using the trapezoidal rule with n = 2 equal divisions, so h = (2-1)/2 = 0.5.
2. The nodes are x = 1, 1.5, 2 with ordinates f0 = 1, f1 = 1/1.5 = 2/3, f2 = 1/2.
3. Trapezoidal rule: T = h[(f0 + f2)/2 + f1].
4. T = 0.5[(1 + 0.5)/2 + 2/3] = 0.5[3/4 + 2/3].
5. 3/4 + 2/3 = 9/12 + 8/12 = 17/12, so T = (1/2)(17/12) = 17/24.

6. Numerically 17/24 = 0.7083, slightly above the true ln 2 = 0.6931 as expected for a convex integrand. Option (b) is correct.

---

6. 2019 · Q72 Simpson's One-Third Rule · Factual

QUESTION

Simpson's one-third rule for evaluation of $\int_a^b f(x)\ dx$ requires the interval $[a,b]$ to be divided into

OPTIONS

(a) an even number of subintervals of equal width

(b) an odd number of subintervals of equal width

(c) any number of subintervals of equal width

(d) any number of subintervals

ANSWER (AI-derived — no official key exists for this paper)

(a) an even number of subintervals of equal width

EXAM SHORTCUT (AI-derived explanation)

Simpson's 1/3 rule fits a parabola over each PAIR of strips, so the number of equal-width subintervals must be even.

TIPS & TRICKS (AI-derived explanation)

- Rule of thumb: Simpson's 1/3 needs n divisible by 2; Simpson's 3/8 needs n divisible by 3; the trapezoidal rule works for any n.
- The word 'equal' matters too - all these Newton-Cotes formulas assume equally spaced nodes, which eliminates option (d).
- The weight pattern 1, 4, 2, 4, 2, ..., 4, 1 only closes correctly when n is even; with odd n the final strip is left without a partner.
- If n happens to be odd, apply the 3/8 rule to the first three strips and the 1/3 rule to the rest - a standard practical fix worth remembering.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Simpson's one-third rule is derived by replacing f over each PAIR of adjacent subintervals by the parabola through the three corresponding points.
2. Each such parabola therefore consumes exactly two subintervals.
3. For the whole interval [a, b] to be covered by complete pairs, the number of subintervals n must be even.
4. The derivation also requires the nodes to be equally spaced with h = (b-a)/n, so the subintervals must be of equal width.
5. Hence [a, b] must be divided into an even number of subintervals of equal width, option (a).

---

7. 2019 · Q73 Simpson's One-Third Rule · Numerical

QUESTION

Consider the table:

x	-1	0	1	2	3	4	5
f(x)	2	1	2	5	10	17	26

The value of $\int_{-1}^{5} f(x)\ dx$, computed by Simpson's $\frac{3}{8}$th rule, is

OPTIONS

(a) 48

(b) 47

(c) 46

(d) 45

ANSWER **(AI-derived — no official key exists for this paper)**

(a) 48

EXAM SHORTCUT **(AI-derived explanation)**

Six strips with h = 1. Composite 3/8 rule: (3/8)[f0 + 3f1 + 3f2 + 2f3 + 3f4 + 3f5 + f6] = (3/8)(128) = 48.

TIPS & TRICKS **(AI-derived explanation)**

- Composite 3/8 weight pattern: 1, 3, 3, 2, 3, 3, 2, ..., 3, 3, 1 - the weight 2 appears at every node where two groups of three strips join.
- Check divisibility first: n = 6 is a multiple of 3, so the 3/8 rule applies cleanly.
- The data are exactly f(x) = x^2 + 1, and Simpson's rules integrate quadratics EXACTLY, so the answer must equal the true value 48 - a decisive cross-check.
- Bracket sum: 2 + 3 + 6 + 10 + 30 + 51 + 26 = 128, and (3/8)(128) = 48. Keeping the weighted terms in a single column reduces slips.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. The tabulated points run from x = -1 to x = 5 with h = 1, giving n = 6 subintervals, which is a multiple of 3 so Simpson's three-eighth rule applies.
2. Ordinates: f0 = 2, f1 = 1, f2 = 2, f3 = 5, f4 = 10, f5 = 17, f6 = 26.
3. Composite three-eighth rule: I = (3h/8)[f0 + 3f1 + 3f2 + 2f3 + 3f4 + 3f5 + f6].
4. Weighted sum: 2 + 3(1) + 3(2) + 2(5) + 3(10) + 3(17) + 26 = 2 + 3 + 6 + 10 + 30 + 51 + 26 = 128.
5. With h = 1, I = (3/8)(128) = 48.
6. Check: the data satisfy f(x) = x^2 + 1, whose exact integral from -1 to 5 is [x^3/3 + x] = (125/3 + 5) - (-1/3 - 1) = 48. Option (a) is correct.

8. 2020 · Q62 Weddle's Rule · Conceptual

QUESTION

To solve $\int_0^1 (1+x)\ dx$ by numerical integration, which method is best?

OPTIONS

(a) Trapezoidal rule

(b) Simpson's one-third rule

(c) Simpson's three-eighth rule

(d) Weddle's rule

ANSWER (AI-derived — no official key exists for this paper)

(a) Trapezoidal rule

EXAM SHORTCUT (AI-derived explanation)

The integrand 1 + x is LINEAR, and the trapezoidal rule is already exact for linear functions - so it gives the exact answer with the least work.

TIPS & TRICKS (AI-derived explanation)

- Match the rule to the degree: trapezoidal is exact for degree 1, Simpson's 1/3 for degree 3, Simpson's 3/8 for degree 3, Weddle's for degree 5.
- 'Best' in numerical work means adequate accuracy for the least effort. Since all four rules would be exact here, the simplest one wins.
- The trapezoidal rule needs only two ordinates; Simpson's 3/8 needs four and Weddle's needs seven - a large amount of extra work for no extra accuracy.
- Geometric check: the region under y = 1 + x from 0 to 1 is a trapezium of area (1 + 2)/2 = 1.5, exactly what the rule returns.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The integrand is f(x) = 1 + x, a polynomial of degree 1.
2. The trapezoidal rule replaces the curve by a straight line joining the endpoints; since the integrand IS a straight line, the replacement introduces no error.
3. Indeed the trapezoidal estimate with a single strip is (1/2)[f(0) + f(1)] = (1/2)(1 + 2) = 1.5, which is exactly the true value of the integral.
4. Simpson's rules and Weddle's rule would also give the exact answer, but they require 3, 4 and 7 ordinates respectively - more computation for no gain.
5. The best method is therefore the simplest one that is exact here.
6. Hence the trapezoidal rule, option (a).

9. 2020 · Q73 Simpson's One-Third Rule · Conceptual

QUESTION

Let $A=\int_0^{10} x^2\ dx$. If B is the value obtained by Simpson's one-third rule dividing $[0,10]$ into 10 equal parts, then $|A\text{-}B|$

OPTIONS

(a) is greater than 10

(b) belongs to $(5,10)$

(c) belongs to $(0,5)$

(d) is 0

ANSWER (AI-derived — no official key exists for this paper)

(d) is 0

EXAM SHORTCUT (AI-derived explanation)

Simpson's one-third rule is exact for polynomials up to degree 3. The integrand x^2 is quadratic, so the numerical value equals the exact one and |A - B| = 0.

TIPS & TRICKS (AI-derived explanation)

- Exactness degrees: trapezoidal 1, Simpson's 1/3 and 3/8 both 3, Weddle's 5. Compare the integrand's degree with these before computing anything.
- Simpson's 1/3 fits parabolas yet is exact for CUBICS too - the odd-degree error term cancels by symmetry. That bonus degree is a favourite examiner point.
- The error term is -(b-a)h^4 f''''(xi)/180; for f(x) = x^2 the fourth derivative is identically zero, so the error is exactly zero.
- Verify if you wish: the exact value is 1000/3 = 333.33, and Simpson's rule with h = 1 returns the same - but the exactness argument settles it in five seconds.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Simpson's one-third rule is derived by fitting a parabola through each consecutive triple of points, so it integrates any quadratic exactly.
2. Its error term is -(b - a) h^4 f''''(xi)/180, which involves the FOURTH derivative of the integrand.
3. Here f(x) = x^2, so f''''(x) = 0 identically and the error term vanishes.
4. Therefore the Simpson estimate B equals the exact value A = the integral of x^2 from 0 to 10 = 1000/3.
5. Hence |A - B| = 0.
6. Option (d) is correct.

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10. 2020 · Q76 Simpson's One-Third Rule · Numerical

QUESTION

$\int_0^1 (3x^2+25)\ dx$ evaluated by Simpson's one-third rule with $h=1/4$ gives

OPTIONS

(a) 23

(b) 24.56

(c) 25.33

(d) 26

ANSWER **(AI-derived — no official key exists for this paper)**

(d) 26

EXAM SHORTCUT **(AI-derived explanation)**

Simpson's 1/3 is exact for quadratics, so the answer equals the exact integral: $[x^3 + 25x]$ from 0 to 1 = 26.

TIPS & TRICKS **(AI-derived explanation)**

- Recognise exactness first: the integrand $3x^2 + 25$ is quadratic and Simpson's 1/3 is exact up to degree 3, so no arithmetic is strictly necessary.
- If you do compute: h = 1/4 gives n = 4 (even, as required), ordinates 25, 25.1875, 25.75, 26.6875, 28 and weighted sum 312, so (h/3)(312) = (1/12)(312) = 26.
- The weight pattern for n = 4 is 1, 4, 2, 4, 1 - check it sums to 12 = 3n/... simply confirm 1+4+2+4+1 = 12, matching 3/h times the interval length.
- Options (b) and (c) are the trapezoidal and mid-point style approximations; recognising exactness protects you from choosing an 'approximate-looking' answer.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. With h = 1/4 on [0,1] there are n = 4 subintervals, an even number, so Simpson's one-third rule applies.
2. Ordinates: f(0) = 25, f(0.25) = 3(0.0625) + 25 = 25.1875, f(0.5) = 3(0.25) + 25 = 25.75, f(0.75) = 3(0.5625) + 25 = 26.6875, f(1) = 28.
3. Simpson's rule: I = (h/3)[f0 + 4f1 + 2f2 + 4f3 + f4].
4. Weighted sum: 25 + 4(25.1875) + 2(25.75) + 4(26.6875) + 28 = 25 + 100.75 + 51.5 + 106.75 + 28 = 312.
5. I = (0.25/3)(312) = 312/12 = 26.
6. This matches the exact value $[x^3 + 25x]$ from 0 to 1 = 1 + 25 = 26, as it must since Simpson's rule is exact for quadratics. Option (d) is correct.

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11. **2020 · Q77** Numerical Integration · Numerical

QUESTION

A numerical integration formula is written as $\int_{-1}^{1} f(x)\ dx = \frac{1}{2}[f(-1) + cf(\frac{1}{3})]$. The value of c for which the method has the highest possible order is

OPTIONS

(a) 1/2

(b) 3

(c) 1

(d) -1

ANSWER **(AI-derived — no official key exists for this paper)**

(b) 3

EXAM SHORTCUT **(AI-derived explanation)**

Force exactness for f = 1: the integral is 2 and the formula gives (1/2)(1 + c), so c = 3. That choice then also integrates x and x^2 exactly.

TIPS & TRICKS **(AI-derived explanation)**

- Method of undetermined coefficients: impose exactness on 1, x, x^2, ... in order. With one unknown, the constant function alone determines it; the remaining checks confirm the ORDER achieved.
- Check f = x: (1/2)[-1 + 3(1/3)] = 0, matching the exact 0. Check f = x^2: (1/2)[1 + 3(1/9)] = 2/3, matching the exact 2/3. So the rule is exact up to degree 2.
- The node 1/3 is not arbitrary - it is exactly the position that, paired with -1, makes a two-point rule reach degree 2 (a Radau-type quadrature).
- Test f = x^3: (1/2)[-1 + 3(1/27)] = -4/9 while the exact value is 0, so degree 2 is the highest order attained - confirming that c = 3 is optimal, not merely adequate.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Require the rule Q(f) = (1/2)[f(-1) + c f(1/3)] to reproduce the integral of f from -1 to 1 for polynomials of as high a degree as possible.
2. Take f(x) = 1: the exact integral is 2, and Q(1) = (1/2)(1 + c). Setting (1/2)(1 + c) = 2 gives c = 3.
3. Verify with f(x) = x: exact integral 0; Q(x) = (1/2)[-1 + 3(1/3)] = (1/2)(0) = 0. Exact.
4. Verify with f(x) = x^2: exact integral 2/3; Q(x^2) = (1/2)[1 + 3(1/9)] = (1/2)(4/3) = 2/3. Exact.
5. With f(x) = x^3: exact integral 0 but Q(x^3) = (1/2)[-1 + 3/27] = -4/9, so degree 2 is the highest order achieved.
6. Hence c = 3 gives the highest possible order, option (b).

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12. **2020 · Q79** Simpson's One-Third Rule · Numerical

QUESTION

A curve passes through:

x	1	1.5	2	2.5	3	3.5	4
y	2	2.4	2.7	2.8	3	2.6	2.1

The area bounded by the curve, the x-axis, and the lines $x=1, x=4$, via Simpson's one-third rule, is

OPTIONS

(a) 7.81

(b) 7.78

(c) 7.74

(d) 7.72

ANSWER **(AI-derived — no official key exists for this paper)**

(b) 7.78

EXAM SHORTCUT **(AI-derived explanation)**

With h = 0.5 and n = 6, Simpson's 1/3 gives (0.5/3)[2 + 4(2.4) + 2(2.7) + 4(2.8) + 2(3) + 4(2.6) + 2.1] = (1/6)(46.7) = 7.783.

TIPS & TRICKS **(AI-derived explanation)**

- Composite Simpson 1/3 weights: 1, 4, 2, 4, 2, 4, 1 - odd-indexed ordinates get 4 and even-indexed interior ones get 2. Check n is even before starting (here n = 6).
- Group the weighted terms: the 4-weighted ordinates are 2.4 + 2.8 + 2.6 = 7.8 giving 31.2, and the 2-weighted are 2.7 + 3 = 5.7 giving 11.4; ends contribute 4.1. Total 46.7.
- h/3 = 0.5/3 = 1/6, so the final division is by 6: 46.7/6 = 7.7833.
- The options differ in the second decimal, so carry the weighted sum exactly (46.7) and divide once at the end rather than rounding intermediate products.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. The abscissae run from x = 1 to x = 4 in steps of h = 0.5, giving n = 6 subintervals - an even number, so Simpson's one-third rule applies.
2. Ordinates: y0 = 2, y1 = 2.4, y2 = 2.7, y3 = 2.8, y4 = 3, y5 = 2.6, y6 = 2.1.
3. Composite rule: Area = (h/3)[y0 + 4(y1 + y3 + y5) + 2(y2 + y4) + y6].
4. Odd-indexed sum: 2.4 + 2.8 + 2.6 = 7.8, contributing 4(7.8) = 31.2. Even-indexed interior sum: 2.7 + 3 = 5.7, contributing 2(5.7) = 11.4. End ordinates: 2 + 2.1 = 4.1.
5. Total weighted sum = 31.2 + 11.4 + 4.1 = 46.7.
6. Area = (0.5/3)(46.7) = 46.7/6 = 7.783, i.e. 7.78. Option (b) is correct.

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13. 2021 · Q45 Simpson's Three-Eighth Rule · Factual

QUESTION

The number of subintervals required in Simpson's three-eighth rule is a multiple of

OPTIONS

(a) 2

(b) 3

(c) 4

(d) 5

ANSWER (AI-derived — no official key exists for this paper)

(b) 3

EXAM SHORTCUT (AI-derived explanation)

Simpson's three-eighth rule fits a cubic over each group of THREE subintervals, so the total number of subintervals must be a multiple of 3.

TIPS & TRICKS (AI-derived explanation)

- Divisibility rules: trapezoidal any n, Simpson's 1/3 needs n even, Simpson's 3/8 needs n a multiple of 3, Weddle's needs n a multiple of 6.
- The name encodes the answer: the '3' in three-eighth refers to the three strips consumed by each cubic arc.
- The weight pattern 1, 3, 3, 1 (repeating with 2 at the joins) only closes correctly when the strips group into complete threes.
- If n is not a multiple of 3, combine rules - for example apply the 3/8 rule to three strips and the 1/3 rule to the remaining even number.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Simpson's three-eighth rule is derived by replacing the integrand over each group of THREE consecutive subintervals by the cubic through the four corresponding points.
2. Each cubic arc therefore consumes exactly three subintervals.
3. For the whole interval to be covered by complete groups, the total number of subintervals n must be divisible by 3.
4. The composite weight pattern 1, 3, 3, 2, 3, 3, 2, ..., 3, 3, 1 closes correctly only under this condition.
5. Hence n must be a multiple of 3, option (b).

---

14. 2021 · Q63 Simpson's Three-Eighth Rule · Theoretical

QUESTION

Consider the following statements: 1. Simpson's three-eighth rule assumes the integrand is a degree-3 polynomial on each sub-interval $[x_0+(i\text{-}1)h,\ x_0+ih]$, $1\leq\ i\leq\ n$. 2. To approximate $\int_a^b f(x)\,dx$ by the Trapezoidal rule, $[a,b]$ is divided into any number of equal sub-intervals.

OPTIONS

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

ANSWER **(AI-derived — no official key exists for this paper)**

(b) 2 only

EXAM SHORTCUT **(AI-derived explanation)**

Statement 1 misstates the 3/8 rule - it fits a cubic over each group of THREE subintervals, not over each single subinterval. Statement 2 is correct: the trapezoidal rule places no divisibility condition on n.

TIPS & TRICKS **(AI-derived explanation)**

- Count the strips per arc: trapezoidal uses 1 strip per line, Simpson's 1/3 uses 2 strips per parabola, Simpson's 3/8 uses 3 strips per cubic.
- A cubic is determined by FOUR points, which span three subintervals - that arithmetic alone exposes the error in statement 1.
- The trapezoidal rule is the only one of the family with no divisibility requirement, which is precisely what statement 2 asserts.
- Read such statements for the exact interval described; the examiner's alteration is often a single word ('each sub-interval' rather than 'each group of three').

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Statement 1: Simpson's three-eighth rule is obtained by replacing the integrand by the cubic through FOUR consecutive tabulated points, which span THREE subintervals.
2. The statement instead asserts that a cubic is assumed on each single subinterval [x0 + (i-1)h, x0 + ih], which is not how the rule is constructed. FALSE.
3. Statement 2: the trapezoidal rule approximates the curve by a straight line on each subinterval, and the composite formula T = h[(f0 + fn)/2 + f1 + ... + f(n-1)] closes correctly for ANY number of equal subintervals.
4. Unlike Simpson's rules, it imposes no requirement that n be even or a multiple of 3. TRUE.
5. Only statement 2 is correct.
6. Hence option (b) is correct.

---

15. **2022 · Q21** Trapezoidal Rule · Numerical

QUESTION

Suppose f is a three-time differentiable function on $[0,2]$. $\int_0^2 f(x)\ dx$ is computed using trapezoidal rule partitioning $[0,2]$ into 10 equal subintervals. The error is given by:

OPTIONS

(a) $-\frac{2}{25}f'(t)$ for some $t \in [0,2]$

(b) $-\frac{1}{150}f''(t)$ for some $t \in [0,2]$

(c) $-\frac{1}{300}f''(t)$ for some $t \in [0,2]$

(d) $-\frac{1}{150}f'''(t)$ for some $t \in [0,2]$

ANSWER (AI-derived — no official key exists for this paper)

(b) $-\frac{1}{150} f''(t)$ for some $t \in [0,2]$

EXAM SHORTCUT (AI-derived explanation)

Composite trapezoidal error = -(b-a)h^2 f''(t)/12. With b - a = 2 and h = 2/10 = 0.2, this is -(2)(0.04)/12 = -1/150 times f''(t).

TIPS & TRICKS (AI-derived explanation)

- Memorise the composite error terms: trapezoidal -(b-a)h^2 f''/12; Simpson's 1/3 -(b-a)h^4 f''''/180. The DERIVATIVE ORDER identifies the rule.
- Trapezoidal error involves the SECOND derivative, so options (a) and (d) with f' and f''' are eliminated before any arithmetic.
- Compute h first: h = (2-0)/10 = 0.2, so h^2 = 0.04 and (2)(0.04)/12 = 0.08/12 = 1/150.
- Option (c) -1/300 comes from using (b-a) = 1 or from halving twice - keep (b-a) = 2 explicitly in the formula.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The composite trapezoidal rule on [a,b] with n equal subintervals has error E = -(b - a) h^2 f''(t)/12 for some t in [a,b], where h = (b-a)/n.
2. Here a = 0, b = 2 and n = 10, so h = 2/10 = 0.2 and h^2 = 0.04.
3. Substituting: E = -(2)(0.04) f''(t)/12 = -0.08 f''(t)/12 = -f''(t)/150.
4. The error therefore involves the second derivative with coefficient -1/150.
5. Option (b) is correct.

---

16. **2022 · Q24** Simpson's One-Third Rule · Numerical

QUESTION

Consider the data:

x	0	1.0	1.5	2.0
f(x)	1.1	2.4	5.7	8.1

Let $I = \int_0^2 f(x)\ dx$. Combining Trapezoidal rule on [0,1] and Simpson's one-third rule on [1,2] gives I as:

OPTIONS

(a) 7.30

(b) 5.55

(c) 3.65

(d) 3.33

ANSWER (AI-derived — no official key exists for this paper)

(a) 7.30

EXAM SHORTCUT (AI-derived explanation)

Trapezoidal on [0,1]: (1/2)(1.1 + 2.4) = 1.75. Simpson's 1/3 on [1,2] with h = 0.5: (0.5/3)[2.4 + 4(5.7) + 8.1] = 5.55. Total 7.30.

TIPS & TRICKS (AI-derived explanation)

- Split the interval to match the data: only two points are available on [0,1] (trapezoidal), but three equally spaced points on [1,2] (Simpson's 1/3).
- Simpson's 1/3 needs h = 0.5 here, so the factor is h/3 = 1/6 - compute this before substituting.
- The weighted sum on [1,2] is 2.4 + 22.8 + 8.1 = 33.3, and 33.3/6 = 5.55. Working the bracket first keeps the arithmetic clean.
- Option (b) 5.55 is the second piece alone and option (c) 3.65 halves the total - both punish stopping short of the final addition.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. On [0,1] only the endpoints x = 0 and x = 1 are tabulated, so the trapezoidal rule applies with h = 1.
2. Trapezoidal contribution: (h/2)[f(0) + f(1)] = (1/2)(1.1 + 2.4) = (1/2)(3.5) = 1.75.
3. On [1,2] three equally spaced points are available - x = 1, 1.5, 2 with h = 0.5 - so Simpson's one-third rule applies.
4. Simpson contribution: (h/3)[f(1) + 4f(1.5) + f(2)] = (0.5/3)[2.4 + 4(5.7) + 8.1].
5. The bracket is 2.4 + 22.8 + 8.1 = 33.3, so the contribution is 33.3/6 = 5.55.
6. Total I = 1.75 + 5.55 = 7.30, option (a).

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17. 2023 · Q47 Trapezoidal Rule · Numerical

QUESTION

The integral $\int_0^1 x^3\, dx$ is evaluated by the Trapezoidal rule with step length h. Then the error is:

OPTIONS

(a) $h^2/12$

(b) $h^2/6$

(c) $h^2/4$

(d) $h^2/3$

ANSWER (AI-derived — no official key exists for this paper)

(c) $h^2/4$

EXAM SHORTCUT (AI-derived explanation)

By Euler-Maclaurin the composite trapezoidal error is $(h^2/12)[f'(1) - f'(0)]$. With $f' = 3x^2$ this is $(h^2/12)(3 - 0) = h^2/4$.

TIPS & TRICKS (AI-derived explanation)

- Euler-Maclaurin form: $T - I = (h^2/12)[f'(b) - f'(a)] + O(h^4)$. It gives an EXPLICIT error rather than one containing an unknown point xi.
- The alternative form $-(b-a)h^2 f''(xi)/12$ is awkward here because $f'' = 6x$ varies over the interval; the derivative-difference form avoids the ambiguity.
- $f'(x) = 3x^2$, so $f'(1) - f'(0) = 3$. Then $3h^2/12 = h^2/4$.
- The trapezoidal rule OVERESTIMATES for a convex integrand, and x^3 is convex on [0,1], so a positive error of $h^2/4$ is exactly what is expected.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For the composite trapezoidal rule the Euler-Maclaurin expansion gives $T - I = (h^2/12)[f'(b) - f'(a)] + O(h^4)$.
2. Here $f(x) = x^3$ on [0,1], so $f'(x) = 3x^2$.
3. $f'(1) = 3$ and $f'(0) = 0$, so $f'(b) - f'(a) = 3$.
4. Hence the error is $(h^2/12)(3) = h^2/4$.
5. The sign is positive, consistent with the trapezoidal rule overestimating the integral of the convex function x^3.
6. Option (c) is correct.

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18. 2023 · Q48 Numerical Integration · Numerical

QUESTION

What is the value of b if $\int_0^h f(x)\ dx = ahf(0)+bhf(\frac{h}{3})+chf(h)$ exists for a polynomial of degree as high as possible?

OPTIONS

(a) 1

(b) 3/4

(c) 1/2

(d) 0

ANSWER (AI-derived — no official key exists for this paper)

(b) 3/4

EXAM SHORTCUT (AI-derived explanation)

Impose exactness on 1, x and x^2. The equations a + b + c = 1, b/3 + c = 1/2 and b/9 + c = 1/3 give b = 3/4 (with c = 1/4 and a = 0).

TIPS & TRICKS (AI-derived explanation)

- Method of undetermined coefficients: force exactness on the basis 1, x, x^2 in turn. Three unknowns require exactly three equations.
- Normalise by h at the start: dividing every equation by the appropriate power of h leaves pure numbers, so h never appears in the answer.
- Subtract the last two equations to eliminate c: b/3 - b/9 = 1/2 - 1/3 gives 2b/9 = 1/6 and b = 3/4 in one step.
- Back-substituting gives c = 1/4 and a = 0, so the node at 0 carries no weight at all - a striking feature worth noting as a check.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Require the rule to be exact for polynomials of the highest possible degree, testing f = 1, x, x^2 in turn.
2. f(x) = 1: the integral from 0 to h is h, and the rule gives (a + b + c)h, so a + b + c = 1.
3. f(x) = x: the integral is $h^2/2$, and the rule gives bh(h/3) + ch(h) = h^2(b/3 + c), so b/3 + c = 1/2.
4. f(x) = x^2: the integral is $h^3/3$, and the rule gives bh(h^2/9) + ch(h^2) = h^3(b/9 + c), so b/9 + c = 1/3.
5. Subtracting the third equation from the second: b/3 - b/9 = 1/6, i.e. 2b/9 = 1/6, so b = 3/4.
6. Then c = 1/2 - (3/4)/3 = 1/4 and a = 1 - 3/4 - 1/4 = 0. Hence b = 3/4, option (b).

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19. 2023 · Q53 Simpson's Three-Eighth Rule · Conceptual

QUESTION

Suppose $t_0=0<t_1<t_2<\cdots<t_n=1$. If f is a continuous function and $f(t_i)$ are given for $0\le i\le n$, which quadrature formula is applicable to compute $\int_0^1 f(x)\ dx$?

OPTIONS

(a) Trapezoidal rule

(b) Simpson's one-third rule

(c) Simpson's three-eighth rule

(d) No quadrature formula

ANSWER (AI-derived — no official key exists for this paper)

(a) Trapezoidal rule

EXAM SHORTCUT (AI-derived explanation)

The nodes are arbitrary and need not be equally spaced. Only the trapezoidal rule works on unequal intervals; both Simpson's rules require equal spacing.

TIPS & TRICKS (AI-derived explanation)

- Read the node specification: t_0 < t_1 < ... < t_n with no equal-spacing assumption. That single observation decides the question.
- The composite trapezoidal rule sums (t_(i+1) - t_i)(f_i + f_(i+1))/2 over the panels, so each panel may have its own width.
- Simpson's rules are derived from equally spaced parabolic and cubic fits, so they break down when the spacing varies.
- Option (d) is wrong because the trapezoidal rule always applies - a quadrature formula does exist for any set of distinct ordered nodes.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The nodes satisfy only 0 = t_0 < t_1 < ... < t_n = 1; nothing is said about equal spacing.
2. Simpson's one-third rule requires an even number of EQUALLY spaced subintervals, and Simpson's three-eighth rule requires the number of equally spaced subintervals to be a multiple of 3. Neither can be applied to arbitrary nodes.
3. The trapezoidal rule, by contrast, approximates the curve by a straight line on each panel independently, so it accommodates panels of different widths.
4. The composite formula is the sum over i of (t_(i+1) - t_i)[f(t_i) + f(t_(i+1))]/2.
5. Since this is always available, option (d) is also wrong.
6. Hence the applicable formula is the trapezoidal rule, option (a).

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20. 2023 · Q54 Simpson's Three-Eighth Rule · Conceptual

QUESTION

Let $f:[0,1]\to\mathbb{R}$ be a continuous function. If $f(\frac{t}{10})$ are given for $t=0,1,2,3,4,5$ and $f(\frac{t}{20})$ are given for $t=11,12,13,\ldots,20$, which quadrature formula will one use to utilize the given data for the computation of $\int_0^1 f(x)\ dx$?

OPTIONS

(a) Trapezoidal rule only

(b) Simpson's one-third rule only

(c) Simpson's three-eighth rule

(d) Trapezoidal rule or combination of Trapezoidal rule and Simpson's one-third rule

ANSWER (AI-derived — no official key exists for this paper)

(d) Trapezoidal rule or combination of Trapezoidal rule and Simpson's one-third rule

EXAM SHORTCUT (AI-derived explanation)

On [0, 0.5] there are 5 equal strips of width 0.1 - an odd number, so only the trapezoidal rule fits; on [0.5, 1] there are 10 equal strips of width 0.05, an even number, so Simpson's one-third rule applies there.

TIPS & TRICKS (AI-derived explanation)

- Translate the data first: t/10 for t = 0..5 gives x = 0, 0.1, ..., 0.5, and t/20 for t = 11..20 gives x = 0.55, 0.60, ..., 1.00, with x = 0.5 shared.
- Count the panels in each half: 5 on the left (odd, so Simpson's 1/3 is barred) and 10 on the right (even, so it is allowed).
- The trapezoidal rule works everywhere, so the honest answer is 'trapezoidal, or trapezoidal on the left combined with Simpson's 1/3 on the right'.
- Simpson's three-eighth rule would need panel counts divisible by 3, which neither half provides - so option (c) is out.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The supplied abscissae are x = t/10 for t = 0,...,5, i.e. 0, 0.1, 0.2, 0.3, 0.4, 0.5, and x = t/20 for t = 11,...,20, i.e. 0.55, 0.60, ..., 1.00.
2. On [0, 0.5] the spacing is uniform at 0.1, giving 5 subintervals - an ODD number, so Simpson's one-third rule cannot be applied to the whole half.
3. On [0.5, 1] the spacing is uniform at 0.05, giving 10 subintervals - an EVEN number, so Simpson's one-third rule applies there.
4. Neither half has a panel count divisible by 3, so Simpson's three-eighth rule is unavailable.
5. The trapezoidal rule can be applied throughout, or one may combine the trapezoidal rule on [0, 0.5] with Simpson's one-third rule on [0.5, 1] for greater accuracy.
6. Hence option (d) is correct.

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21. 2024 · Q63 Simpson's One-Third Rule · Numerical

QUESTION

Use Simpson's one-third rule to evaluate $\int_1^3 x^4\, dx$ by dividing [1,3] into two parts. What is the value of the error?

OPTIONS

(a) 0.12

(b) 0.19

(c) 0.27

(d) 0.33

ANSWER (AI-derived — no official key exists for this paper)

(c) 0.27

EXAM SHORTCUT (AI-derived explanation)

With h = 1 and nodes 1, 2, 3, Simpson gives (1/3)[1 + 64 + 81] = 48.667, while the exact value is 242/5 = 48.4. The error is 0.267, i.e. about 0.27.

TIPS & TRICKS (AI-derived explanation)

- Simpson's 1/3 is exact only up to CUBICS, so for x^4 there is a genuine error - that is the point of the question.
- Compute both values: Simpson's bracket is 1 + 4(16) + 81 = 146, so the estimate is 146/3 = 48.6667; the exact integral is $(3^5 - 1)/5 = 48.4$.
- Error formula check: $-(b-a)h^4$ f''''(xi)/180 with f'''' = 24 gives (2)(1)(24)/180 = 0.2667 in magnitude - matching exactly, since the fourth derivative is constant.
- Simpson OVERestimates here (the fourth derivative is positive), so the error 48.667 - 48.4 is positive - a useful sign check.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Dividing [1,3] into two parts gives h = 1 with nodes x = 1, 2, 3 and ordinates f = 1, 16, 81.
2. Simpson's one-third rule: I = (h/3)[f0 + 4f1 + f2] = (1/3)[1 + 64 + 81] = 146/3 = 48.6667.
3. The exact value is the integral of x^4 from 1 to 3 = $[x^5/5]$ from 1 to 3 = (243 - 1)/5 = 48.4.
4. Error = 48.6667 - 48.4 = 0.2667.
5. This agrees with the error formula $-(b-a)h^4$ f''''(xi)/180 = -(2)(1)(24)/180 = -0.2667 in magnitude, since f''''(x) = 24 is constant for x^4.
6. The error is approximately 0.27, option (c).

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22. 2024 · Q65 Trapezoidal Rule · Numerical

QUESTION

$y=x^2+2x+3$ and y_0, y_1, y_2 are the values of y for $x=a, a+h, a+2h$ respectively. What is the value of $\int_a^{a+2h} y\, dx$? (Use trapezoidal rule)

OPTIONS

(a) $\frac{h}{2}(y_0+2y_1+y_2)$

(b) $\frac{h}{2}(2y_0+y_1+2y_2)$

(c) $\frac{h}{2}(y_0+2y_1+2y_2)$

(d) $\frac{h}{2}(y_0+y_1+y_2)$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{h}{2}(y_0+2y_1+y_2)$

EXAM SHORTCUT (AI-derived explanation)

Composite trapezoidal over two strips: (h/2)[y0 + 2y1 + y2] - the interior ordinate carries weight 2 and the ends weight 1.

TIPS & TRICKS (AI-derived explanation)

- Composite trapezoidal weights are 1, 2, 2, ..., 2, 1 with prefactor h/2 - only the two END ordinates are halved.

- The weights must total 2n for n strips so that a constant integrates correctly: here 1 + 2 + 1 = 4 = 2(2). Options (b), (c) and (d) fail this test.
- The specific function $x^2 + 2x + 3$ is a distractor - the trapezoidal FORMULA does not depend on it (though the rule is not exact for a quadratic).
- Contrast with Simpson's 1/3 over the same two strips, which would be (h/3)[y0 + 4y1 + y2] and WOULD be exact for this quadratic.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The interval [a, a+2h] is divided into two equal strips of width h, with ordinates y0, y1, y2 at x = a, a+h, a+2h.
2. The trapezoidal rule approximates the curve by a straight line on each strip.
3. First strip: (h/2)(y0 + y1). Second strip: (h/2)(y1 + y2).
4. Adding: (h/2)(y0 + y1 + y1 + y2) = (h/2)(y0 + 2y1 + y2).
5. The weights 1, 2, 1 sum to 4 = 2 x 2, as required for two strips.
6. Hence the value is (h/2)(y0 + 2y1 + y2), option (a).

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23. 2024 · Q68 Weddle's Rule · Conceptual

QUESTION

To find the approximate value of $\int_a^b y\,dx$, where $b=a+8h$, which of the following rules can be applied?

OPTIONS

(a) Trapezoidal rule and Simpson's three-eighths rule

(b) Trapezoidal rule and Simpson's one-third rule

(c) Simpson's one-third rule and Weddle rule

(d) Simpson's three-eighths rule and Weddle rule

ANSWER (AI-derived — no official key exists for this paper)

(b) Trapezoidal rule and Simpson's one-third rule

EXAM SHORTCUT (AI-derived explanation)

b = a + 8h means 8 strips. Trapezoidal needs no condition and Simpson's 1/3 needs an even count - 8 qualifies. Simpson's 3/8 needs a multiple of 3 and Weddle a multiple of 6, neither of which 8 is.

TIPS & TRICKS (AI-derived explanation)

- Divisibility table: trapezoidal any n; Simpson's 1/3 needs n even; Simpson's 3/8 needs n divisible by 3; Weddle's needs n divisible by 6.
- Read the number of strips straight from b - a = 8h, so n = 8. Everything else is a divisibility check.
- 8 is even but not divisible by 3 or 6, so exactly two of the four rules apply - which pins the answer without further thought.

- In practice one could still apply the 3/8 rule to a sub-range and the 1/3 rule to the rest, but the question asks which rules apply to the whole interval.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Since b = a + 8h, the interval [a, b] is divided into n = 8 subintervals of equal width h.
2. The trapezoidal rule places no condition on n, so it applies.
3. Simpson's one-third rule requires n to be even; 8 is even, so it applies.
4. Simpson's three-eighth rule requires n to be a multiple of 3; 8 is not, so it does not apply.
5. Weddle's rule requires n to be a multiple of 6; 8 is not, so it does not apply.
6. Hence the applicable rules are the trapezoidal rule and Simpson's one-third rule, option (b).

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24. 2024 · Q69 Simpson's One-Third Rule · Numerical

QUESTION

The value of the integral $\int_1^3 f(x)\ dx$ is 5 by using trapezoidal rule and 3 by using Simpson's one-third rule dividing $[1,3]$ into minimum number of intervals. What is the value of $f(2)$?

OPTIONS

(a) 1/2

(b) 1

(c) 3/2

(d) 2

ANSWER (AI-derived — no official key exists for this paper)

(b) 1

EXAM SHORTCUT (AI-derived explanation)

The minimum for the trapezoidal rule is one strip: f(1) + f(3) = 5. The minimum for Simpson's 1/3 is two strips: (1/3)[f(1) + 4f(2) + f(3)] = 3, so f(1) + 4f(2) + f(3) = 9 and 4f(2) = 4, giving f(2) = 1.

TIPS & TRICKS (AI-derived explanation)

- 'Minimum number of intervals' differs by rule: 1 for the trapezoidal rule (h = 2) but 2 for Simpson's 1/3 (h = 1, since n must be even).
- The trapezoidal equation supplies f(1) + f(3) = 5 with no f(2) in it - substituting that into Simpson's equation isolates f(2) immediately.
- Simpson's bracket must equal 9 because the prefactor is h/3 = 1/3 and the stated value is 3.
- Consistency check: with f(1) + f(3) = 5 and f(2) = 1, Simpson gives (1/3)(5 + 4) = 3, exactly as stated.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For the trapezoidal rule the minimum number of intervals is 1, so h = 2 and the estimate is (h/2)[f(1) + f(3)] = f(1) + f(3).
2. Setting this equal to 5 gives f(1) + f(3) = 5.

3. For Simpson's one-third rule the number of intervals must be even, so the minimum is 2, giving h = 1 with nodes 1, 2, 3.
4. The estimate is (h/3)[f(1) + 4f(2) + f(3)] = (1/3)[f(1) + 4f(2) + f(3)], and setting this equal to 3 gives f(1) + 4f(2) + f(3) = 9.
5. Substituting f(1) + f(3) = 5: 5 + 4f(2) = 9, so 4f(2) = 4.
6. Hence f(2) = 1, option (b).

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25. 2024 · Q77 Numerical Integration · Numerical

QUESTION

If the formula

$$\int_{x_0}^{x_2} f(x)\ dx = \alpha\ f(x_0)+\beta\ f(x_0+h)+\gamma\ f(x_0+2h)$$

is exact for polynomial of as high order as possible for any $h>0$, then what are the values of α, β, γ respectively?

OPTIONS

(a) $h/3,\ 4h/3,\ h/3$

(b) $4h/3,\ h/3,\ h/3$

(c) $h,\ h/2,\ h$

(d) $h/2,\ h,\ h/2$

ANSWER (AI-derived — no official key exists for this paper)

(a) $h/3,\ 4h/3,\ h/3$

EXAM SHORTCUT (AI-derived explanation)

Three equally spaced nodes with maximum exactness is Simpson's one-third rule, whose weights are h/3, 4h/3, h/3.

TIPS & TRICKS (AI-derived explanation)

- Recognise the pattern: a three-point rule on [x0, x0+2h] with the highest possible order IS Simpson's 1/3 rule - no derivation needed.
- If you derive it: exactness for f = 1 gives alpha + beta + gamma = 2h, and symmetry gives alpha = gamma, so exactness for x^2 completes the system.
- The weights must sum to the interval length 2h: h/3 + 4h/3 + h/3 = 6h/3 = 2h. Only option (a) satisfies this AND the symmetry alpha = gamma.
- Bonus fact: this rule is exact for cubics too, because the odd-degree error term cancels by symmetry.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The three nodes x0, x0 + h, x0 + 2h are equally spaced across the interval of length 2h, and we require exactness for polynomials of the highest possible degree.
2. Exactness for f = 1: alpha + beta + gamma = 2h.

3. Exactness for f = x - x0 (odd about the midpoint): 0(alpha) + h(beta) + 2h(gamma) = $2h^2$, i.e. beta + 2gamma = 2h.
4. Exactness for f = $(x - x0)^2$: h^2 beta + $4h^2$ gamma = $8h^3/3$, i.e. beta + 4 gamma = 8h/3.
5. Subtracting the second from the third: 2 gamma = 2h/3, so gamma = h/3; then beta = 2h - 2h/3 = 4h/3 and alpha = 2h - 4h/3 - h/3 = h/3.
6. These are Simpson's one-third weights h/3, 4h/3, h/3, option (a).

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26. 2024 · Q78 Simpson's Three-Eighth Rule · Theoretical

QUESTION

For $\int_0^1 f(x)\ dx$, where $f(x)$ is cubic polynomial, consider the following statements: I. Trapezoidal rule will give the exact value of the integral. II. Simpson's one-third rule will give the exact value of the integral. III. Simpson's three-eighths rule will give the exact value of the integral. Which of the statements given above are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(b) II and III only

EXAM SHORTCUT (AI-derived explanation)

Both Simpson's rules are exact up to degree 3, but the trapezoidal rule is exact only up to degree 1. So II and III hold and I fails.

TIPS & TRICKS (AI-derived explanation)

- Degrees of exactness: trapezoidal 1, Simpson's 1/3 = 3, Simpson's 3/8 = 3, Weddle 5. Compare each with the integrand's degree 3.
- Simpson's 1/3 fits parabolas yet is exact for CUBICS because its odd-degree error term cancels by symmetry - a bonus degree worth remembering.
- The trapezoidal error involves f'', which is generally non-zero for a cubic, so statement I must fail.
- Simpson's 3/8 fits cubics directly, so its exactness for a cubic needs no special argument.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: the trapezoidal rule replaces the integrand by a straight line, and its error term involves f''. For a cubic f'' is not identically zero, so the rule is not exact. FALSE.
2. Statement II: Simpson's one-third rule has error term proportional to the FOURTH derivative, which vanishes for a cubic, so it integrates cubics exactly - a bonus degree beyond the parabola it fits. TRUE.

3. Statement III: Simpson's three-eighth rule is derived by fitting a cubic through four points, so it reproduces any cubic exactly; its error term also involves the fourth derivative. TRUE.
4. Only statements II and III are correct.
5. Hence option (b) is correct.

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27. 2024 · Q79 Simpson's One-Third Rule · Numerical

QUESTION

Consider the following data with regard to depth (y) at a distance (x) of a river 40 m wide:

Distance from one side, x	**0**	**10**	**20**	**30**	**40**
Depth, y (metre)	0	4	7	9	12

What is the approximate area of cross-section of the river? (Use Simpson's one-third rule)

OPTIONS

(a) 780 square metre

(b) 390 square metre

(c) 260 square metre

(d) 195 square metre

ANSWER (AI-derived — no official key exists for this paper)

(c) 260 square metre

EXAM SHORTCUT (AI-derived explanation)

With h = 10 and n = 4, Simpson's 1/3 gives (10/3)[0 + 4(4) + 2(7) + 4(9) + 12] = (10/3)(78) = 260 square metres.

TIPS & TRICKS (AI-derived explanation)

- Check n first: five ordinates give n = 4 subintervals, an even number, so Simpson's 1/3 applies.
- Weight pattern 1, 4, 2, 4, 1: the odd-indexed depths get 4 and the single interior even one gets 2.
- Weighted sum: 0 + 16 + 14 + 36 + 12 = 78, then multiply by h/3 = 10/3 to get 260.
- Plausibility: the average depth is roughly 6.5 m over 40 m, giving about 260 square metres - the answer is exactly of that order, unlike options (a) and (b).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The distances are 0, 10, 20, 30, 40 with h = 10, giving n = 4 subintervals - an even number, so Simpson's one-third rule applies.
2. Ordinates (depths): y0 = 0, y1 = 4, y2 = 7, y3 = 9, y4 = 12.
3. Simpson's rule: Area = (h/3)[y0 + 4y1 + 2y2 + 4y3 + y4].
4. Weighted sum: 0 + 4(4) + 2(7) + 4(9) + 12 = 0 + 16 + 14 + 36 + 12 = 78.
5. Area = (10/3)(78) = 780/3 = 260.

6. Hence the cross-sectional area is about 260 square metres, option (c).

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28. 2025 · Q24 Simpson's One-Third Rule · Numerical

QUESTION

Simpson's one-third rule applied to $\int_0^2 f(x)\ dx$ gives the value 2, where $f(1)=0.5$. If the trapezoidal rule is applied instead, what is the value of $\int_0^2 f(x)\ dx$?

OPTIONS

(a) 8

(b) 4

(c) 2

(d) 1

ANSWER (AI-derived — no official key exists for this paper)

(b) 4

EXAM SHORTCUT (AI-derived explanation)

Simpson's rule with h = 1 gives (1/3)[f(0) + 4(0.5) + f(2)] = 2, so f(0) + f(2) = 4. The single-strip trapezoidal rule then gives (2/2)[f(0) + f(2)] = 4.

TIPS & TRICKS (AI-derived explanation)

- Simpson's 1/3 on [0,2] needs two strips, so h = 1 and the nodes are 0, 1, 2 - which is why f(1) is supplied.
- Solve for the end ordinates: 3 x 2 = 6 = f(0) + 4(0.5) + f(2), so f(0) + f(2) = 4. That single quantity is all the trapezoidal rule needs.
- The trapezoidal rule over the whole interval as ONE trapezium uses only the endpoints, so f(1) drops out entirely - the design of the question.
- With h = 2 the prefactor h/2 = 1, so the answer is just f(0) + f(2) = 4.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Simpson's one-third rule on [0,2] requires an even number of strips, so the minimum is two, giving h = 1 with nodes 0, 1, 2.
2. The rule states (h/3)[f(0) + 4f(1) + f(2)] = 2, i.e. (1/3)[f(0) + 4(0.5) + f(2)] = 2.
3. Multiplying by 3: f(0) + 2 + f(2) = 6, so f(0) + f(2) = 4.
4. The trapezoidal rule applied over the whole interval as a single trapezium uses h = 2 and only the end ordinates: (h/2)[f(0) + f(2)] = (2/2)(4) = 4.
5. Option (b) is correct.

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29. 2025 · Q25 Simpson's One-Third Rule · Numerical

QUESTION

Consider the table:

x	0	1/4	1/2	3/4	1
f(x)	1	4/5	2/3	4/7	1/2

Simpson's one-third rule with 4 equal intervals approximates the area bounded by $f(x)$ and the x-axis from $x=0$ to $x=1$. What is the approximate area?

OPTIONS

(a) 0.6951

(b) 0.6944

(c) 0.6932

(d) 0.70

ANSWER (AI-derived — no official key exists for this paper)

(c) 0.6932

EXAM SHORTCUT (AI-derived explanation)

With h = 1/4 and n = 4, Simpson gives (1/12)[1 + 4(0.8) + 2(0.6667) + 4(0.5714) + 0.5] = (1/12)(8.3190) = 0.6932.

TIPS & TRICKS (AI-derived explanation)

- Weight pattern 1, 4, 2, 4, 1 with prefactor h/3 = 1/12. Check n = 4 is even before applying the rule.
- Convert the fractions to decimals carefully: 4/5 = 0.8, 2/3 = 0.6667, 4/7 = 0.5714. Four-decimal precision is needed since the options differ in the third decimal.
- Weighted sum: 1 + 3.2 + 1.3333 + 2.2857 + 0.5 = 8.3190, and 8.3190/12 = 0.69325.
- The tabulated values are f(x) = 1/(1+x), whose exact integral over (0,1) is ln 2 = 0.6931 - the near-agreement confirms the arithmetic and identifies option (c).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The abscissae 0, 1/4, 1/2, 3/4, 1 give h = 1/4 and n = 4 strips - an even number, so Simpson's one-third rule applies.
2. Ordinates: f0 = 1, f1 = 0.8, f2 = 0.66667, f3 = 0.57143, f4 = 0.5.
3. Simpson's rule: I = (h/3)[f0 + 4f1 + 2f2 + 4f3 + f4] with h/3 = 1/12.
4. Weighted sum: 1 + 4(0.8) + 2(0.66667) + 4(0.57143) + 0.5 = 1 + 3.2 + 1.33333 + 2.28571 + 0.5 = 8.31905.
5. I = 8.31905/12 = 0.69325.
6. The data are f(x) = 1/(1+x), whose exact integral is ln 2 = 0.69315 - so the estimate 0.6932 is correct, option (c).

30. 2025 · Q26 Trapezoidal Rule · Numerical

QUESTION

The trapezoidal rule is used to find the absolute error in evaluating $\int_a^{a+h} f(x)\ dx$. If the absolute error is $\frac{2}{3} f''(\xi),\ a<\xi<b$, what is the value of h?

OPTIONS

(a) 2

(b) 3

(c) 4

(d) 8

ANSWER (AI-derived — no official key exists for this paper)

(a) 2

EXAM SHORTCUT (AI-derived explanation)

The single-interval trapezoidal error is h^3 f''(xi)/12 in magnitude. Setting $h^3/12 = 2/3$ gives $h^3 = 8$ and h = 2.

TIPS & TRICKS (AI-derived explanation)

- For ONE strip the trapezoidal error is $-(h^3/12)$ f''(xi); the composite form $-(b-a)h^2$ f''/12 applies only to many strips.
- The integral runs from a to a + h, i.e. a single strip - read the limits before choosing the error formula.
- Equate coefficients of f''(xi): $h^3/12 = 2/3$, so $h^3 = 8$ and h = 2. The cube root is the key step.
- Sanity check: with h = 2 the coefficient is 8/12 = 2/3, matching the stated error exactly.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The limits a to a + h span a single subinterval of width h, so the appropriate error formula is the single-strip one.
2. For one strip, the trapezoidal rule has error E = $-(h^3/12)$ f''(xi) for some xi in the interval.
3. In absolute value, |E| = $(h^3/12)$|f''(xi)|.
4. Equating the coefficient with the given (2/3) f''(xi): $h^3/12 = 2/3$.
5. So $h^3 = 24/3 = 8$, giving h = 2.
6. Option (a) is correct.

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31. 2025 · Q27 Simpson's One-Third Rule · Theoretical

QUESTION

Consider the statements about $\int_a^b f(x)\ dx$: I. Trapezoidal rule gives exact result for a linear polynomial. II. Trapezoidal rule gives exact result for a quadratic polynomial. III. Simpson's one-third rule gives exact result for a cubic polynomial. IV. Simpson's one-third rule gives exact result for a quadratic polynomial. Which are correct?

OPTIONS

(a) I, III and IV

(b) I and IV only

(c) I and III only

(d) II, III and IV

ANSWER **(AI-derived — no official key exists for this paper)**

(a) I, III and IV

EXAM SHORTCUT **(AI-derived explanation)**

Trapezoidal is exact for degree 1 but not degree 2, so I is true and II false. Simpson's 1/3 is exact up to degree 3, so both III and IV are true.

TIPS & TRICKS **(AI-derived explanation)**

- Exactness degrees: trapezoidal 1, Simpson's 1/3 = 3, Simpson's 3/8 = 3, Weddle 5. Every statement here is a direct comparison against these numbers.
- Exactness for degree 3 IMPLIES exactness for degree 2, so statements III and IV must stand or fall together - both are true.
- The trapezoidal error involves f'', which is non-zero for a quadratic, so statement II is definitively false.
- Simpson's 1/3 fits parabolas yet is exact for cubics because the odd-degree error term cancels by symmetry - the classic bonus degree.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Statement I: the trapezoidal rule replaces the curve by a straight line, so it reproduces any polynomial of degree 1 exactly. TRUE.
2. Statement II: its error term is proportional to f'', which is a non-zero constant for a quadratic, so the rule is NOT exact for degree 2. FALSE.
3. Statement IV: Simpson's one-third rule is constructed by fitting a parabola through three points, so it integrates any quadratic exactly. TRUE.
4. Statement III: its error term is proportional to the FOURTH derivative, which vanishes for a cubic, so Simpson's one-third rule is exact for cubics as well - a bonus degree beyond the parabola it fits. TRUE.
5. The correct statements are I, III and IV.
6. Hence option (a) is correct.

32. 2025 · Q69 Numerical Integration · Numerical

QUESTION

What is the approximate value of $\int_0^1 x^3\ dx$ using the quadrature formula $\int_0^1 f(x)\ dx = \alpha\ f(0) + \beta\ f(x_1)$, exact for polynomials of as high a degree as possible?

OPTIONS

(a) 1/4

(b) 1/9

(c) 2/3

(d) 2/9

ANSWER (AI-derived — no official key exists for this paper)

(d) 2/9

EXAM SHORTCUT (AI-derived explanation)

Three free parameters (alpha, beta, x_1) means exactness up to degree 2. Force f = 1, x, x^2: alpha + beta = 1, beta*x_1 = 1/2, beta*x_1^2 = 1/3. Dividing gives x_1 = 2/3, then beta = 3/4, alpha = 1/4. So the integral of x^3 is (3/4)(2/3)^3 = 2/9.

TIPS & TRICKS (AI-derived explanation)

- Count the unknowns: an n-parameter rule can be made exact for polynomials of degree n-1, so three unknowns give exactness through degree 2.
- Set up the moment equations with the monomials 1, x, x^2 - always the fastest route for 'as high a degree as possible' problems.
- Divide the x^2 equation by the x equation to eliminate beta and get the node x_1 in one step.
- Cross-check the error: the exact value of the integral is 1/4, while the rule gives 2/9 = 0.222, so the rule is not exact for cubics - consistent with degree of precision 2. The distractor 1/4 is exactly the exact value, which is not what was asked.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The rule integral from 0 to 1 of f(x) dx = alpha f(0) + beta f(x_1) has three free parameters alpha, beta, x_1, so we can impose exactness for f = 1, x, x^2.
2. f = 1: alpha + beta = integral from 0 to 1 of 1 dx = 1.
3. f = x: alpha*0 + beta*x_1 = integral from 0 to 1 of x dx = 1/2, so beta*x_1 = 1/2.
4. f = x^2: beta*x_1^2 = integral from 0 to 1 of x^2 dx = 1/3.
5. Dividing the third equation by the second: x_1 = (1/3)/(1/2) = 2/3.
6. Then beta = (1/2)/x_1 = (1/2)/(2/3) = 3/4, and alpha = 1 - 3/4 = 1/4.
7. The rule is integral from 0 to 1 of f = (1/4) f(0) + (3/4) f(2/3).
8. Apply it to f(x) = x^3: (1/4)(0) + (3/4)(2/3)^3 = (3/4)(8/27) = 24/108 = 2/9.
9. Hence option (d) is correct.

33. 2026 · Q50 Simpson's One-Third Rule · Numerical

QUESTION

Consider the table:

x	0	1/4	1/2	3/4	1
f(x)	1	4/5	2/3	4/7	1/2

Apply Simpson's one-third rule with 4 equal subintervals to estimate the area A bounded by the curve and x-axis from $x=0$ to $x=1$. What is $2520A$?

OPTIONS

(a) 1547

(b) 1650

(c) 1747

(d) 1850

ANSWER (AI-derived — no official key exists for this paper)

(c) 1747

EXAM SHORTCUT (AI-derived explanation)

Simpson: A = (h/3)[f0 + 4(f1+f3) + 2f2 + f4] with h = 1/4. Sum = 1 + 4(4/5 + 4/7) + 4/3 + 1/2 = 1747/210, so A = 1747/2520 and 2520A = 1747.

TIPS & TRICKS (AI-derived explanation)

- The multiplier 2520 in the question is a hint: it equals 12 * 210, i.e. (3/h) times the LCD, so the answer is designed to be an exact integer - work in fractions, never decimals.
- Simpson's 1/3 pattern with 4 subintervals is 1, 4, 2, 4, 1 - odd-indexed ordinates get 4, the single interior even one gets 2.
- 4/5 + 4/7 = 48/35, and multiplying by 4 gives 192/35. Putting everything over 210 gives 210 + 1152 + 280 + 105 = 1747.
- Cross-check: the data are f(x) = 1/(1+x^2)... in fact f(x) = 4/(4+...) - numerically A = 1747/2520 = 0.6933, a plausible area under a curve falling from 1 to 0.5.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. There are 4 equal subintervals on [0, 1], so h = 1/4, with ordinates f0 = 1, f1 = 4/5, f2 = 2/3, f3 = 4/7, f4 = 1/2.
2. Simpson's one-third rule: A = (h/3)[f0 + 4(f1 + f3) + 2 f2 + f4] = (1/12)[f0 + 4(f1 + f3) + 2 f2 + f4].
3. f1 + f3 = 4/5 + 4/7 = (28 + 20)/35 = 48/35, so 4(f1 + f3) = 192/35; and the even-position ordinate gives 2 f2 = 4/3.
4. Sum inside the bracket = 1 + 192/35 + 4/3 + 1/2. Over the common denominator 210: 210/210 + 1152/210 + 280/210 + 105/210 = 1747/210.
5. A = (1/12)(1747/210) = 1747/2520.
6. Therefore 2520 A = 1747.

7. Hence option (c) is correct.

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34. 2026 · Q51 Trapezoidal Rule · Numerical

QUESTION

Given $f(1)=1,\ f(2)=0.5,\ f(3)=0.3,\ f(4)=0.25,\ f(5)=0.20$. What is $\int_1^5 f(x)\ dx$ estimated by the trapezoidal rule?

OPTIONS

(a) 1.45

(b) 1.55

(c) 1.65

(d) 1.75

ANSWER (AI-derived — no official key exists for this paper)

(c) 1.65

EXAM SHORTCUT (AI-derived explanation)

h = 1, so T = (1/2)[first + last + 2(interior)] = 0.5[1 + 0.2 + 2(0.5 + 0.3 + 0.25)] = 0.5[1.2 + 2.1] = 1.65.

TIPS & TRICKS (AI-derived explanation)

- Trapezoidal weights are 1, 2, 2, ..., 2, 1 multiplied by h/2 - only the two end ordinates get weight 1.
- Add the interior ordinates first (0.5 + 0.3 + 0.25 = 1.05), then double: that single grouping avoids most arithmetic slips.
- The spacing here is h = 1 (nodes 1 to 5), so the multiplier is 1/2 exactly.
- Since f is convex and decreasing, the trapezoidal rule overestimates the integral - consistent with 1.65 exceeding Simpson's 1.60 in the next question.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The nodes 1, 2, 3, 4, 5 are equally spaced with h = 1, and there are n = 4 subintervals.
2. Trapezoidal rule: integral = (h/2)[f(1) + 2(f(2) + f(3) + f(4)) + f(5)].
3. Interior sum: f(2) + f(3) + f(4) = 0.5 + 0.3 + 0.25 = 1.05, so twice it is 2.10.
4. End values: f(1) + f(5) = 1 + 0.20 = 1.20.
5. Total inside the bracket = 1.20 + 2.10 = 3.30.
6. Integral = (1/2)(3.30) = 1.65.
7. Hence option (c) is correct.

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35. 2026 · Q52 Simpson's One-Third Rule · Numerical

QUESTION

With the same data as Q51, what is $\int_1^5 f(x)\ dx$ estimated by Simpson's one-third rule?

OPTIONS

(a) 1.60

(b) 1.62

(c) 1.65

(d) 1.68

ANSWER (AI-derived — no official key exists for this paper)

(a) 1.60

EXAM SHORTCUT (AI-derived explanation)

Simpson with h = 1: (1/3)[f1 + 4(f2 + f4) + 2 f3 + f5] = (1/3)[1 + 4(0.75) + 0.6 + 0.2] = (1/3)(4.8) = 1.60.

TIPS & TRICKS (AI-derived explanation)

- Simpson's 1/3 rule needs an even number of subintervals; here n = 4, so it applies.
- The 1, 4, 2, 4, 1 pattern maps to f(1), f(2), f(3), f(4), f(5): the second and fourth ordinates get weight 4, the middle one gets 2.
- f(2) + f(4) = 0.5 + 0.25 = 0.75, and 4(0.75) = 3 exactly - the numbers are chosen to be clean.
- Compare with the trapezoidal answer 1.65: Simpson's rule is exact for cubics and hence far more accurate, and for this convex decreasing data it sits below the trapezoidal value.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The five ordinates at x = 1, 2, 3, 4, 5 with h = 1 give n = 4 subintervals, an even number, so Simpson's one-third rule applies.
2. Simpson's one-third rule: integral = (h/3)[f(1) + 4 f(2) + 2 f(3) + 4 f(4) + f(5)].
3. Odd-position interior ordinates (weight 4): f(2) + f(4) = 0.5 + 0.25 = 0.75, contributing 4(0.75) = 3.00.
4. Even-position interior ordinate (weight 2): 2 f(3) = 2(0.3) = 0.60.
5. End ordinates: f(1) + f(5) = 1 + 0.20 = 1.20.
6. Bracket total = 3.00 + 0.60 + 1.20 = 4.80.
7. Integral = (1/3)(4.80) = 1.60.
8. Hence option (a) is correct.

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36. 2026 · Q53 Trapezoidal Rule · Theoretical

QUESTION

The global truncation error estimate for the trapezoidal rule on $\int_a^b f(x)\ dx$, partitioned into n subintervals with $h = \frac{b-a}{n}$, $\xi \in [a,b]$, is:

OPTIONS

(a) $-\frac{1}{2}\ nh^2 f'(\xi)$

(b) $-\frac{1}{12} nh^3 f''(\xi)$

(c) $-\frac{3}{80} nh^5 f^{iv}(\xi)$

(d) $\frac{1}{2} nh^2 f''(\xi)$

ANSWER (AI-derived — no official key exists for this paper)

(b) $-\frac{1}{12} nh^3 f''(\xi)$

EXAM SHORTCUT (AI-derived explanation)

Local error per strip is -(h^3/12) f''; summing over n strips gives -(1/12) n h^3 f''(xi). Answer (b).

TIPS & TRICKS (AI-derived explanation)

- Local (one-strip) error of the trapezoidal rule is -(h^3/12) f''(xi); the global error over n strips is n times that, i.e. -(nh^3/12) f''(xi) = -((b-a)h^2/12) f''(xi).
- The order of the derivative must be 2 for the trapezoidal rule (it is exact for linear functions), so option (a) with f' is wrong on principle.
- The sign is negative for a convex f, expressing the familiar fact that trapezoids overestimate the area under a convex curve.
- Option (c) with h^5 and f'''' is the error term for Simpson's 3/8 rule family (Simpson's 1/3 rule global error is -(1/180) n h^5 f''''), not for the trapezoidal rule.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. On a single strip [x_i, x_{i+1}] of width h, Taylor expansion gives the trapezoidal error as -(h^3/12) f''(xi_i) for some xi_i in the strip.
2. Summing over the n strips that partition [a, b] and applying the intermediate value theorem to the average of the f''(xi_i) yields a single xi in [a, b].
3. Global truncation error E = -(1/12) n h^3 f''(xi).
4. Equivalently, since nh = b - a, E = -((b - a)/12) h^2 f''(xi), showing the method is second order accurate in h.
5. Option (a) uses the wrong derivative order, option (c) is a Simpson-type h^5 term, and option (d) has the wrong sign and coefficient.
6. Hence option (b) is correct.

N7 · NUMERICAL ANALYSIS

Summation of Series

3 questions · 2018: 0 2019: 0 2020: 1 2021: 0 2022: 0 2023: 0 2024: 0 2025: 1 2026: 1

Weight. 3 questions (1.7% of Numerical Analysis) · appeared in 3 of 9 papers · peak year 2020 (1)

Examiner's pattern. The thinnest numerical topic (3 items), and both patterns from the syllabus have now been sampled: recovering f from its first difference (2020, 2025) and multiplying two geometric series (2026). Cheap marks if you recognise the antidifference.

Must-know. Sum from a to b-1 of Delta f(x) = f(b) - f(a) (telescoping). To invert Delta on a polynomial, integrate in factorial-polynomial form: Delta^(-1) x^(n) = x^(n+1)/((n+1)h).

1. 2020 · Q70

Finite Differences of Different Orders · Numerical

QUESTION

The function whose first difference is $9x^2+11x+5$, with $h=1$, is

OPTIONS

(a) x^3+x^2+x+c

(b) x^3+2x^2+2x+c

(c) $3x^3+x^2+2x+c$

(d) $3x^3+x^2+x+c$

ANSWER (AI-derived — no official key exists for this paper)

(d) $3x^3+x^2+x+c$

EXAM SHORTCUT (AI-derived explanation)

Try F = ax^3 + bx^2 + cx + d. Then Delta F = 3a x^2 + (3a+2b)x + (a+b+c). Matching 9, 11, 5 gives a = 3, b = 1, c = 1, so F = 3x^3 + x^2 + x + c.

TIPS & TRICKS (AI-derived explanation)

- Since the first difference is quadratic, F must be CUBIC - that alone eliminates nothing here, so proceed by matching coefficients.
- Expand once and reuse: Delta x^3 = 3x^2 + 3x + 1, Delta x^2 = 2x + 1, Delta x = 1. Memorise this short table.
- Match from the highest power downwards: 3a = 9 gives a = 3, then 3a + 2b = 11 gives b = 1, then a + b + c = 5 gives c = 1.
- Verify by testing at x = 0: Delta F(0) = F(1) - F(0) = (3 + 1 + 1) - 0 = 5, which matches 9$(0)^2$ + 11(0) + 5 = 5.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let F(x) = a x^3 + b x^2 + c x + d and take h = 1.
2. Delta x^3 = $(x+1)^3$ - x^3 = 3x^2 + 3x + 1; Delta x^2 = 2x + 1; Delta x = 1; Delta d = 0.

3. So Delta F = $a(3x^2 + 3x + 1) + b(2x + 1) + c = 3a\,x^2 + (3a + 2b)\,x + (a + b + c)$.
4. Matching with $9x^2 + 11x + 5$: $3a = 9$ gives $a = 3$; $3a + 2b = 11$ gives $9 + 2b = 11$, so $b = 1$; $a + b + c = 5$ gives $3 + 1 + c = 5$, so $c = 1$.
5. The constant d is arbitrary, giving $F(x) = 3x^3 + x^2 + x + c$.
6. Hence option (d) is correct.

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2. 2025 · Q61 Sub-division of Intervals · Numerical

QUESTION

What is the function whose first difference is $3x^2+x+4$? (interval of differencing = unity)

OPTIONS

(a) $4x^2+7x+\lambda$

(b) $2x^3+x^2+\lambda$

(c) $2x^3+x^2+8x+\lambda$

(d) $x^3-x^2+4x+\lambda$

ANSWER (AI-derived — no official key exists for this paper)

(d) $x^3-x^2+4x+\lambda$

EXAM SHORTCUT (AI-derived explanation)

Anti-difference of a degree-2 polynomial is degree 3, so drop (a) at once. Only (b) and (d) are cubics with the right leading term test: Delta of x^3 gives $3x^2+3x+1$, so you need the x-coefficient to come out as +1 after correction. Test (d): Delta$(x^3-x^2+4x) = (3x^2+3x+1)-(2x+1)+4 = 3x^2+x+4$. Match on the first try.

TIPS & TRICKS (AI-derived explanation)

- Delta of a degree-n polynomial is degree n-1, so the answer must be one degree higher than the given first difference.
- Memorise Delta $x^3 = 3x^2+3x+1$, Delta $x^2 = 2x+1$, Delta $x = 1$ (with $h = 1$). Everything else is bookkeeping.
- The arbitrary constant lambda is present in every option, so it never discriminates - ignore it.
- Faster alternative: express $3x^2+x+4$ in factorial notation, 3x^(2)+4x^(1)+4x^(0) with x^(2)=x(x-1), then anti-difference term by term using Delta^{-1} x^(n) = x^(n+1)/(n+1).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. We need F(x) with Delta $F(x) = F(x+1) - F(x) = 3x^2 + x + 4$, taking the interval of differencing $h = 1$.
2. Basic differences: Delta $x^3 = (x+1)^3 - x^3 = 3x^2 + 3x + 1$; Delta $x^2 = (x+1)^2 - x^2 = 2x + 1$; Delta $x = 1$.
3. Try $F(x) = x^3 - x^2 + 4x$ + lambda (option d).

4. Delta F = $(3x^2 + 3x + 1) - (2x + 1) + 4(1) = 3x^2 + 3x + 1 - 2x - 1 + 4 = 3x^2 + x + 4$. This is exactly the required first difference.
5. Cross-check (b) F = $2x^3 + x^2$ + lambda: Delta F = $2(3x^2+3x+1) + (2x+1) = 6x^2 + 8x + 3$, which is wrong.
6. Cross-check (c) F = $2x^3 + x^2 + 8x$ + lambda: Delta F = $6x^2 + 8x + 3 + 8 = 6x^2 + 8x + 11$, also wrong.
7. Option (a) is a quadratic, whose first difference is linear, so it cannot equal a quadratic.
8. Hence option (d) is correct.

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3. 2026 · Q58 Summation of Series · Numerical

QUESTION

If $-1<x<1$, what is the product $(1+x+x^2+x^3+\cdots)(1-x+x^2-x^3+\cdots)$?

OPTIONS

(a) $1+x^2+x^4+x^6+\cdots$

(b) $1-x^2+x^4-x^6+\cdots$

(c) $1-x^2+x^6-x^{10}+\cdots$

(d) $1+x^2+x^6+x^{10}+\cdots$

ANSWER (AI-derived — no official key exists for this paper)

(a) $1+x^2+x^4+x^6+\cdots$

EXAM SHORTCUT (AI-derived explanation)

The two series sum to 1/(1-x) and 1/(1+x). Their product is $1/(1-x^2)$, whose expansion is $1 + x^2 + x^4 + x^6 + \ldots$

TIPS & TRICKS (AI-derived explanation)

- Recognise both series as geometric with ratios x and -x respectively; |x| < 1 guarantees convergence and justifies multiplying them term by term.
- The product of sums telescopes cleanly through the closed forms - never attempt a direct Cauchy product here.
- $1/(1 - x^2)$ expands by substituting x^2 into the standard geometric series, so all signs are positive - option (b) with alternating signs is the trap.
- Numerical check at x = 1/2: (2)(2/3) = 4/3, and 1/(1 - 1/4) = 4/3. Correct in five seconds.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For -1 < x < 1, the first series is geometric with first term 1 and ratio x: $1 + x + x^2 + \ldots = 1/(1 - x)$.
2. The second is geometric with first term 1 and ratio -x: $1 - x + x^2 - \ldots = 1/(1 + x)$.
3. Both converge absolutely for |x| < 1, so their product is the product of the sums: $[1/(1-x)][1/(1+x)] = 1/(1 - x^2)$.

4. Expanding $1/(1 - x^2)$ as a geometric series in x^2 (valid since $|x^2| < 1$) gives $1 + x^2 + x^4 + x^6 + \ldots$
5. Check at $x = 1/2$: $1/(1 - 0.25) = 4/3$, matching $(1/(1-0.5))(1/(1+0.5)) = 2 * (2/3) = 4/3$.
6. Hence option (a) is correct.

N8 · NUMERICAL ANALYSIS

Numerical Solution of Differential Equations - Euler, Milne, Picard, Runge-Kutta

26 questions · 2018: 2 2019: 3 2020: 1 2021: 4 2022: 1 2023: 2 2024: 4 2025: 4 2026: 5

Weight. 26 questions (14.4% of Numerical Analysis) · appeared in 9 of 9 papers · peak year 2026 (5)

Examiner's pattern. Steady 2-5 marks per paper and rising. Recurring: one or two Euler steps by hand, the absolute-stability interval for forward Euler, successive Picard iterates (first through fourth), and the observation that RK4 applied to y' = -y reproduces the Taylor series through h^4.

Must-know. Euler: y_(n+1) = y_n + h f(x_n, y_n); absolute stability for y' = lambda y needs |1 + hlambda| < 1, i.e. h < 2/|lambda|. Backward Euler is unconditionally stable. RK4: local error O(h^5), global O(h^4). Picard: y_(k+1)(x) = y0 + integral of f(t, y_k(t)) dt.

1. 2018 · Q44 Runge-Kutta Method · Theoretical

QUESTION

RK4 solves $\frac{dy}{dx}=f(x),\ y(0)=0$ with step h. The solution at $x=h$ is given by

OPTIONS

(a) $y(h)=\frac{h}{6}[3f(0)+f(\frac{h}{2})+3f(h)]$

(b) $y(h)=\frac{h}{6}[f(0)+5f(h)]$

(c) $y(h)=\frac{h}{6}[f(0)+3f(\frac{h}{2})+2f(h)]$

(d) $y(h)=\frac{h}{6}[f(0)+4f(\frac{h}{2})+f(h)]$

ANSWER (AI-derived — no official key exists for this paper)

(d) $y(h)=\frac{h}{6}[f(0)+4f(\frac{h}{2})+f(h)]$

EXAM SHORTCUT (AI-derived explanation)

When f does not involve y, RK4 has k2 = k3 = h f(h/2), so y(h) = (k1 + 2k2 + 2k3 + k4)/6 = (h/6) [f(0) + 4f(h/2) + f(h)] - exactly Simpson's one-third rule.

TIPS & TRICKS (AI-derived explanation)

- The weights 1, 2, 2, 1 divided by 6 collapse to 1, 4, 1 divided by 6 when the two midpoint slopes coincide. Recognising this makes the whole question a one-liner.
- Every option has the same 1/6 prefactor, so just check whether the interior weights sum correctly: 1 + 4 + 1 = 6, which only option (d) satisfies.
- This equivalence is worth remembering conceptually: RK4 applied to a pure quadrature problem IS Simpson's rule, which is why both are fourth-order accurate.
- The weights in any such formula must sum to 6 (so that a constant f is integrated exactly to h f); options (a), (b) and (c) sum to 7, 6 and 6 respectively, so use the x-symmetry check to finish.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The RK4 increments for dy/dx = f(x, y) are k1 = h f(x0, y0), k2 = h f(x0 + h/2, y0 + k1/2), k3 = h f(x0 + h/2, y0 + k2/2), k4 = h f(x0 + h, y0 + k3), and y1 = y0 + (k1 + 2k2 + 2k3 + k4)/6.
2. Here f depends on x only, so the y-arguments are irrelevant: k1 = h f(0), k2 = h f(h/2), k3 = h f(h/2), k4 = h f(h).
3. Since y(0) = 0, y(h) = (1/6)[h f(0) + 2h f(h/2) + 2h f(h/2) + h f(h)].
4. Combining the two equal middle terms: y(h) = (h/6)[f(0) + 4 f(h/2) + f(h)].
5. This is precisely Simpson's one-third rule applied on [0, h].
6. Hence option (d) is correct.

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2. 2018 · Q60 Picard's Method · Numerical

QUESTION

For $\frac{dy}{dx}=\frac{y}{x^2}$, $y(1)=1$, approximations y_1, y_2 by Picard's method with $y_0=y(1)$ have constant terms respectively

OPTIONS

(a) $1,1$

(b) $2,2$

(c) $2,\frac{3}{2}$

(d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(d) None of the above

EXAM SHORTCUT (AI-derived explanation)

y1 = 1 + integral of 1/t^2 = 2 - 1/x, constant term 2. y2 = 1 + integral of (2/t^2 - 1/t^3) = 5/2 - 2/x + 1/(2x^2), constant term 5/2. The pair (2, 5/2) is not listed, so 'None of the above'.

TIPS & TRICKS (AI-derived explanation)

- Picard's iteration is y_(n+1)(x) = y(x0) + integral from x0 to x of f(t, y_n(t)) dt. The lower limit is x0 = 1 here, NOT 0 - using 0 is the commonest error and makes the integrals divergent.
- Track only the constant term if that is all the question wants: each iterate's constant is 1 minus the value of the antiderivative at x0.
- y1 has constant 2 and y2 has constant 5/2, so the answer differs from every listed pair. Do not force a fit - 'None of the above' is a legitimate key in this paper.
- Cross-check with the exact solution y = exp(1 - 1/x): its expansion in powers of 1/x is 1 + (1 - 1/x) + (1 - 1/x)^2/2 + ..., whose constant terms build up towards e, consistent with the increasing sequence 2, 5/2, ...

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Picard's method for dy/dx = y/x^2 with y(1) = 1 uses y_(n+1)(x) = 1 + integral from 1 to x of y_n(t)/t^2 dt, starting from y_0 = 1.
2. First approximation: y1(x) = 1 + integral from 1 to x of 1/t^2 dt = 1 + [-1/t] from 1 to x = 1 + (1 - 1/x) = 2 - 1/x. Its constant term is 2.
3. Second approximation: y2(x) = 1 + integral from 1 to x of (2 - 1/t)/t^2 dt = 1 + integral from 1 to x of (2/t^2 - 1/t^3) dt.
4. The antiderivative is -2/t + 1/(2t^2), so the integral equals (-2/x + 1/(2x^2)) - (-2 + 1/2) = -2/x + 1/(2x^2) + 3/2.
5. Hence y2(x) = 5/2 - 2/x + 1/(2x^2), whose constant term is 5/2.
6. The constant terms are therefore 2 and 5/2, which matches neither (a), (b) nor (c). Option (d), None of the above, is correct.

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3. 2019 · Q74 Euler's Method · Numerical

QUESTION

Consider the IVP $y'=y+2x-1$, $y(0)=1$, over $0\leq x\leq 1$. The value of $y(0.2)$, obtained using Euler's method with $h=0.1$, is

OPTIONS

(a) 0

(b) 1

(c) 1.020

(d) 1.062

ANSWER (AI-derived — no official key exists for this paper)

(c) 1.020

EXAM SHORTCUT (AI-derived explanation)

At (0,1) the slope is 1 + 0 - 1 = 0, so y(0.1) = 1. At (0.1, 1) the slope is 1 + 0.2 - 1 = 0.2, so y(0.2) = 1 + 0.1(0.2) = 1.020.

TIPS & TRICKS (AI-derived explanation)

- Euler's method: y_(n+1) = y_n + h f(x_n, y_n). Two steps of h = 0.1 are needed to reach x = 0.2 - stopping after one step gives 1 and lands on option (b).
- The first slope is exactly zero here, which makes the first step trivial and is clearly deliberate on the examiner's part.
- Note 2x means 2 times x, so at x = 0.1 the term is 0.2, not 0.1. Misreading it gives 1.010 and no matching option.
- The exact solution is y = 2e^x - 2x - 1, giving y(0.2) = 1.0428; Euler's value 1.020 is the expected underestimate for this convex solution.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Euler's method for y' = f(x, y) is y_(n+1) = y_n + h f(x_n, y_n), with f(x,y) = y + 2x - 1, y(0) = 1 and h = 0.1.
2. Step 1 at (x0, y0) = (0, 1): f = 1 + 2(0) - 1 = 0, so y1 = 1 + 0.1(0) = 1, giving y(0.1) = 1.
3. Step 2 at (x1, y1) = (0.1, 1): f = 1 + 2(0.1) - 1 = 0.2.
4. So y2 = 1 + 0.1(0.2) = 1 + 0.02 = 1.02.
5. Hence y(0.2) = 1.020, option (c). (The exact solution y = 2e^{x} - 2x - 1 gives 1.0428, so Euler slightly underestimates, as expected.)

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4. **2019 · Q75** Picard's Method · Numerical

QUESTION

The third approximation of the IVP $\frac{dy}{dx}=x+y,\ y(0)=1$, obtained using Picard's method, is

OPTIONS

(a) $1+x+x^2+\frac{x^3}{3}+\frac{x^4}{24}$

(b) $1+x+x^2+\frac{x^3}{6}$

(c) $1+x^2+\frac{x^3}{6}$

(d) $x+\frac{x^2}{2}+\frac{x^3}{3}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $1+x+x^2+\frac{x^3}{3}+\frac{x^4}{24}$

EXAM SHORTCUT (AI-derived explanation)

Picard from y0 = 1: y1 = 1 + x + x^2/2; y2 = 1 + x + x^2 + x^3/6; y3 = 1 + x + x^2 + x^3/3 + x^4/24.

TIPS & TRICKS (AI-derived explanation)

- Picard iteration: y_(n+1)(x) = y(0) + integral from 0 to x of f(t, y_n(t)) dt. Each round raises the degree by one, so the THIRD approximation is a quartic.
- Degree check alone decides the item: only option (a) has an x^4 term, so it is the only candidate for a third approximation.
- Option (b) is y2, the second approximation - the classic 'one iteration short' trap.
- Cross-check with the exact solution y = 3e^{x} - x - 1 = 1 + x + x^2 + x^3/3 + x^4/12 + ...; Picard's terms agree up to x^3, as they must after three iterations.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Picard's method for dy/dx = x + y with y(0) = 1 uses y_(n+1)(x) = 1 + integral from 0 to x of (t + y_n(t)) dt, starting from y0 = 1.
2. First approximation: y1 = 1 + integral from 0 to x of (t + 1) dt = 1 + x + x^2/2.

3. Second approximation: y2 = 1 + integral from 0 to x of (t + 1 + t + t^2/2) dt = 1 + integral of (1 + 2t + t^2/2) dt = 1 + x + x^2 + x^3/6.
4. Third approximation: y3 = 1 + integral from 0 to x of (t + 1 + t + t^2 + t^3/6) dt = 1 + integral of (1 + 2t + t^2 + t^3/6) dt.
5. Integrating term by term: y3 = 1 + x + x^2 + x^3/3 + x^4/24.
6. Hence option (a) is correct.

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5. 2019 · Q78 Runge-Kutta Method · Numerical

QUESTION

What is the approximate value of y at x=1.1 for $\frac{dy}{dx}=3x+y^2$, obtained by Runge-Kutta method of second-order formula? [Given y=1.2 at x=1]

OPTIONS

(a) 1.722

(b) 1.644

(c) 1.044

(d) 1.022

ANSWER (AI-derived — no official key exists for this paper)

(a) 1.722

EXAM SHORTCUT (AI-derived explanation)

k1 = 0.1(3 + 1.44) = 0.444; k2 = 0.1(3.3 + 1.644^2) = 0.60027. Then y = 1.2 + (k1 + k2)/2 = 1.2 + 0.5221 = 1.722.

TIPS & TRICKS (AI-derived explanation)

- Second-order RK (Heun's method): k1 = h f(x0, y0), k2 = h f(x0 + h, y0 + k1), y1 = y0 + (k1 + k2)/2. Write the three lines out before substituting.
- Option (b) 1.644 is exactly y0 + k1, the plain EULER value - the trap for candidates who stop after k1.
- Watch the square: y0 + k1 = 1.644, and 1.644^2 = 2.7027. Squaring the updated y, not the original, is the essential second step.
- Averaging the two slopes is what makes the method second order; the final increment 0.5221 sits between k1 = 0.444 and k2 = 0.600, as it must.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The problem is dy/dx = f(x,y) = 3x + y^2 with x0 = 1, y0 = 1.2 and h = 0.1.
2. Second-order Runge-Kutta (Heun's form): k1 = h f(x0, y0), k2 = h f(x0 + h, y0 + k1), y1 = y0 + (k1 + k2)/2.
3. k1 = 0.1[3(1) + $(1.2)^2$] = 0.1[3 + 1.44] = 0.1(4.44) = 0.444.

4. y0 + k1 = 1.2 + 0.444 = 1.644, so k2 = 0.1[3(1.1) + $(1.644)^2$] = 0.1[3.3 + 2.7027] = 0.1(6.0027) = 0.60027.
5. y1 = 1.2 + (0.444 + 0.60027)/2 = 1.2 + 0.52214 = 1.72214.
6. Hence y(1.1) is approximately 1.722, option (a).

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6. 2020 · Q64 Euler's Method · Numerical

QUESTION

$y(x)$ solves $\frac{dy}{dx}=x+y,\ y(0)=0$. Euler's method estimates $y(0.5)$ in five steps. The estimate belongs to

OPTIONS

(a) $(0,0.1)$

(b) $(0.1,0.2)$

(c) $(0.2,0.3)$

(d) $(0.3,0.4)$

ANSWER (AI-derived — no official key exists for this paper)

(b) $(0.1,0.2)$

EXAM SHORTCUT (AI-derived explanation)

With h = 0.1 the Euler iterates are 0, 0, 0.01, 0.031, 0.0641, 0.11051 - so y(0.5) is about 0.1105, which lies in (0.1, 0.2).

TIPS & TRICKS (AI-derived explanation)

- Euler: y_(n+1) = y_n + h(x_n + y_n). Tabulate x_n and y_n side by side and work down; five short lines are quicker than any shortcut.
- The first step gives zero because both x and y vanish at the start - the solution only begins to move on the second step.
- The exact solution is y = e^x - x - 1, so y(0.5) = 0.1487. Euler's 0.1105 is the expected underestimate for a convex solution, and both lie in the same interval.
- Keep four decimals throughout: rounding to two decimals early can push the running total across an interval boundary.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Euler's method for dy/dx = x + y with y(0) = 0 and h = 0.1 is y_(n+1) = y_n + 0.1(x_n + y_n).
2. Step 1: at (0, 0), slope 0, so y(0.1) = 0 + 0.1(0) = 0.
3. Step 2: at (0.1, 0), slope 0.1, so y(0.2) = 0 + 0.1(0.1) = 0.01.
4. Step 3: at (0.2, 0.01), slope 0.21, so y(0.3) = 0.01 + 0.021 = 0.031.
5. Step 4: at (0.3, 0.031), slope 0.331, so y(0.4) = 0.031 + 0.0331 = 0.0641. Step 5: at (0.4, 0.0641), slope 0.4641, so y(0.5) = 0.0641 + 0.04641 = 0.11051.

6. Since 0.1 < 0.11051 < 0.2, the estimate lies in (0.1, 0.2), option (b).

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7. 2021 · Q42 Runge-Kutta Method · Numerical

QUESTION

Approximate y for $\frac{dy}{dx}=x+y$ by Runge–Kutta 4th order, at $x=0.1$, given $y(0)=1$:

OPTIONS

(a) 1.1103

(b) 1.2124

(c) 1.3236

(d) 1.4125

ANSWER (AI-derived — no official key exists for this paper)

(a) 1.1103

EXAM SHORTCUT (AI-derived explanation)

k1 = 0.1, k2 = 0.11, k3 = 0.1105, k4 = 0.12105. Then y = 1 + (k1 + 2k2 + 2k3 + k4)/6 = 1 + 0.66205/6 = 1.1103.

TIPS & TRICKS (AI-derived explanation)

- RK4 recipe: k1 = h f(x0,y0); k2 = h f(x0+h/2, y0+k1/2); k3 = h f(x0+h/2, y0+k2/2); k4 = h f(x0+h, y0+k3); then y1 = y0 + (k1 + 2k2 + 2k3 + k4)/6.
- Write the four k values in a column and keep five decimals - the options differ in the second decimal, so a single rounding slip is survivable but sloppiness is not.
- The k values should be close to one another for a smooth problem (0.100, 0.110, 0.1105, 0.121 here); a wildly different k signals an arithmetic error.
- Exact solution y = 2e^{x} - x - 1 gives y(0.1) = 1.110342, matching RK4 to six decimals - a striking illustration of fourth-order accuracy.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For dy/dx = f(x,y) = x + y with x0 = 0, y0 = 1 and h = 0.1:
2. k1 = h f(0, 1) = 0.1(0 + 1) = 0.1.
3. k2 = h f(0.05, 1.05) = 0.1(0.05 + 1.05) = 0.11.
4. k3 = h f(0.05, 1 + 0.055) = 0.1(0.05 + 1.055) = 0.1105.
5. k4 = h f(0.1, 1 + 0.1105) = 0.1(0.1 + 1.1105) = 0.12105.
6. y(0.1) = 1 + (0.1 + 0.22 + 0.221 + 0.12105)/6 = 1 + 0.66205/6 = 1.11034, i.e. 1.1103, option (a).

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8. 2021 · Q47 Runge-Kutta Method · Factual

QUESTION

RK4's local truncation error e is

OPTIONS

(a) $e = ch^4 + O(h^5)$

(b) $e = ch^5 + O(h^6)$

(c) $e = ch^3 + O(h^4)$

(d) $e = ch^2 + O(h^3)$

ANSWER (AI-derived — no official key exists for this paper)

(b) $e = ch^5 + O(h^6)$

EXAM SHORTCUT (AI-derived explanation)

RK4 is a fourth-order method, meaning GLOBAL error O(h^4); the LOCAL truncation error is one power higher, O(h^5).

TIPS & TRICKS (AI-derived explanation)

- Local versus global: a method of order p has local truncation error O(h^(p+1)) and global error O(h^p). The one-power gap arises because errors accumulate over about 1/h steps.
- The question says LOCAL, so the answer must carry h^5 - option (a)'s h^4 is the global error, the intended trap.
- Comparison table: Euler is order 1 (local h^2), RK2 order 2 (local h^3), RK4 order 4 (local h^5).
- Practical consequence: halving h reduces RK4's global error by a factor of about 16, which is why it remains the workhorse single-step method.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The classical fourth-order Runge-Kutta method matches the Taylor expansion of the solution through the h^4 term.
2. The first unmatched term is of order h^5, so the error committed in a SINGLE step - the local truncation error - is of the form e = c h^5 + O(h^6).
3. Accumulating this over the roughly (b-a)/h steps needed to traverse the interval gives a global error of order h^4, which is why RK4 is called a fourth-order method.
4. The question asks specifically for the LOCAL truncation error.
5. Hence e = c h^5 + O(h^6), option (b).

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9. 2021 · Q50 Picard's Method · Numerical

QUESTION

Second approximation to y at $x=0.2$ for $\frac{dy}{dx} = x\text{-}y$, $y(0)=1$, by Picard's method:

OPTIONS

(a) 0.8278

(b) 0.8387

(c) 0.8327

(d) 0.8297

ANSWER (AI-derived — no official key exists for this paper)

(b) 0.8387

EXAM SHORTCUT (AI-derived explanation)

y1 = 1 - x + x^2/2 and y2 = 1 - x + x^2 - x^3/6. At x = 0.2 that is 1 - 0.2 + 0.04 - 0.001333 = 0.8387.

TIPS & TRICKS (AI-derived explanation)

- Picard: y_(n+1)(x) = y(0) + integral from 0 to x of f(t, y_n(t)) dt. Each iteration raises the degree by one, so the SECOND approximation is a cubic.
- Integrate carefully: f(t, y1) = t - (1 - t + t^2/2) = 2t - 1 - t^2/2, whose integral is x^2 - x - x^3/6.
- Evaluate in one pass: 1 - 0.2 = 0.8, plus 0.04 gives 0.84, minus 0.008/6 = 0.00133 gives 0.83867.
- The options differ in the third decimal, so keep x^3/6 = 0.001333 to five places rather than rounding early.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Picard's method for dy/dx = x - y with y(0) = 1 uses y_(n+1)(x) = 1 + integral from 0 to x of (t - y_n(t)) dt, starting from y0 = 1.
2. First approximation: y1 = 1 + integral from 0 to x of (t - 1) dt = 1 + x^2/2 - x.
3. Second approximation: the integrand is t - y1(t) = t - (1 - t + t^2/2) = 2t - 1 - t^2/2.
4. Integrating: the integral from 0 to x is x^2 - x - x^3/6, so y2 = 1 - x + x^2 - x^3/6.
5. At x = 0.2: 1 - 0.2 + 0.04 - (0.008)/6 = 0.84 - 0.001333 = 0.838667.
6. Hence y(0.2) is approximately 0.8387, option (b).

---

10. 2021 · Q68 Euler's Method · Numerical

QUESTION

For $\frac{dy}{dx} = x^2 + y^2$, $h = 0.1$, $y(0) = 0$, $0 \le x \le 0.4$, Euler's method gives $y_4 \approx$

OPTIONS

(a) 0.002

(b) 0.005

(c) 0.009

(d) 0.014

ANSWER (AI-derived — no official key exists for this paper)

(d) 0.014

EXAM SHORTCUT (AI-derived explanation)

Euler with h = 0.1 gives y1 = 0, y2 = 0.001, y3 = 0.0050, y4 = 0.0140.

TIPS & TRICKS (AI-derived explanation)

- Euler: y_(n+1) = y_n + h(x_n^2 + y_n^2). The y^2 term is negligible here because y stays tiny, so each step is essentially 0.1 x_n^2.
- Running the x^2 contributions: 0, 0.001, 0.004, 0.009 - their cumulative sum 0.014 is the answer to three decimals.
- Four steps are needed to reach x = 0.4; stopping at y3 = 0.005 lands on option (b), the planted 'one step short' trap.
- Keep four decimals: the y^2 contributions (about 2.5 x 10^{-5} at the last step) are genuinely negligible but should be acknowledged rather than assumed.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Euler's method for dy/dx = x^2 + y^2 with y(0) = 0 and h = 0.1 is y_(n+1) = y_n + 0.1(x_n^2 + y_n^2).
2. Step 1 at (0, 0): slope 0, so y(0.1) = 0.
3. Step 2 at (0.1, 0): slope 0.01, so y(0.2) = 0 + 0.001 = 0.001.
4. Step 3 at (0.2, 0.001): slope 0.04 + 0.000001 = 0.040001, so y(0.3) = 0.001 + 0.0040001 = 0.0050001.
5. Step 4 at (0.3, 0.0050001): slope 0.09 + 0.000025 = 0.090025, so y(0.4) = 0.0050001 + 0.0090025 = 0.0140026.
6. Hence y4 is approximately 0.014, option (d).

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11. 2022 · Q22 Euler's Method · Numerical

QUESTION

Euler's method is used to solve $\frac{dy}{dt} = \sin y,\ y(0)=1,\ t \in (0,7)$. The local truncation error is always bounded by:

OPTIONS

(a) $\frac{h}{2}$

(b) $\frac{h^2}{2}$

(c) $\frac{h^2}{4}$

(d) $\frac{3h}{2}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{h^2}{4}$

EXAM SHORTCUT (AI-derived explanation)

Euler's local truncation error is $(h^2/2)y''(xi)$. Here y'' = (cos y)(sin y) = (1/2)sin 2y, so |y''| <= 1/2 and the bound is $h^2/4$.

TIPS & TRICKS (AI-derived explanation)

- Euler LTE = $(h^2/2)$ y''(xi), so bounding the error means bounding the SECOND derivative of the solution.
- Differentiate the ODE itself: y' = sin y gives y'' = (cos y) y' = sin y cos y = (1/2) sin 2y, whose modulus never exceeds 1/2.
- Combining: |LTE| <= $(h^2/2)(1/2) = h^2/4$. Missing the double-angle simplification gives the looser and wrong bound $h^2/2$.
- Note the error is $O(h^2)$ LOCALLY and O(h) globally, which is why options (a) and (d) with a single power of h are the global-error distractors.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For Euler's method the local truncation error over one step is E = $(h^2/2)$ y''(xi) for some xi in the step.
2. The differential equation is y' = sin y, so differentiating with respect to t gives y'' = (cos y) y' = (cos y)(sin y).
3. Using the double-angle identity, y'' = (1/2) sin(2y).
4. Since |sin(2y)| <= 1 for all y, we have |y''| <= 1/2 everywhere.
5. Therefore |E| <= $(h^2/2)(1/2) = h^2/4$.
6. Option (c) is correct.

---

12. 2023 · Q57 Euler's Method · Theoretical

QUESTION

Euler's method is applied to the initial value problem $\frac{dy}{dx}=\lambda\ y;\ y(0)=1,\ \lambda<0$. Then the method is stable (absolutely) for step size h that satisfies:

OPTIONS

(a) $-2<\lambda\ h<2$

(b) $0<\lambda\ h<2$

(c) $-1<\lambda\ h<1$

(d) $-2<\lambda\ h<0$

ANSWER (AI-derived — no official key exists for this paper)

(d) $-2<\lambda\ h<0$

EXAM SHORTCUT (AI-derived explanation)

Euler gives y_(n+1) = (1 + lambda h) y_n, so stability needs |1 + lambda h| < 1, i.e. -2 < lambda h

< 0.

TIPS & TRICKS (AI-derived explanation)

- Absolute stability means the numerical solution DECAYS, which requires the amplification factor to satisfy |1 + lambda h| < 1.
- Solve the modulus inequality: -1 < 1 + lambda h < 1 gives -2 < lambda h < 0. The upper limit is 0, not 2.
- Since lambda < 0 is given, lambda h is automatically negative, so only the lower bound bites: h < 2/|lambda| - the practical step-size restriction.
- Options (a) and (b) admit positive lambda h, which would make the numerical solution grow while the true solution decays - impossible for a stable method.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Applying Euler's method to y' = lambda y gives y_(n+1) = y_n + h lambda y_n = (1 + lambda h) y_n.
2. Hence y_n = (1 + lambda h)^n y_0, and the numerical solution decays to zero - matching the true solution e^(lambda x) for lambda < 0 - precisely when |1 + lambda h| < 1.
3. The inequality |1 + lambda h| < 1 means -1 < 1 + lambda h < 1.
4. Subtracting 1 throughout: -2 < lambda h < 0.
5. Since lambda < 0 and h > 0, lambda h is automatically negative, so the binding condition is lambda h > -2, i.e. h < 2/|lambda|.
6. Hence the region of absolute stability is -2 < lambda h < 0, option (d).

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13. 2023 · Q58 Runge-Kutta Method · Numerical

QUESTION

The classical Runge-Kutta fourth order method is applied to $\frac{dy}{dx}$=-y; $y(0)=1$. The result obtained after applying one step of length h is:

OPTIONS

(a) $1\text{-}h+\frac{h^2}{2}$

(b) $1+h+\frac{h^2}{2}+\frac{h^3}{6}+\frac{h^4}{24}$

(c) $1\text{-}h+\frac{h^2}{2}\text{-}\frac{h^3}{6}+\frac{h^4}{24}$

(d) $1\text{-}h\text{-}\frac{h^2}{2}\text{-}\frac{h^3}{6}\text{-}\frac{h^4}{24}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $1\text{-}h+\frac{h^2}{2}\text{-}\frac{h^3}{6}+\frac{h^4}{24}$

EXAM SHORTCUT (AI-derived explanation)

For a linear problem y' = lambda y, RK4 reproduces the Taylor series to fourth order: y1 = 1 + (lambda h) + (lambda h)^2/2 + (lambda h)^3/6 + (lambda h)^4/24. With lambda = -1 the signs alternate.

TIPS & TRICKS (AI-derived explanation)

- RK4 applied to y' = lambda y gives exactly the first five terms of e^(lambda h) - that is what 'fourth order' means for a linear test problem.
- With lambda = -1 the powers of (-h) alternate in sign: 1 - h + h^2/2 - h^3/6 + h^4/24. Option (d), with all signs negative, cannot be a truncated exponential.
- Compare with the exact solution e^(-h) = 1 - h + h^2/2 - h^3/6 + h^4/24 - ...; the method matches through h^4 and errs at h^5, confirming the local error O(h^5).
- Option (a) is the second-order (RK2) result and option (b) is the series for e^(+h) - both are one step away from the correct answer.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For the linear test equation y' = lambda y, the four RK4 increments are k1 = h lambda y0, k2 = h lambda(y0 + k1/2), k3 = h lambda(y0 + k2/2), k4 = h lambda(y0 + k3).
2. Substituting and collecting, y1 = y0[1 + (lambda h) + (lambda h)^2/2 + (lambda h)^3/6 + (lambda h)^4/24].
3. This is exactly the Taylor polynomial of e^(lambda h) truncated after the h^4 term, which is why the method is of fourth order.
4. Here lambda = -1 and y0 = 1, so lambda h = -h.
5. y1 = 1 - h + h^2/2 - h^3/6 + h^4/24.
6. Option (c) is correct.

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14. 2024 · Q61 Picard's Method · Numerical

QUESTION

If y satisfies the initial value problem

$$\frac{dy}{dx}=1+xy; \quad y(2)=0$$

then what is the approximation of y at $x=3$? (Use second approximation of Picard's method)

OPTIONS

(a) 3.0

(b) 2.33

(c) 1.33

(d) 0

ANSWER (AI-derived — no official key exists for this paper)

(b) 2.33

EXAM SHORTCUT (AI-derived explanation)

Picard gives y1 = x - 2 and then y2 = x + x^3/3 - x^2 - 2/3. At x = 3 this is 3 + 9 - 9 - 0.667 = 2.33.

TIPS & TRICKS (AI-derived explanation)

- Picard iteration starts at the GIVEN point: y_(n+1)(x) = y(2) + integral from 2 to x of f(t, y_n(t)) dt. Using 0 as the lower limit is the standard error.
- First iterate: with y0 = 0 the integrand is just 1, so y1 = x - 2 - a clean start.
- Second iterate: the integrand becomes 1 + t(t - 2) = t^2 - 2t + 1, whose antiderivative is t^3/3 - t^2 + t; evaluate between 2 and x.
- The constant is -(8/3 - 4 + 2) = -2/3; keeping it as a fraction avoids rounding drift before the final substitution.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Picard's method for dy/dx = 1 + xy with y(2) = 0 uses y_(n+1)(x) = 0 + integral from 2 to x of (1 + t y_n(t)) dt, starting from y0 = 0.
2. First approximation: y1(x) = integral from 2 to x of 1 dt = x - 2.
3. Second approximation: the integrand is 1 + t(t - 2) = t^2 - 2t + 1, whose antiderivative is t^3/3 - t^2 + t.
4. Evaluating from 2 to x: y2(x) = (x^3/3 - x^2 + x) - (8/3 - 4 + 2) = x^3/3 - x^2 + x - 2/3.
5. At x = 3: y2(3) = 27/3 - 9 + 3 - 2/3 = 9 - 9 + 3 - 0.667 = 2.333, i.e. about 2.33, option (b).

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15. 2024 · Q62 Runge-Kutta Method · Theoretical

QUESTION

Consider the following statements: I. Forward Euler method and backward Euler method for initial value problem have same order. II. Classical fourth order Runge-Kutta method for initial value problem has higher order than forward Euler method. III. Classical fourth order Runge-Kutta method for initial value problem has higher order than backward Euler method. Which of the statements given above are correct?

OPTIONS

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II and III

EXAM SHORTCUT (AI-derived explanation)

Forward and backward Euler are both first-order methods, and RK4 is fourth order, so it exceeds both. All three statements hold.

TIPS & TRICKS (AI-derived explanation)

- Order table: forward Euler 1, backward Euler 1, RK2 (Heun/midpoint) 2, RK4 4. Memorise it and all three statements answer themselves.

- Forward and backward Euler differ in STABILITY, not in order: backward Euler is implicit and unconditionally stable, but still first order.
- Order refers to the GLOBAL error exponent: O(h) for Euler, $O(h^4)$ for RK4 - a difference of three powers of h.
- Do not confuse order with stability or with cost per step; the question asks only about order.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Forward Euler, y_(n+1) = y_n + h f(x_n, y_n), matches the Taylor expansion through the h term, so its local error is $O(h^2)$ and its global error O(h) - first order.
2. Backward Euler, y_(n+1) = y_n + h f(x_(n+1), y_(n+1)), is implicit but likewise matches only through the h term, so it is also first order.
3. Statement I: both Euler methods therefore have the same order, namely 1. TRUE. (They differ in stability, not order.)
4. The classical fourth-order Runge-Kutta method matches the Taylor expansion through h^4, giving a global error $O(h^4)$ - fourth order.
5. Statements II and III: RK4's order 4 exceeds the order 1 of both forward and backward Euler. Both TRUE.
6. All three statements are correct, option (d).

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16. 2024 · Q64 Runge-Kutta Method · Numerical

QUESTION

Runge-Kutta method of order four is used to find the solution of the differential equation $\frac{dy}{dx}=f(x)$; $y(0)=0$ with step size h. If the solution is

$$y(h) = h[af(0)+bf(\frac{h}{2})+cf(h)]$$

then what are the values of a, b, c respectively?

OPTIONS

(a) 1/6, 2/3, 2/3

(b) 2/3, 2/3, 1/6

(c) 2/3, 1/6, 1/6

(d) 1/6, 2/3, 1/6

ANSWER (AI-derived — no official key exists for this paper)

(d) 1/6, 2/3, 1/6

EXAM SHORTCUT (AI-derived explanation)

For f independent of y, RK4 reduces to Simpson's rule: y(h) = (h/6)[f(0) + 4f(h/2) + f(h)] = h[(1/6)f(0) + (2/3)f(h/2) + (1/6)f(h)].

TIPS & TRICKS (AI-derived explanation)

- When f does not involve y, the two midpoint stages coincide, so the weights 1, 2, 2, 1 over 6 collapse to 1, 4, 1 over 6.
- Divide by 6 to read off the coefficients: 1/6, 4/6 = 2/3, 1/6. The end weights must be EQUAL by symmetry, which only option (d) satisfies.
- The weights must sum to 1 so that a constant f integrates to h f: 1/6 + 2/3 + 1/6 = 1. Options (a), (b) and (c) sum to 1.5, 1.5 and 1 respectively - so use the symmetry test.
- This is the same identity as 2018 Q44: RK4 applied to a pure quadrature problem IS Simpson's one-third rule.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For dy/dx = f(x) the RK4 increments are k1 = h f(0), k2 = h f(h/2), k3 = h f(h/2) and k4 = h f(h), since the y-arguments are irrelevant.
2. With y(0) = 0, y(h) = (k1 + 2k2 + 2k3 + k4)/6.
3. The two identical midpoint terms combine: y(h) = (h/6)[f(0) + 4f(h/2) + f(h)].
4. Writing this as h[a f(0) + b f(h/2) + c f(h)] gives a = 1/6, b = 4/6 = 2/3 and c = 1/6.
5. The weights are symmetric and sum to 1, as any such quadrature rule requires.
6. Hence (a, b, c) = (1/6, 2/3, 1/6), option (d).

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17. 2024 · Q80 Euler's Method · Numerical

QUESTION

Consider the initial value problem $\frac{dy}{dx}=-8y;\ y(0)=1$. What is the largest interval for step size h so that the forward Euler method to solve the initial value problem is absolutely stable?

OPTIONS

(a) $(0,\ 1/8)$

(b) $(0,\ 1/4)$

(c) $(0,\ 1/3)$

(d) $(0,\ 1/2)$

ANSWER (AI-derived — no official key exists for this paper)

(b) $(0,\ 1/4)$

EXAM SHORTCUT (AI-derived explanation)

Forward Euler is absolutely stable when |1 + lambda h| < 1. With lambda = -8 this gives -2 < -8h < 0, i.e. 0 < h < 1/4.

TIPS & TRICKS (AI-derived explanation)

- Stability condition: |1 + lambda h| < 1, which for real negative lambda becomes 0 < h < 2/|lambda|.
- Here |lambda| = 8, so h < 2/8 = 1/4 - one division once the general rule is known.

- The step must be positive, giving the open interval (0, 1/4); h = 1/4 exactly puts the amplification factor at -1, the boundary of stability.
- Contrast with backward Euler, which is unconditionally stable for this problem - no restriction on h at all.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Applying forward Euler to y' = lambda y gives y_(n+1) = (1 + lambda h) y_n, so y_n = (1 + lambda h)^n.
2. The numerical solution decays, matching the true solution e^(lambda x) for lambda < 0, precisely when |1 + lambda h| < 1.
3. This means -1 < 1 + lambda h < 1, i.e. -2 < lambda h < 0.
4. Here lambda = -8, so -2 < -8h < 0, which gives 0 < h < 1/4.
5. Equivalently h < 2/|lambda| = 2/8 = 1/4.
6. Hence the largest interval is (0, 1/4), option (b).

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18. 2025 · Q28 Euler's Method · Numerical

QUESTION

Forward Euler method finds the numerical solution y_1 at x=0.2 with h=0.2 for $\frac{dy}{dx}$=-2xy^2, y(0)=1. What is |y(0.2)-y_1| approximately?

OPTIONS

(a) 0.0215

(b) 0.0385

(c) 0.9615

(d) 1

ANSWER (AI-derived — no official key exists for this paper)

(b) 0.0385

EXAM SHORTCUT (AI-derived explanation)

Euler gives y1 = 1 + 0.2(-2 x 0 x 1) = 1, while the exact solution y = 1/(1+x^2) gives y(0.2) = 1/1.04 = 0.9615. The difference is 0.0385.

TIPS & TRICKS (AI-derived explanation)

- Solve the ODE exactly: dy/y^2 = -2x dx integrates to -1/y = -x^2 + C, and y(0) = 1 gives y = 1/(1 + x^2).
- The Euler step is trivially 1 here because the slope at x = 0 is zero - so the whole error is the curvature Euler misses.
- 1/1.04 = 0.961538, so the error is 1 - 0.961538 = 0.038462, i.e. about 0.0385.
- Option (c) 0.9615 is the exact VALUE, not the error - read what is being asked before selecting.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Euler's method: y1 = y0 + h f(x0, y0) with f(x,y) = -2xy^2, x0 = 0, y0 = 1, h = 0.2.
2. f(0, 1) = -2(0)(1)^2 = 0, so y1 = 1 + 0.2(0) = 1.
3. The exact solution: separating variables, dy/y^2 = -2x dx gives -1/y = -x^2 + C, and y(0) = 1 gives C = -1.
4. Hence -1/y = -x^2 - 1, i.e. y = 1/(1 + x^2).
5. At x = 0.2: y(0.2) = 1/(1 + 0.04) = 1/1.04 = 0.961538.
6. |y(0.2) - y1| = |0.961538 - 1| = 0.038462, i.e. about 0.0385, option (b).

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19. 2025 · Q29 Picard's Method · Numerical

QUESTION

The Picard method finds $y^{(1)}(x)$ (first iterate) for $\frac{dy}{dx}$=-2xy^2, $y(0)=1$. What is $|y^{(1)}(x)-y(x)|$ at $x=0.2$?

OPTIONS

(a) 0.9915

(b) 0.9615

(c) 0.0015

(d) 0.0005

ANSWER (AI-derived — no official key exists for this paper)

(c) 0.0015

EXAM SHORTCUT (AI-derived explanation)

Picard's first iterate is y = 1 - x^2, giving 0.96 at x = 0.2, while the exact value is 0.961538. The difference is 0.0015.

TIPS & TRICKS (AI-derived explanation)

- Picard: y1(x) = 1 + integral from 0 to x of (-2t)(1)^2 dt = 1 - x^2, using the constant y0 = 1 inside the integrand.
- The exact solution y = 1/(1+x^2) expands as 1 - x^2 + x^4 - ..., so Picard's first iterate matches through the x^2 term - the error is O(x^4).
- 0.961538 - 0.96 = 0.001538, i.e. about 0.0015. Option (d) 0.0005 is a third of that and does not arise.
- Compare with the previous part: Picard's error 0.0015 is 25 times smaller than Euler's 0.0385 - a good illustration of why series methods beat first-order stepping.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Picard's iteration for dy/dx = -2xy^2 with y(0) = 1 is y_(n+1)(x) = 1 + integral from 0 to x of (-2t y_n(t)^2) dt, starting from y0 = 1.
2. First iterate: y^(1)(x) = 1 + integral from 0 to x of (-2t) dt = 1 - x^2.
3. At x = 0.2: y^(1)(0.2) = 1 - 0.04 = 0.96.

4. The exact solution is $y = 1/(1 + x^2)$, so $y(0.2) = 1/1.04 = 0.961538$.
5. |y^(1)(0.2) - y(0.2)| = |0.96 - 0.961538| = 0.001538.
6. This is approximately 0.0015, option (c). (Note $1/(1+x^2) = 1 - x^2 + x^4 - \ldots$, so the first iterate is correct to order x^2.)

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20. 2025 · Q30 Euler's Method · Numerical

QUESTION

If $y(x)$ solves $\frac{dy}{dx}=1+x+y,\ y(0)=1$ using Euler's method with step $h=0.1$, then $y(0.3)$ equals:

OPTIONS

(a) 0.963

(b) 1.125

(c) 1.693

(d) 1.963

ANSWER **(AI-derived — no official key exists for this paper)**

(c) 1.693

EXAM SHORTCUT **(AI-derived explanation)**

Euler with h = 0.1: y(0.1) = 1.2, y(0.2) = 1.43, y(0.3) = 1.693.

TIPS & TRICKS **(AI-derived explanation)**

- Euler: y_(n+1) = y_n + h(1 + x_n + y_n). Tabulate x_n, y_n and the slope side by side and work down three lines.
- Slopes are 2.0, 2.3 and 2.63; each increment is one tenth of the slope, so the running values are 1.2, 1.43 and 1.693.
- Keep three decimals: the options differ in the first decimal, but rounding early can still shift the third step.
- The exact solution $y = 3e^x - x - 2$ gives y(0.3) = 1.7493, so Euler's 1.693 underestimates - as expected for a convex solution.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. Euler's method for dy/dx = 1 + x + y with y(0) = 1 and h = 0.1 is y_(n+1) = y_n + 0.1(1 + x_n + y_n).
2. Step 1 at (0, 1): slope = 1 + 0 + 1 = 2, so y(0.1) = 1 + 0.2 = 1.2.
3. Step 2 at (0.1, 1.2): slope = 1 + 0.1 + 1.2 = 2.3, so y(0.2) = 1.2 + 0.23 = 1.43.
4. Step 3 at (0.2, 1.43): slope = 1 + 0.2 + 1.43 = 2.63, so y(0.3) = 1.43 + 0.263 = 1.693.
5. The exact solution $y = 3e^x - x - 2$ gives y(0.3) = 1.7493, so Euler underestimates slightly, as expected.

6. Hence y(0.3) = 1.693, option (c).

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21. 2025 · Q65 Picard's Method · Conceptual

QUESTION

If $y_4(x)$ is the 4^{th} Picard iterate for $\frac{dy}{dx}=1+x^2+y^2;\ y(1)=2$, what is $y_4(1)$?

OPTIONS

(a) 1

(b) 2

(c) 3

(d) 4

ANSWER (AI-derived — no official key exists for this paper)

(b) 2

EXAM SHORTCUT (AI-derived explanation)

Picard's iterate is y_n(x) = y_0 + integral from x_0 to x of f(t, y_{n-1}(t)) dt. At x = x_0 the integral is zero, so every iterate equals the initial value. Here x_0 = 1 and y(1) = 2, so y_4(1) = 2 without computing a single integral.

TIPS & TRICKS (AI-derived explanation)

- Every Picard iterate reproduces the initial condition exactly - this is what makes the scheme consistent, and it is a favourite one-line trap.
- The same idea works for Taylor-series and Runge-Kutta starts: at the initial abscissa the numerical value is the exact initial value.
- Do not be intimidated by the nonlinear right-hand side 1 + x^2 + y^2 (a Riccati equation with no elementary closed form) - it is irrelevant at x = 1.
- If asked for y_4 at some other point, you would iterate y_n(x) = 2 + integral from 1 to x of (1 + t^2 + y_{n-1}$(t)^2$) dt starting with y_0(x) = 2.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Picard's method of successive approximations for dy/dx = f(x,y) with y(x_0) = y_0 defines y_n(x) = y_0 + integral from x_0 to x of f(t, y_{n-1}(t)) dt.
2. Here x_0 = 1, y_0 = 2 and f(x,y) = 1 + x^2 + y^2.
3. For every n >= 1, setting x = x_0 = 1 makes the integral run from 1 to 1, which is zero.
4. Therefore y_n(1) = y_0 + 0 = 2 for all n, and in particular y_4(1) = 2.
5. Hence option (b) is correct.

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22. 2026 · Q48 Picard's Method · Numerical

QUESTION

Which is the second Picard iterate of $\frac{dy}{dx}=1+x+y,\ y(0)=1$?

OPTIONS

(a) $1+3x+x^2+\frac{x^3}{6}$

(b) $1+2x+\frac{3}{2}x^2+\frac{x^3}{6}$

(c) $1+x+\frac{3}{2}x^2+\frac{x^3}{6}$

(d) $1+2x-\frac{3}{2}x^2+\frac{x^3}{6}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $1+2x+\frac{3}{2}x^2+\frac{x^3}{6}$

EXAM SHORTCUT (AI-derived explanation)

y_0 = 1; y_1 = 1 + integral of (2 + t) = 1 + 2x + x^2/2; y_2 = 1 + integral of (2 + 3t + t^2/2) = 1 + 2x + (3/2)x^2 + x^3/6.

TIPS & TRICKS (AI-derived explanation)

- Picard recursion: y_{n+1}(x) = y_0 + integral from x_0 to x of f(t, y_n(t)) dt, starting from the constant y_0 = y(x_0).
- Every iterate must satisfy the initial condition, so the constant term is always 1 - and the coefficient of x is always f(0,1) = 1 + 0 + 1 = 2, which immediately eliminates options (a) and (c).
- The x^3/6 term appears in every option, so the discriminating terms are those in x and x^2; concentrate the arithmetic there.
- Cross-check against the exact solution y = 3e^x - x - 2 = 1 + 2x + (3/2)x^2 + (1/2)x^3 + ...; the second iterate agrees through x^2, as Picard's theory predicts.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Picard's scheme for dy/dx = f(x,y) = 1 + x + y with y(0) = 1 is y_{n+1}(x) = 1 + integral from 0 to x of [1 + t + y_n(t)] dt.
2. Zeroth iterate: y_0(x) = 1.
3. First iterate: y_1(x) = 1 + integral from 0 to x of (1 + t + 1) dt = 1 + integral from 0 to x of (2 + t) dt = 1 + 2x + x^2/2.
4. Second iterate: y_2(x) = 1 + integral from 0 to x of [1 + t + 1 + 2t + t^2/2] dt = 1 + integral from 0 to x of (2 + 3t + t^2/2) dt.
5. Integrating: y_2(x) = 1 + 2x + (3/2)x^2/... more precisely 1 + 2x + 3x^2/2 + x^3/6.
6. Comparison with the exact solution y = 3e^x - x - 2 = 1 + 2x + 1.5x^2 + 0.5x^3 + ... confirms agreement up to the x^2 term.
7. Hence option (b) is correct.

23. 2026 · Q54 Euler's Method · Numerical

QUESTION

$\frac{dy}{dx} = \frac{y\text{-}x}{y+x}$; $y(0)=1$, step $h=0.05$. Using forward Euler, what is the approximate $y(0.1)$?

OPTIONS

(a) 1.005

(b) 1.05

(c) 1.10

(d) 1.50

ANSWER (AI-derived — no official key exists for this paper)

(c) 1.10

EXAM SHORTCUT (AI-derived explanation)

Step 1: slope (1-0)/(1+0) = 1, so y(0.05) = 1.05. Step 2: slope (1.05-0.05)/(1.05+0.05) = 1/1.1 = 0.909, so y(0.1) = 1.05 + 0.05(0.909) = 1.0955, which rounds to 1.10.

TIPS & TRICKS (AI-derived explanation)

- h = 0.05 but the target is x = 0.1, so TWO Euler steps are required - taking one step gives 1.05, which is exactly distractor (b).
- Euler formula: y_{n+1} = y_n + h f(x_n, y_n), always evaluating the slope at the point you are leaving.
- Update x as well as y between steps: the second slope uses x = 0.05, not x = 0.
- The exact solution of this homogeneous ODE gives y(0.1) about 1.0954, so the two-step Euler value is accurate to four decimals here - the options are far apart, so light rounding is safe.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Forward Euler: y_{n+1} = y_n + h f(x_n, y_n), with f(x, y) = (y - x)/(y + x), h = 0.05, x_0 = 0, y_0 = 1.
2. Step 1: f(0, 1) = (1 - 0)/(1 + 0) = 1. So y_1 = 1 + 0.05(1) = 1.05 at x = 0.05.
3. Step 2: f(0.05, 1.05) = (1.05 - 0.05)/(1.05 + 0.05) = 1/1.1 = 0.90909.
4. y_2 = 1.05 + 0.05(0.90909) = 1.05 + 0.04545 = 1.09545 at x = 0.10.
5. To two decimal places, y(0.1) is approximately 1.10.
6. Hence option (c) is correct.

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24. 2026 · Q55 Euler's Method · Theoretical

QUESTION

Backward Euler method solves $\frac{dy}{dx}$=-8y; $y(0)=1$. Which is correct?

OPTIONS

(a) For any $h>0$, absolutely stable

(b) Absolutely stable only for $h<1/8$

(c) Absolutely stable only for $h<1/4$

(d) Absolutely stable only for $h<1$

ANSWER (AI-derived — no official key exists for this paper)

(a) For any $h>0$, absolutely stable

EXAM SHORTCUT (AI-derived explanation)

Backward Euler gives y_{n+1} = y_n/(1 + 8h), and |1/(1+8h)| < 1 for every h > 0. Backward Euler is A-stable, so no step restriction exists. Answer (a).

TIPS & TRICKS (AI-derived explanation)

- Backward Euler applied to y' = lambda y gives the amplification factor 1/(1 - h lambda); with lambda = -8 this is 1/(1 + 8h), always inside the unit disc for h > 0.
- A-stability means the stability region contains the whole left half-plane - true for backward Euler and the trapezoidal (Crank-Nicolson) method, false for forward Euler and explicit Runge-Kutta.
- The step restrictions in options (b), (c), (d) are the sort produced by EXPLICIT methods; for forward Euler here the condition would be h < 2/8 = 0.25, which is exactly the trap in option (c).
- Price of A-stability: backward Euler is implicit, so each step needs an algebraic solve (trivial here because the equation is linear).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Backward (implicit) Euler for dy/dx = f(x,y) is y_{n+1} = y_n + h f(x_{n+1}, y_{n+1}).
2. With f = -8y: y_{n+1} = y_n - 8h y_{n+1}, so y_{n+1}(1 + 8h) = y_n, giving y_{n+1} = y_n/(1 + 8h).
3. The amplification factor is R(h) = 1/(1 + 8h).
4. Absolute stability requires |R(h)| < 1, i.e. |1 + 8h| > 1, which holds for every h > 0 (indeed 1 + 8h > 1).
5. Thus the method is absolutely stable for all positive step sizes - it is A-stable, and no restriction on h is needed.
6. (For comparison, forward Euler here would give R = 1 - 8h and require h < 0.25.)
7. Hence option (a) is correct.

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25. 2026 · Q57 Euler's Method · Numerical

QUESTION

$\frac{du}{dt}=-2tu^2$; $u(0)=1$, step $h=0.2$. Using forward Euler, what is the estimated $u(0.4)$?

OPTIONS

(a) 0.84

(b) 0.86

(c) 0.92

(d) 0.96

ANSWER (AI-derived — no official key exists for this paper)

(c) 0.92

EXAM SHORTCUT (AI-derived explanation)

At t = 0 the slope $-2tu^2$ is 0, so the first step leaves u unchanged at 1. Second step: slope = $-2(0.2)(1)^2 = -0.4$, so u(0.4) = 1 + 0.2(-0.4) = 0.92.

TIPS & TRICKS (AI-derived explanation)

- Notice the free first step: f(0, 1) = 0 because of the factor t, so u(0.2) = 1 with no arithmetic at all.
- Two steps are needed to reach t = 0.4 with h = 0.2 - stopping after one step would leave you at 1, which is not on the option list, a useful warning sign.
- Update t before evaluating the second slope: the derivative depends explicitly on t, so using t = 0 twice would give 1 again.
- Exact solution: $u = 1/(1 + t^2)$, so u(0.4) = 1/1.16 = 0.862. Euler's 0.92 overshoots, as expected for a convex decreasing solution with a coarse step - the option 0.86 is the exact value planted as a distractor.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Forward Euler: $u_{n+1} = u_n + h f(t_n, u_n)$, with $f(t, u) = -2 t u^2$, h = 0.2, $t_0 = 0$, $u_0 = 1$.
2. Step 1 (t = 0 to t = 0.2): $f(0, 1) = -2(0)(1)^2 = 0$, so $u_1 = 1 + 0.2(0) = 1$.
3. Step 2 (t = 0.2 to t = 0.4): $f(0.2, 1) = -2(0.2)(1)^2 = -0.4$, so $u_2 = 1 + 0.2(-0.4) = 1 - 0.08 = 0.92$.
4. Therefore the Euler estimate of u(0.4) is 0.92.
5. (For reference the exact solution is $u(t) = 1/(1 + t^2)$, giving u(0.4) = 0.8621; the Euler value is an overestimate because of the coarse step.)
6. Hence option (c) is correct.

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26. 2026 · Q60 Euler's Method · Numerical

QUESTION

Forward Euler solves $\frac{dy}{dx} = -20y;\ y(0) = 1$ with step h. For which value of h is the method absolutely stable?

OPTIONS

(a) 0.05

(b) 0.15

(c) 0.25

(d) 0.35

ANSWER **(AI-derived — no official key exists for this paper)**

(a) 0.05

EXAM SHORTCUT **(AI-derived explanation)**

Euler's amplification factor is 1 + h*lambda = 1 - 20h, and |1 - 20h| < 1 requires 0 < h < 0.1. Only h = 0.05 qualifies.

TIPS & TRICKS **(AI-derived explanation)**

- Forward Euler on y' = lambda y is absolutely stable when |1 + h lambda| < 1; for real negative lambda this means 0 < h < 2/|lambda|.
- Here |lambda| = 20, so h < 2/20 = 0.1 - compute this bound first, then simply read off which option fits.
- The other three options all exceed 0.1 and would produce growing oscillations: for h = 0.15 the factor is 1 - 3 = -2, so the numerical solution alternates in sign and doubles in magnitude each step.
- Contrast with backward Euler, which is A-stable and imposes no restriction on h - the pairing of these two questions is deliberate.

STEP-BY-STEP SOLUTION **(AI-derived explanation)**

1. For the test equation dy/dx = lambda y with lambda = -20, forward Euler gives y_{n+1} = y_n + h(lambda y_n) = (1 + h lambda) y_n.
2. The amplification factor is R = 1 + h lambda = 1 - 20h.
3. Absolute stability requires |R| < 1, i.e. -1 < 1 - 20h < 1.
4. The right inequality gives h > 0; the left gives 20h < 2, i.e. h < 0.1.
5. So the method is absolutely stable exactly for 0 < h < 0.1.
6. Checking the options: h = 0.05 gives R = 1 - 1 = 0, stable. h = 0.15 gives R = -2, h = 0.25 gives R = -4, h = 0.35 gives R = -6 - all unstable.
7. Hence option (a) is correct.